S130 PDK User Guide

Spec No. 002-22691

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Revision History

Release Date	Page	Description
September 28, 2021	–	Rev. ** (initial release).

1 Overview

The S130 process technology replaces the S8 family of related semiconductor manufacturing processes and uses a base set of design rules referred to as the 130 nm platform. SkyWater developed this technology primarily for programmable system-on-a-chip (PSoC) products, but it can also be adapted for multiple mixed-signal applications.

This document describes how to install and use the S130 process design kit (PDK), device parameterized cells (pcells), and physical verification and simulation tools. An appendix that describes density checking is also included.

1.1 Supported Tools

SkyWater used the electronic design automation (EDA) tools listed in *Table 1-1* to develop and test the S130 PDK. *Table 1-2* lists the PDK physical verification tools used.

Table 1-1. S130 PDK Supported Tool Versions

Tool Name	Minimum Version[1]
Cadence Physical Verification System (PVS)	19.12.003
Cadence Quantus Extraction Solution	19.13.000
Cadence Spectre Simulation Platform	19.12.003
Cadence Virtuoso Layout Suite XL (VXL)	IC6.1.7-64b
Siemens Calibre Design Solutions, including parasitic extraction	2019.1_18

1. These tool versions are the required minimum versions, and only these or subsequent versions are supported.

Table 1-2. Physical Verification Tools

Physical Verification Step	Cadence PVS	Siemens Cailbre
Design rule checking (DRC)	✓	✓
Layout versus schematic checking (LVS)	✓	✓
Parasitic extraction (PEX)	✓ (QRC)	✓ (xRC)

For more information, refer to the following sections:

- *Section 5 Physical Verification* on page 84
- *Section 6 Parasitic Extraction* on page 90

1.2 PDK Installation

SkyWater delivers the PDK as a downloadable, compressed tape archive (TAR) file: s130_<Vx.x.x>.tgz. The compressed files can be extracted using the following command: tar -xvzf s130_<Vx.x.x>.tgz. Execute the tar command in the SkyWater PDK root directory ($PDK_HOME) to create the <Vx.x.x> directory structure shown in *Figure 1-1*. The <Vx.x.x> directory is referred to as $PDK_HOME, and an environment variable with this name points to the s130/pdk/Vx.x.x path.

```
$SW_PDK_ROOT/
└pdk/
  └<Vx.x.x>/                       #$PDK_HOME
     ├DOC/                         #PDF files of design rules and release notes
     ├IO/
     ├LIBS/
     │ └S130/                      #S130 device library
     ├MODELS/
     │ └common/                    #S130 Spectre simulation models
     ├PROGS/
     ├PV/                          #Physical verification tool files
     ├SAMPLES/
     └TECH/
```

Figure 1-1. S130 PDK Directory Structure

Note that all PDK directory and file names are case-sensitive.

Follow these steps to finish installing the S130 PDK:

1. Copy the files proj_start_s130, cds.lib, display.drf, .cdsinit, pvtech.lib, and .simrc from the $PDK_HOME/SAMPLES directory to your local Cadence project directory.
2. A sample project start bash script (proj_start_s130) is included in the $PDK_HOME/SAMPLES directory. The PROJ_HOME environment variable is used in the PDK and set in the proj_start_s130 script. This variable points to the project directory where the script can be started. Its definition improves the ability of the PDK to point to the original working directory for tools such as Calibre when it establishes the run directories for DRC or LVS checking. Set this variable to the same directory as your project directory.
3. Carefully edit the script file to reflect the paths to your PDK installation, license, project, and tool directories.
4. Source the script to start VXL (bash proj_start_s130).

1.3 Layout Data Streaming

The user can stream layout databases to GDSII files and GDSII files to layout databases using the default stream file layer map, $PDK_HOME/LIBS/S130/S130.layermap. This layer map file is used by default whenever the user streams data in or out of the PDK.

Note: Consider using a via map when streaming out a layout. Without a via map, a separate cell for each via placement is generated in the GDS stream file, resulting in an excessive amount of unnecessary data. Be sure to include the via map with the GDSII file before the data is transferred to another PDK user.

1.4 S8 Compatibility

An S8 layout can be converted to an S130 layout using a special layer map file provided in the PDK: `$PDK_HOME/LIBS/S130/S8_to_S130.layermap`. This file merges the S8 diff and tap layers to the S130 diff layer. Data on other layers are also mapped accordingly, for example, the S8 li1 layer is mapped to the S130 li layer.

Table 1-3 summarizes the layer differences between S8 and S130.

Table 1-3. S8 and S130 Layer Differences

S8 Layer	S8 Purpose	S130 Layer	S130 Purpose	Comments
li1	drawing	li	drawing	
licon1	drawing	licon	drawing	
via	drawing	via1	drawing	
tap	drawing	diff	drawing	In S130, tap is determined by a Boolean operation.
tap	label			Obsolete.
tap	boundary			Obsolete.
hvi	drawing	v5	drawing	
		thkox	drawing	Added new layer for thick oxide.
vhvi	drawing	v12	drawing	
uhvi	drawing	v20	drawing	
areaid	padCenter	areaid	padLength	Changed name/use from pad center to pad length indicator.
areaid	extDrain20			Use areaid:extendedDrain.
areaid	contRes			Obsolete.
areaid	sigPadWell			Obsolete.
areaid	sigPadDiff			Obsolete.
areaid	sigPadMetNtr			Obsolete.
		areaid	pad_pwr	Added for latchup DRC.
		areaid	pad_gnd	Added for latchup DRC.
		areaid	pad_io	Added for latchup DRC.
all layers	short			Obsolete.
all layers	net			Obsolete.
all layers	cut			Obsolete.
		all layers	fill	Added for fill.
		areaid	guard_boundary	Added for latchup DRC, but not currently used.
		annotate	drawing	Added for DRC quality assurance (QA) automation.

S8 schematics cannot be brought directly into the S130 environment because the device names have all been changed. To perform this conversion task, create a CDL netlist from the S8 schematic and update the device names to the corresponding S130 names. The resulting edited netlist can be read into the S130 PDK using CDL in.

Note: In S8, the hvi layer was both a marker and a mask layer for thick oxide. In S130, v5 replaces the hvi layer, but v5 is only a marker layer. After converting an S8 layout to an S130 layout, the hvi layer automatically becomes the v5 layer, but the thkox layer must be manually added to high-voltage regions where required. LVS checking does not recognize high-voltage devices without the thkox layer present.

Some S8 layers, such as the areaid:padCenter layer, are no longer supported in S130. Consequently, some S130 DRC rules might not perform checks correctly on a converted layout. For example, the S130 latchup rules use a new method to determine which pads are signal, power, or ground and depend on the use of the new pad pcells. Therefore, the S130 latchup DRC rules might not perform properly on a converted database unless pad markers are added for signal, power, and ground. The user must ensure that the database to be converted to S130 is DRC correct before the conversion occurs to minimize DRC issues.

Note: Databases created in S130 are not backward compatible with S8 because the S130 diff layer cannot be split and remapped back to the S8 diff and tap layers.

2 Layout Environment Setup

2.1 Database Units

Only databases with 1000:1 units (1000 internal units equal 1 µm) are supported.

2.2 Layout Grid

The default layout grid enforced through DRC rules is 0.005 µm. Data cannot generally be created off-grid because the default snap grid is also set to 0.005 µm. In some cases, data can be drawn off-grid. For example, a path can be created with the center line on-grid, but a non-90° bend in the path might force a corner vertex to be off-grid. Editing might also force vertices off-grid, and careful use of alignment commands is recommended to ensure that resulting geometries are on the default grid. Users can also reset the snap grid to the smallest value of 0.001 µm, but this practice is typically discouraged.

2.3 Angles

DRC angle checks restrict some layers to 45° angles and other layers to 90° angles, and bends in gates are prohibited.

2.4 Layer Purpose Pairs

The S130 technology library is compiled using the technology file `$PDK_HOME/TECH/S130.tf`, which defines the available layers. All other libraries, including design libraries, must be attached to the main technology library.

Layers in the Cadence environment have purposes, and the layer and purpose are referenced as a layer purpose pair (LPP). Abbreviations for layer name and layer purpose are written as <layer name>:<layer purpose>. In external formats, such as a GDSII stream, this format is equivalent to a layer number and data type (as mapped in the PDK layer map file). Abbreviations are written as <layer number>:<data type>. For example, the LPP for diff:drawing is 63:20.

Certain layers are defined as “areaid” layers. They are used to identify areas that are checked to different design rules or to label devices for DRC and LVS purposes. In some cases, areaid layers are also used to generate mask layers.

2.5 Layer and Purpose Identification

Every layout layer has a purpose. *Table 2-1* lists the layer and purpose of each layer purpose pair used in the S130 PDK.

Table 2-1. S130 Layers and Layer Purposes (Page 1 of 3)

Layer Name	Purpose	GDSII Layer No.: Data Type	Layer Used to Identify
<layer>	mask	<layer>:0	Mask layer (post mask generation)
<layer>	maskDrop	<layer>:22, <layer>:42	Region for manual feature editing during mask generation
<layer>	res	<layer>:13	Resistor type (diff, poly, well, or metal)
<metlayer>	blockage	<layer>:10	Metal blockage layer used for place and route
<metlayer>	fill	<layer>:99	Reserved for generated or manual fill shapes and dummy patterns
<metlayer>	fill_block	<layer>:98	Region blocked from generated fill shapes
areaid	analog	81:79	Analog circuit region
areaid	core	81:2	Memory core that allows tighter design rules for exempt cells (for DRC)
areaid	critCorner	81:51	Critical corner area in the seal ring (for metal stress relief DRC)
areaid	critSid	81:52	Critical side area in the seal ring (for metal stress relief DRC)
areaid	deadZon	81:50	Dead zone area in the seal ring (for metal stress relief DRC)
areaid	dieCut	81:11	Die location within the frame
areaid	diode	81:23	Diode (for LVS checking)
areaid	esd	81:19	Electrostatic discharge (ESD) protection device (for DRC)
areaid	etest	81:101	E-test module (for DRC)
areaid	extendedDrain	81:57	Extended-drain device (for DRC and LVS checking)
areaid	fabBlock	82:82	Nikon marks, optical proximity correction (OPC) calibration structures, revision letters, critical dimension (CD) bars, and so on
areaid	frame	81:3	Pad in the frame (for DRC)
areaid	frameRect	81:12	Frame boundary and fill delimiter (for DRC and created layer DRC)
areaid	guard_boundary	81:73	Guard ring (for DRC); not currently used
areaid	hvnwell	81:63	Area where a low-voltage, field-effect transistor (FET) can be placed and the v5 drawing layer is prohibited (for DRC)
areaid	injection	81:17	Circuit susceptible to injection; marked area must be more than 100 μm from a signal pad diffusion (for DRC)
areaid	low_vt	81:108	20 V diode (dnwdiodehv_psub); not currently used
areaid	lowTapDensity	81:14	Low-density tap area with a low risk of latchup; must be > 50 μm from signal pad diffusion (for DRC)
areaid	lvNative	81:60	3.3 V native NMOS device (for LVS checking)
areaid	moduleCut	81:10	E-test module within the frame (for frame planning tool); reserved for SkyWater use only

Table 2-1. S130 Layers and Layer Purposes (Page 2 of 3)

Layer Name	Purpose	GDSII Layer No.: Data Type	Layer Used to Identify
areaid	nonCritSide	81:15	Area where critical side stress rules are not checked (for DRC); rarely used
areaid	opcDrop	81:54	Area where OPC does not automatically occur (for fab blocks and lithography calibration structures)
areaid	pad_gnd	81:72	Ground pad (for latchup checking)
areaid	pad_io	81:70	Signal pad (for latchup checking)
areaid	pad_pwr	81:71	Power pad (for latchup checking)
areaid	padLength	81:67	Pad orientation and length (for DRC and LVS checking)
areaid	photo	81:81	Deep n-well photodiode (for DRC)
areaid	rdlprobepad	81:27	Probe pad on the optional rdl drawing layer
areaid	rfdiode	81:125	RF diode that requires series resistance extraction (for LVS checking)
areaid	seal	81:1	Seal ring area; also a fill delimiter
areaid	standardc	81:4	Cell in the standard cell library (for DRC)
areaid	substrateCut	81:53	Isolated substrate regions with two different resistive connections > 100 µm apart (for latchup DRC and soft connection LVS checking)
bump	drawing	127:22	Solder bump region
cap2m	drawing	97:44	Metal-insulator-metal (MIM) capacitor plate between met4 and met5
capacitor	drawing	82:64	Interdigitated capacitor (for DRC and LVS checking)
capm	drawing	89:44	MIM capacitor plate between met3 and met4
diff	drawing	65:20	Field-oxide region for active devices
dnwell	drawing	64:18	Deep n-well region
hvntm	drawing	125:20	High-voltage n-tip implant region
hvtp	drawing	78:44	High-V_t, low-voltage (1.8 V) PMOS implant region
hvtr	drawing	18:20	High-V_t RF PMOS transistor implant region
inductor	drawing	82:24	Inductor region (for DRC)
inductor	term1	82:26	Inductor terminal label (for LVS checking)
inductor	term2	82:27	Inductor terminal label (for LVS checking)
inductor	term3	82:28	Inductor terminal label (for LVS checking)
ldntm	drawing	11:44	N-tip or n-type lightly doped drain (LDD) implant region inside silicon-oxide-nitride-oxide-silicon (SONOS) devices
li	drawing	67:20	Local interconnect layer between licon and mcon
licon	drawing	66:44	Local interconnect contact to diffusion and poly regions
LVS_exclude	drawing	84:44	Area within an e-test module exempt from LVS checking
lvtn	drawing	125.44	Low-V_t NMOS device implant region
mcon	drawing	67:44	Metal contact between li and met1

Table 2-1. S130 Layers and Layer Purposes (Page 3 of 3)

Layer Name	Purpose	GDSII Layer No.: Data Type	Layer Used to Identify
met1	drawing	68:20	First metal layer
met2	drawing	69:20	Second metal layer
met3	drawing	70:20	Third metal layer
met4	drawing	71:20	Fourth metal layer
met5	drawing	72:20	Fifth metal layer
npc	drawing	95:20	Nitride poly cut region for local interconnect contacts to poly
npn	drawing	82:20	NPN bipolar junction transistor (for DRC and LVS checking)
nsdm	drawing	93:44	N+ source/drain implant region
nsm	drawing	61:20	Nitride seal mask
ntm	drawing	26:20	N-tip or n-type LDD implant region
nwell	drawing	64:20	N-type well diffused on a p-type substrate active area
pad	drawing	76:20	Pad connection region
pmm	drawing	85:44	Polyimide region
pmm2	drawing	77:20	Alternate polyimide region
pnp	drawing	82:44	PNP bipolar junction transistor (for DRC and LVS checking)
poly	drawing	66:20	Polysilicon line or region
psdm	drawing	94:20	P+ source/drain implant region
pwbm	drawing	19:44	Region blocked from p-well implants for drain-extended MOS devices
pwde	drawing	124:20	P-well region for drain-extended MOS devices
rdl	drawing	74:20	Redistribution layer that connects a customer's top metal to bump terminals
rpm	drawing	86:20	Poly resistor; blocked from n+ poly implant
rrpm	drawing	102:20	300 Ω/sq. p+ poly resistor implant region
thkox	drawing	75:21	High-voltage (5 V or higher), thick gate oxide region
tunm	drawing	80:20	Tunnel oxide implant region for SONOS devices
ubm	drawing	127:21	Under solder bump metal layer
urpm	drawing	79:20	2000 Ω/sq. p- poly resistor implant region
v5	drawing	75:20	5 V region or device
v12	drawing	74:21	12–16 V region or device
v20	drawing	74:22	20 V region or device
via1	drawing	68:44	Contact between met1 and met2
via2	drawing	69:44	Contact between met2 and met3
via3	drawing	70:44	Contact between met3 and met4
via4	drawing	71:44	Contact between met4 and met5

2.6 S130 Layer Cross Section

This PDK uses an n-well and deep n-well (not shown in *Figure 2-1),* diffusion (diff), n+ and p+ implants (nsdm/psdm), poly, local interconnect (li), and up to five metal layers (met1–met5) plus a redistribution layer (rdl) to form devices and interconnects.

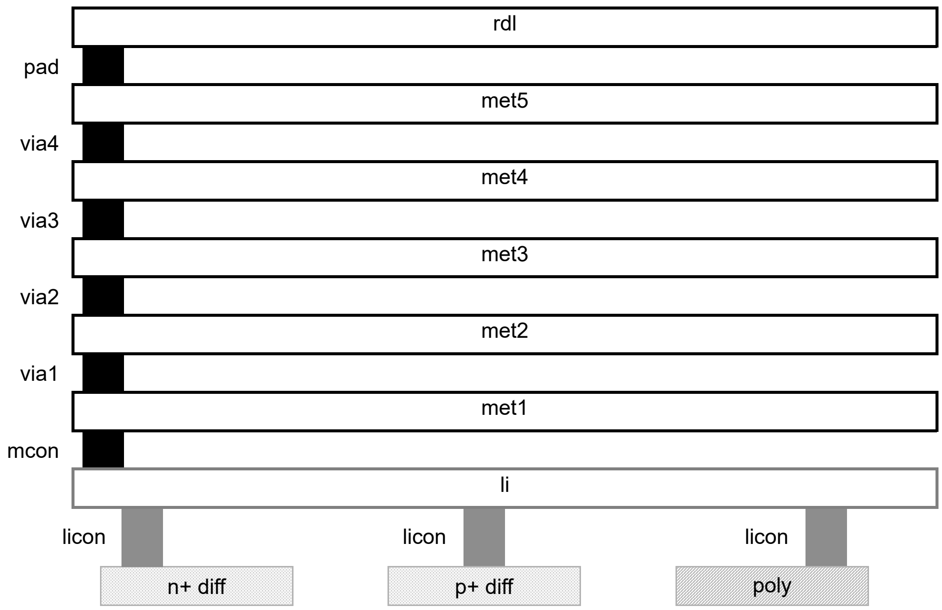


Figure 2-1. 5LM Stack Cross Section

2.7 Via Definitions

Table 2-2 defines the S130 vias, and *Figure 2-2* on page 20 shows the vias in the layout view.

Table 2-2. S130 Via Definitions (Page 1 of 2)

Via Name	Definition	Layers
DFLI	Diffusion to local interconnect	diff, licon, li
LIM1	Local interconnect to met1	li, mcon, met1
M1M2	First metal to second metal	met1, via1, met2
M2M3	Second metal to third metal	met2, via2, met3
M3M4	Third metal to fourth metal	met3, via3, met4
M4M5	Fourth metal to fifth metal	met4, via4, met5
NPDF	N+ diffusion to local interconnect	nsdm, diff, licon, li
NTAP[1]	N+ diffusion to local interconnect	psdm, diff, licon, li

1. Separate NTAP and PTAP vias are defined to accommodate larger implant enclosures required for tap vias.

Table 2-2. S130 Via Definitions (Page 2 of 2)

Via Name	Definition	Layers
PPDF	P+ diffusion to local interconnect	psdm, diff, licon, li
PTAP[1]	P+ diffusion to local interconnect	nsdm, diff, licon, li
PYLI	Poly to local interconnect	poly, npc, licon, li

1. Separate NTAP and PTAP vias are defined to accommodate larger implant enclosures required for tap vias.

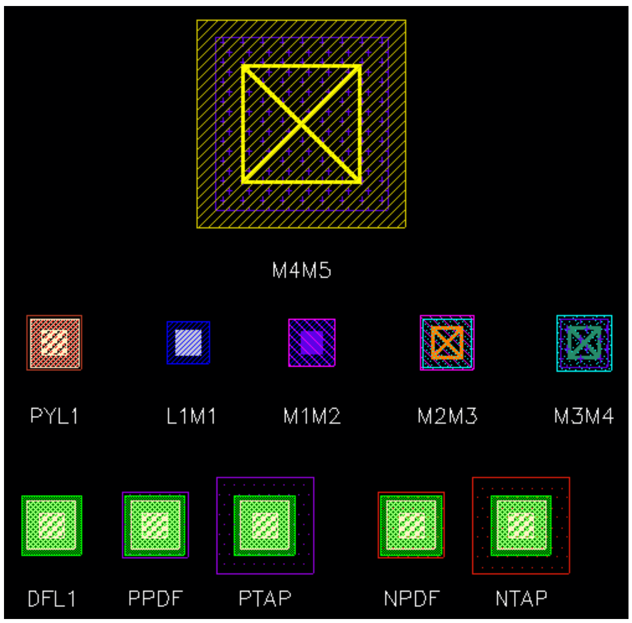


Figure 2-2. S130 Vias in Layout View

Note: Because vias do not appear as cells in the S130 library, via mapping must be used on streaming to and from GDSII to preserve the vias. Failure to use a via map results in an excessive number pf via cells created in the resulting stream file (one for each via encountered) with names like M1M2_CDNS_583334386101. Select the option to use a via map file on stream out to create the via map needed to stream in the resulting GDSII file.

2.8 Multipart Path Definitions

Figure 2-3 on page 21 shows the 12 basic multipart paths (MPPs) included in the S130 PDK.

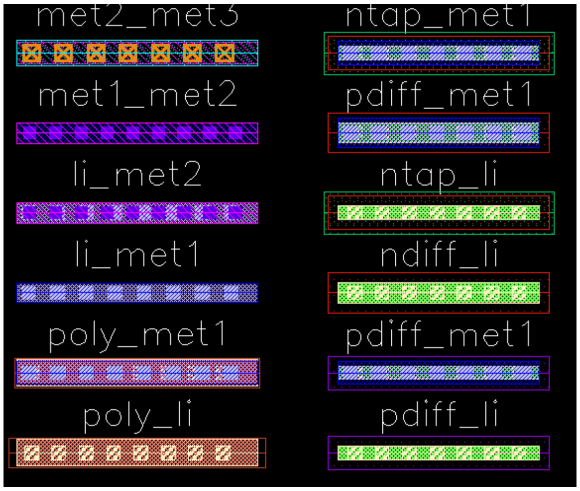


Figure 2-3. S130 Multipart Paths in Layout View

2.9 High-Voltage Markers and Layers

Three marker layers are used to indicate voltages higher than nominal (1.8 V): v5:drawing, v12:drawing, and v20:drawing. These layers are placed around devices and wells to indicate that they operate at higher than nominal voltage. The thkox:drawing layer is required over diffusion marked as a high-voltage region. The thkox layer can also be used as a mask layer for thick gate oxide.

A lower level voltage device placed in a higher voltage n-well, such as a 5 V device placed in a 12 V n-well, is an exception. In this case, a band of the required high-voltage marker layer is placed around the n-well that is also marked with a lower level voltage marker layer, as shown in *Figure 2-4*. This example marks the n-well as 12 V and the devices placed in the n-well as 5 V. The thkox layer is required over the 5 V devices.

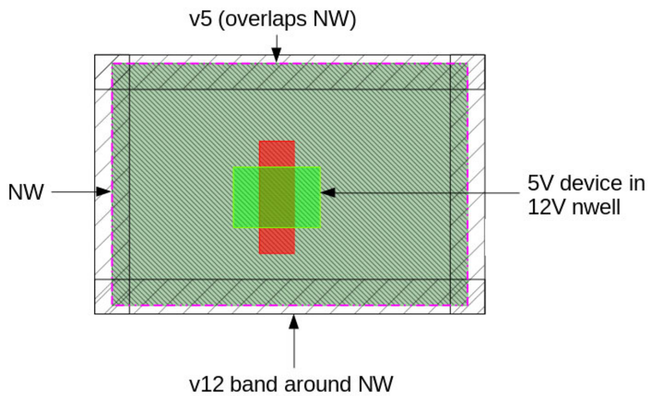


Figure 2-4. Example of a High-Voltage Marker Layer Exception

2.10 Other Layers of Interest

Layers pwbm (p-well blocking layer) and pwde (p-well exclusion layer) are used in the 20 V drain-extended pcells only. Do not draw on these layers.

Layers ldntm, hvntm, and lvntm appear in the layer selection window, but they are only visible because they are used in specific pcells, such as low- or high-voltage threshold devices.

Layer npc (nitride poly cut) is used whenever local interconnect (li) connects through a contact (licon) to poly.

Layers rpm and urpm are used in high-precision and high-resistance poly resistor pcells.

A pad is marked with an overlay rectangle of one of three layers:

- areaid:pad_io indicates that the pad and net connected to the pad are considered a signal net during latchup DRC.
- areaid:pad_pwr indicates that the pad and net connected to the pad are considered a power net during latchup DRC.
- areaid:pad_gnd indicates that the pad and net connected to the pad are considered a ground net during latchup DRC.

The type of marker placed with the pad is selected in the Create Instance form for bond-pad pcells as shown in *Figure 2-5* on page 23.

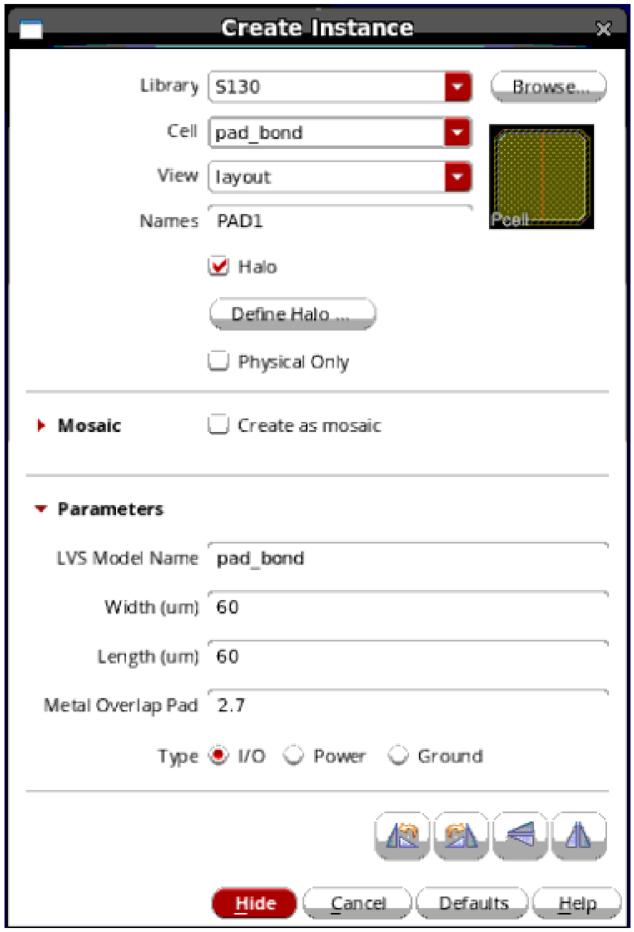


Figure 2-5. Pad Marker Selection in Bond-Pad Pcell Create Instance Form

2.11 Labels and Pins

To identify nets, place a shape on the layer purpose pin inside the appropriate layer purpose drawing. Then use the layer purpose label to place a net name with the origin or the label inside the pin shape as shown in *Figure 2-6*. As an alternative, use the **Create > Pin** command.

Figure 2-6. Create Pin Example

2.12 Resistors

Resistors can be created by placing a shape on the resistor purpose layer over an appropriate drawing purpose layer as shown in *Figure 2-7*. However, SkyWater recommends using the **Create > Instance** command to place a defined pcell from the S130 library, even for dummy resistors used to break a net. All resistors in the layout must have an equivalent symbol in the schematic for clean LVS checking results.

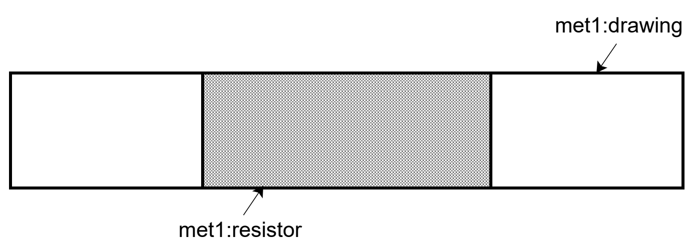


Figure 2-7. Draw Resistor Example

2.13 Deep N-Well Diode Use

Any time a deep n-well is used, one or two deep n-well diodes are recognized during LVS checking:

- Deep n-well to the substrate (`dnw_sub`)
- P+ tub to the deep n-well (`dipw_dnw`)

Figure 2-8 is a example of an n+ transistor in a deep n-well surrounded by an n-well ring. This layout forms an isolated p+ tub with a p+ tap. Contacts (licon and mcon) are omitted for clarity, and the n+ tap on the n-well ring is also not shown.

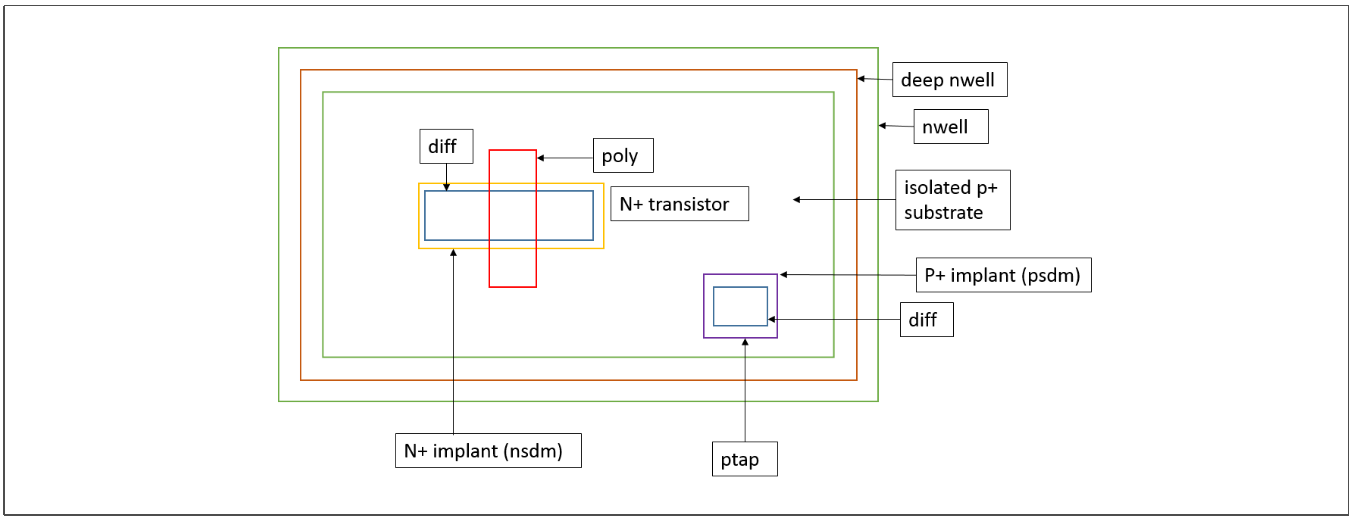


Figure 2-8. Layout Example of an N+ Transistor in an Isolated P+ Substrate

When placed in a schematic (see *Figure 2-9* on page 26), the area and perimeter values of two deep n-well diodes might be difficult to measure or calculate. By running LVS checking and examining the parameter mismatch, the layout area and perimeter values are easily back-annotated to the schematic.

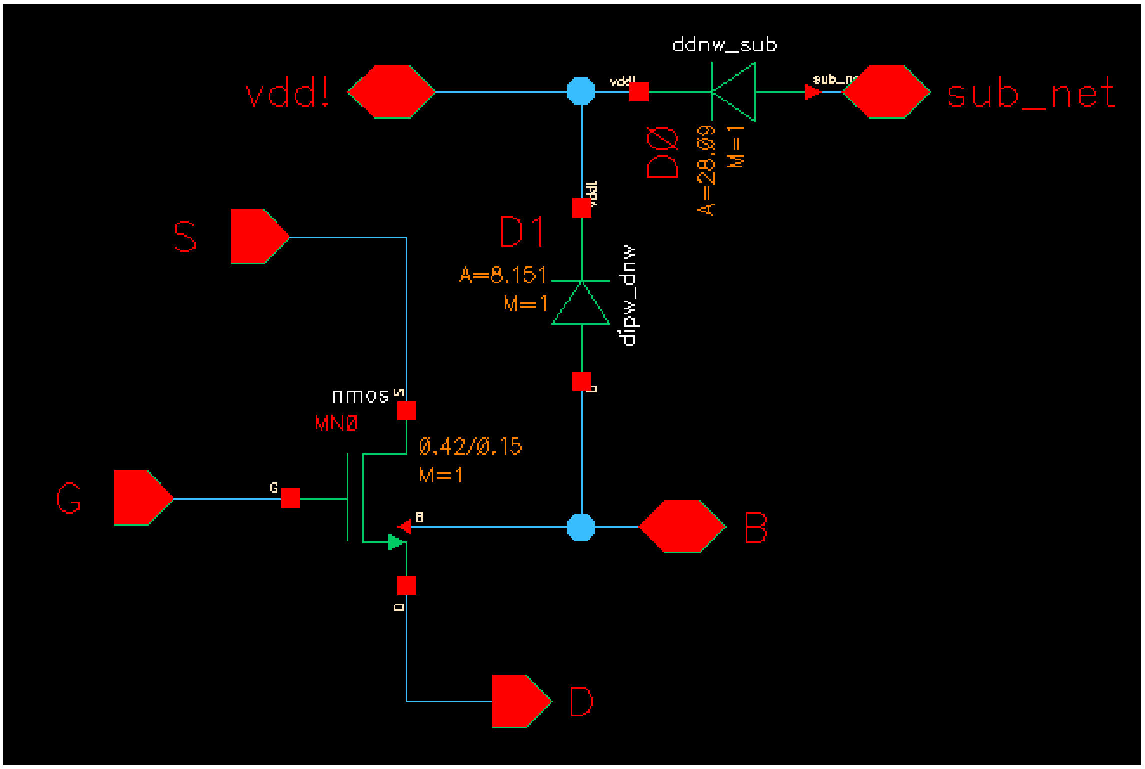


Figure 2-9. Schematic Example of an N+ Transistor in an Isolated P+ Substrate

2.14 P-Well Formation and Use

In general, a p-well is formed wherever n-well is not present. The S130 PDK includes layers pwell:pin and pwell:label, but it does not include layer pwell:drawing. The pwell:pin and pwell:label layers can be used to label an isolated p-well or other p-well; however, these layers do not replace the placement of a p+ tap.

An isolated p-well can be created in one of two ways:

- Create a deep n-well surrounded by a deep n-well ring.
- Place an areaid:substrateCut shape over the p-well.

2.15 Design Guidelines

1. Do not create your own devices because the PDK only supports the devices provided in the S130 device library (see *Section 3 Device Library* on page 28).
2. Be aware that flattening existing S130 library device placements or editing the resulting shapes can cause unwanted LVS checking issues. For example, a flattened device that is not modeled according to supported parameters can affect simulation results. Devices are only recognized during LVS checking if the required layers are present in the correct configuration.

3. Use multiple contacts and vias whenever possible to improve reliability.
4. Do not draw on the areaid:moduleCut layer; this layer is reserved for SkyWater use only.
5. Take care when drawing on fill layers and using fill blocking layers. Chip pattern density must pass DRC before manufacturing.

2.16 CMP Fill and Fill Generation

SkyWater performs chemical-mechanical polishing (CMP) or planarization on poly, diffusion, and all metal layers. The foundry also runs chip pattern density checks on all submitted designs and generates fill patterns for poly and diffusion before manufacturing. Customers must generate metal fill before tapeout. SkyWater informs the designer if a density error is detected on a chip delivered to the foundry. The DRC local pattern density design rules can be used to generate predictive fill to help identify any density issues before tapeout. For information on layout density checking using Calibre, refer to *Appendix A.1 Calibre DRC Density Checking* on page 103.

For designers who find the foundry automated fill generation tool inadequate for sensitive areas of the chip, they can manually place layer purpose fill using the PDK utility for metal fill generation. This utility can be found in `$PDK_HOME/PV/Calibre/DRC/calibre_generate_metal_fill.rul` and requires a Siemens Calibre Design for Manufacturing (DFM) tool suite license. A comparable utility for poly or diff fill generation is not provided.

2.17 Layer Display Editor

Often users want to change the way drawing layers are displayed in Virtuoso XL (VXL) according to their own preferences. To change the default settings, either edit a local copy of the default `display.drf` file in the `$PDK_HOME/SAMPLES` directory or invoke the VXL Display Resource Editor shown in *Figure 2-10*. To change the settings in VXL, right-click over the AV/NV/AS/NS bar above the layer definitions and select **Edit Display Resources** to open the Display Resource Editor form.

Figure 2-10. VXL Display Resource Editor

3 Device Library

In electronic circuit designs, a cell is the basic unit of functionality. A given cell might be placed or instantiated many times. A parameterized cell (pcell) represents a circuit component whose structure depends on one or more physical parameters. Most pcells are physical cells, but device symbols in circuit schematics can also be defined as pcells.

EDA tools automatically generate pcells based on parameter values. For example, a user can define different length and width parameters for a transistor pcell to create pcell instances that represent transistors of different sizes but otherwise similar characteristics. By using pcells, a circuit designer can easily generate a large number of similar structures and increase design productivity and consistency.

This section describes the various pcells available in the S130 PDK. The pcells are grouped by device type and presented as follows:

- MOS field-effect transistors
- Bipolar junction transistors
- Capacitors
- Pads
- Resistors
- Diodes
- Miscellaneous devices

Each device type section includes the cell names, operating ranges, device cross sections, pcell layout views and parameters, and LVS checking parameters.

3.1 MOS Field-Effect Transistors

Table 3-1. S130 MOSFET Devices (Page 1 of 2)

Device Type	Cell Name	Operating Voltage Ranges		
		V_{DS} (V)	V_{GS} (V)	V_{BS} (V)
1.8 V FET	nmos	0 to 1.95	0 to 1.95	0.3 to -1.95
	pmos	0 to -1.95	0 to -1.95	-0.1 to 1.95
Low-V_t FET	nmos_lvt	0 to 1.95	0 to 1.95	0.3 to -1.95
	pmos_lvt	0 to 1.95	0 to 1.95	-0.1 to 1.95
High-V_t PFET	pmos_hvt	0 to 1.95	0 to 1.95	-0.1 to 1.95
Native NFET	nmos_nat_v3	0 to 3.3	0 to 3.3	0 to -3.3
	nmos_nat_v5	0 to 5.5	0 to 5.5	0.3 to -5.5
5 V FET	nmos_v5	0 to 11	0 to 5.5	0 to -5.5
	pmos_v5	0 to -11	0 to -5.5	0 to 5.5

Table 3-1. S130 MOSFET Devices (Page 2 of 2)

Device Type	Cell Name	Operating Voltage Ranges		
		V_{DS} (V)	V_{GS} (V)	V_{BS} (V)
ESD FET	nmos_esd_nat_v5	0 to 11	0 to 5	0.3 to -5.5
	nmos_esd_v5	0 to 11	0 to 5	0 to -5.5
	pmos_esd_v5	To be provided in a future release		
12 V drain-extended FET	nmos_de_v12	0 to 16 (V_{GS} = 0) 0 to 11 (V_{GS} > 0)	0 to 5.5	0 to -2
	pmos_de_v12	0 to -16 (V_{GS} = 0) 0 to -10 (V_{GS} < 0)	0 to -5.5	0 to 2
20 V drain-extended NFET	nmos_de_v20	0 to 22	0 to 5.5	0 to -2
	nmos_de_iso_v20	0 to 22	0 to 5.5	0 to -2
	nmos_de_nat_v20	0 to 22	0 to 5.5	0 to -2
	nmos_de_zvt_v20	0 to 22	0 to 5.5	0 to -2
20 V drain-extended PFET	pmos_de_v20	0 to -22	0 to -5.5	0 to 2

3.1.1 NFET Cross Sections

Figure 3-1 is a cross section of a 1.8 V NFET with the p-well inside an optional deep n-well.

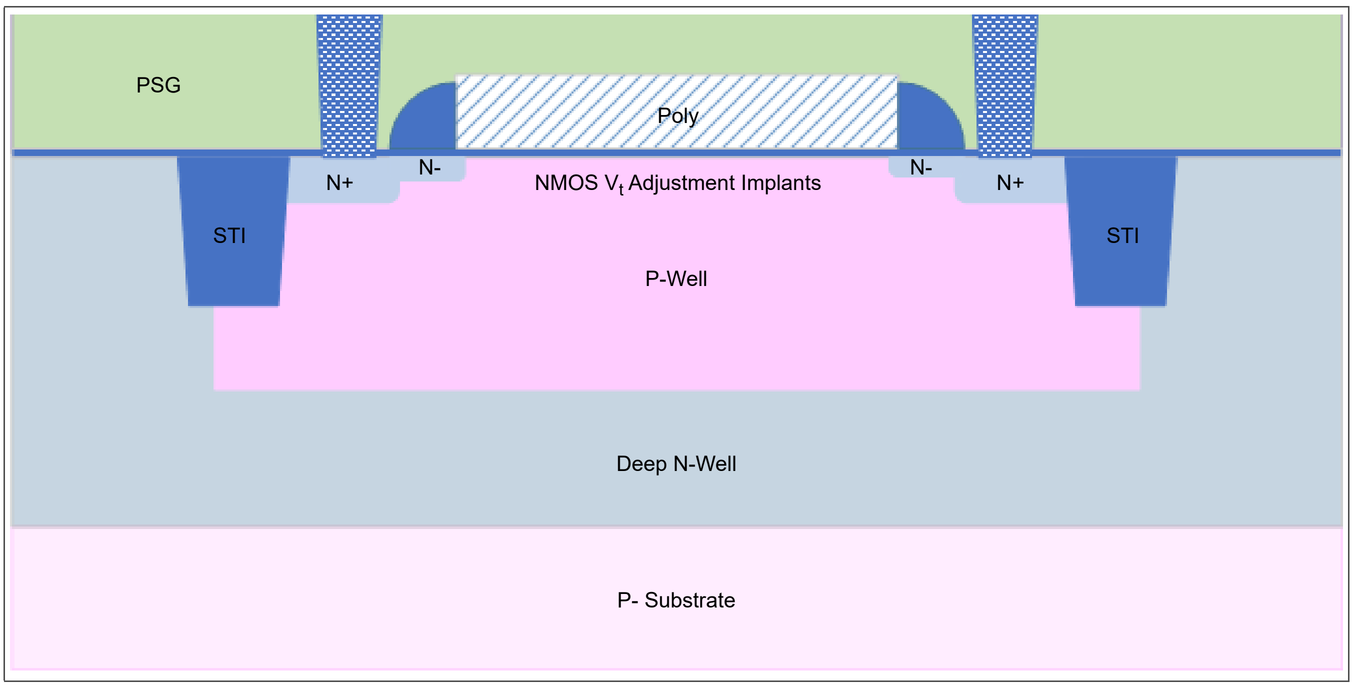


Figure 3-1. 1.8 V NFET Cross Section

The low-V_t NFET shown in *Figure 3-2* has threshold voltage implants to achieve a lower threshold voltage but is otherwise identical to the 1.8 V NFET.

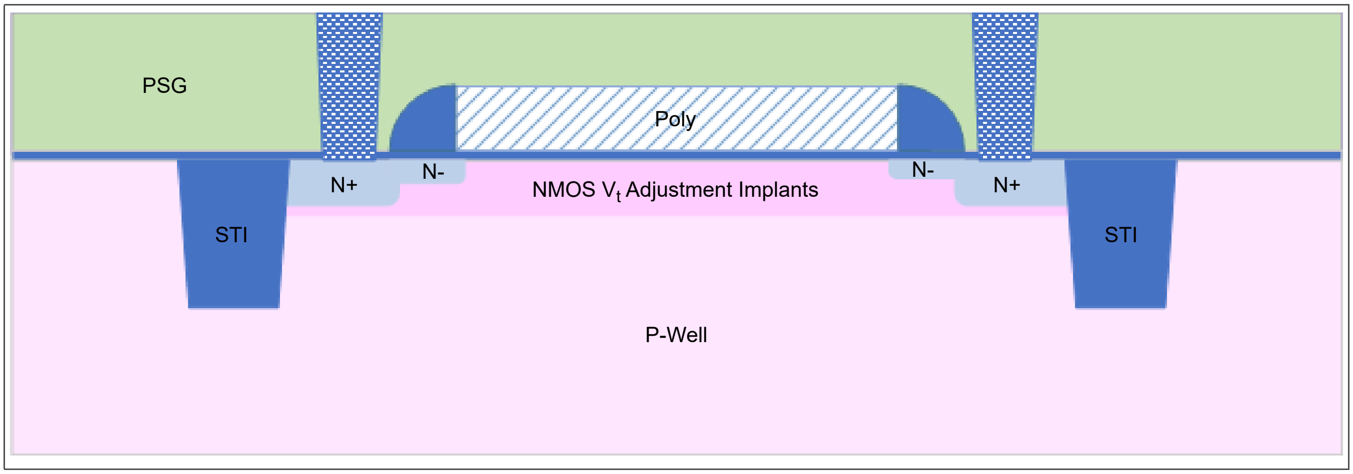


Figure 3-2. Low-V_t NFET Cross Section

Figure 3-3 is a cross section of a native NFET. The only differences between the 3 V and 5 V native NFET devices are the minimum gate length and V_{DS} limits.

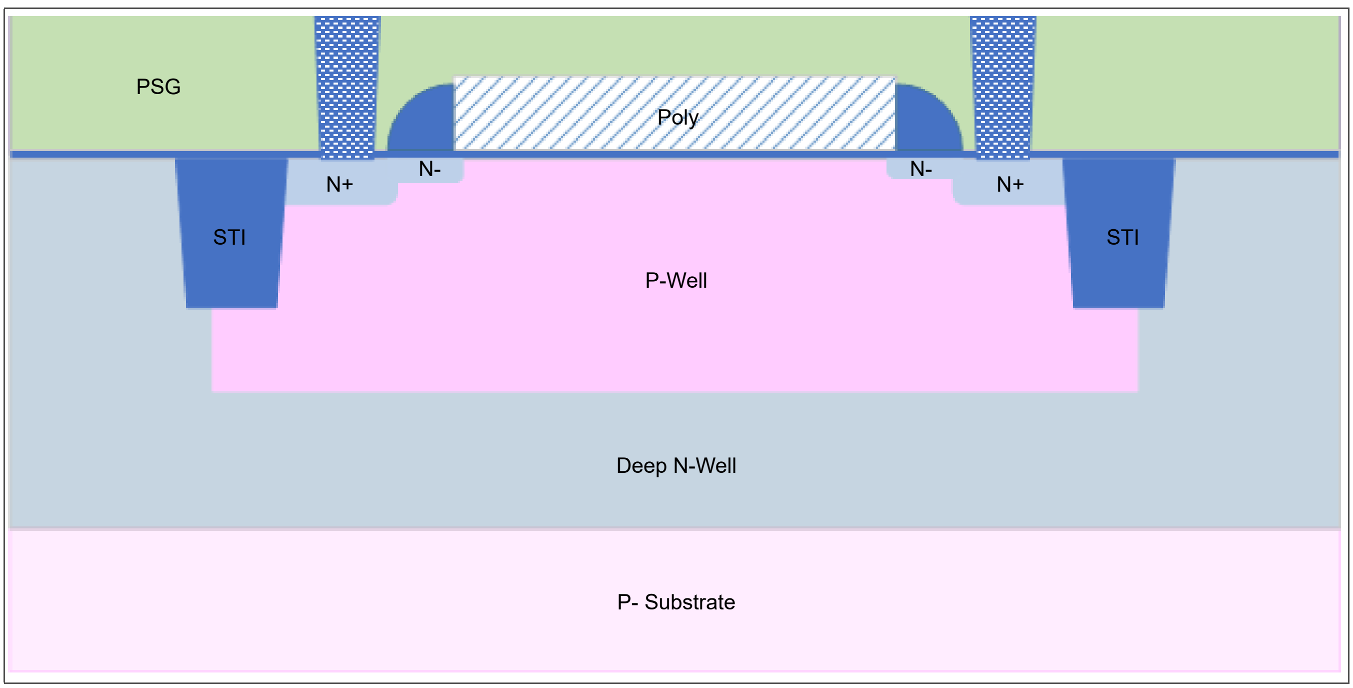


Figure 3-3. Native NFET Cross Section

Figure 3-4 is a cross section of a 5 V NFET.

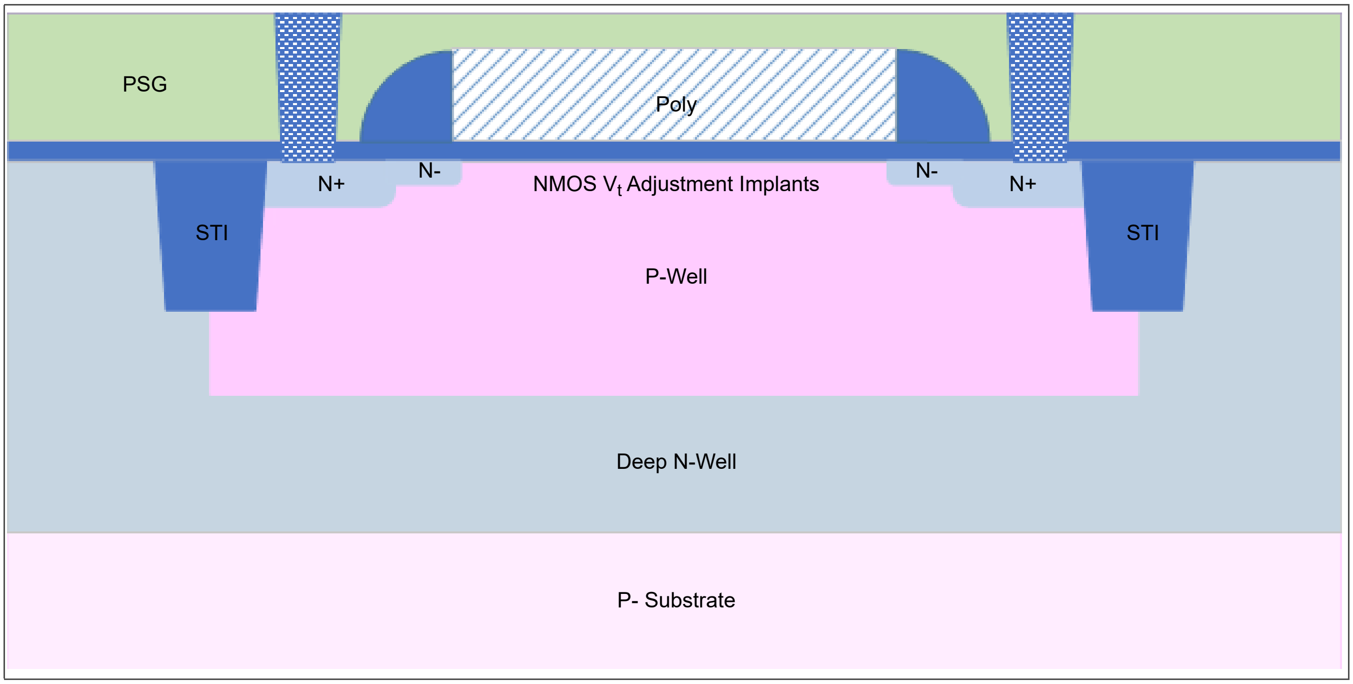


Figure 3-4. 5 V NFET Cross Section

Figure 3-5 is a cross section of a 5 V ESD-protected NFET. The NMOS threshold voltage adjustment implants do not apply to the native ESD-protected NFET.

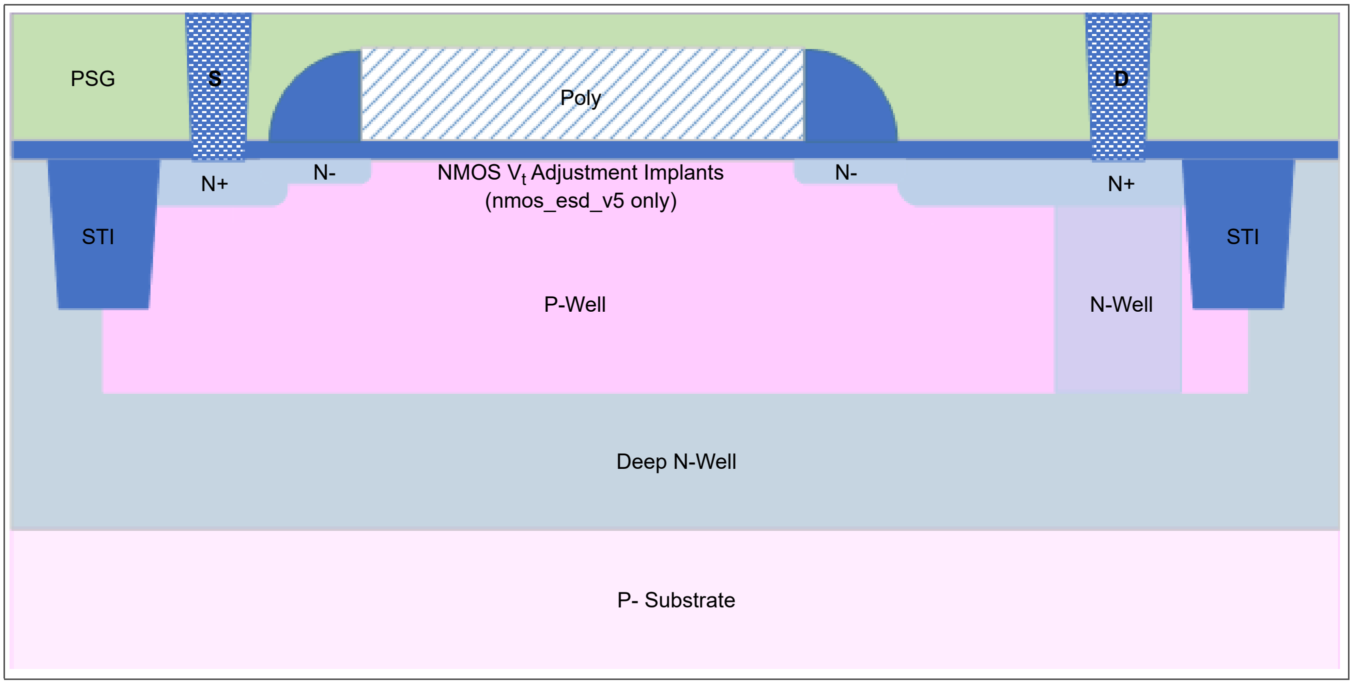


Figure 3-5. ESD-Protected NFET Cross Section

Figure 3-6 is a cross section of a 12 V drain-extended NFET.

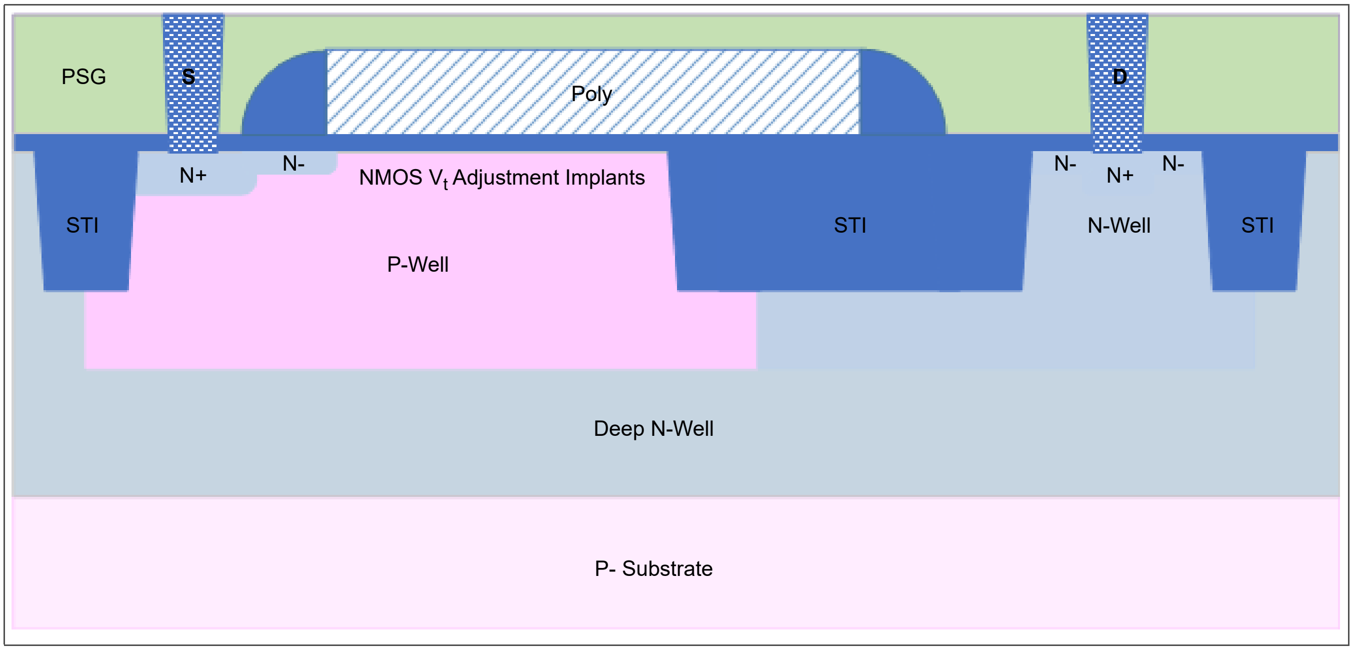


Figure 3-6. 12 V Drain-Extended NFET Cross Section

Figure 3-7 is a cross section of a 20 V drain-extended NFET.

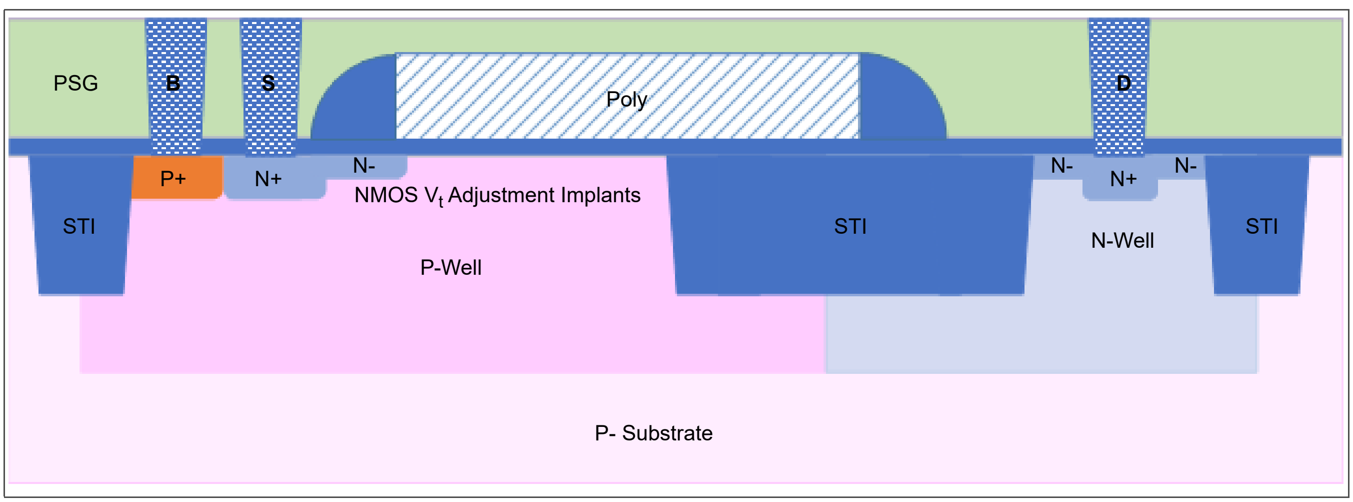


Figure 3-7. 20 V Drain-Extended NFET Cross Section

Figure 3-8 is a cross section of an isolated 20 V drain-extended NFET.

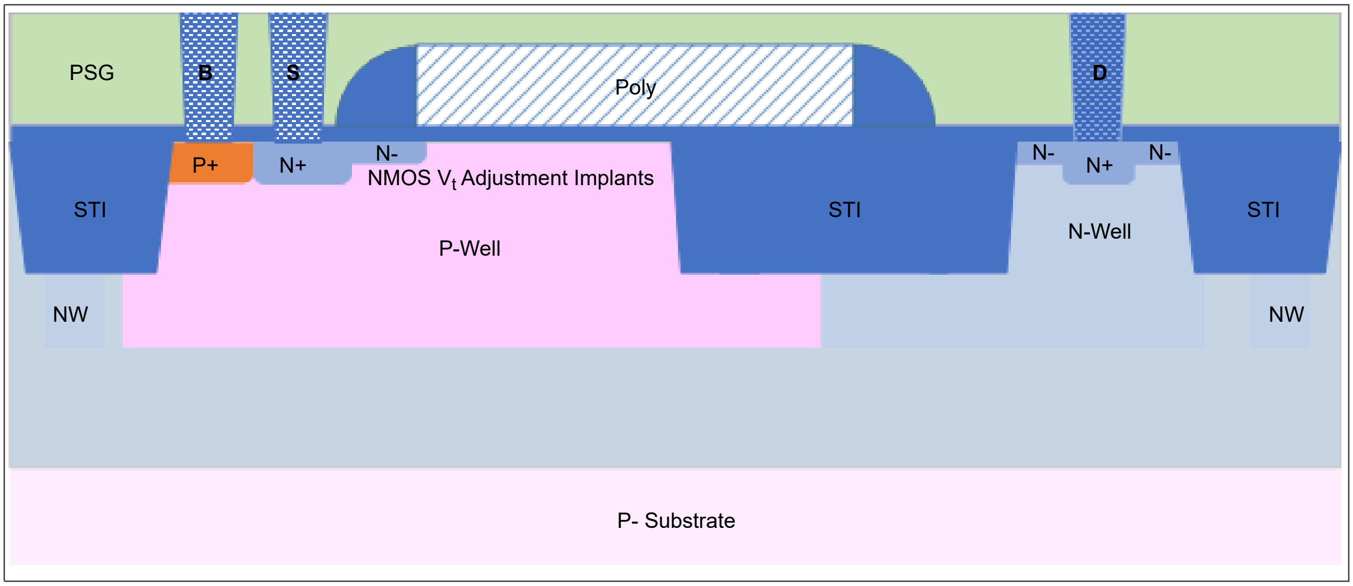


Figure 3-8. Isolated 20 V Drain-Extended NFET Cross Section

Figure 3-9 is a cross section of a native 20 V drain-extended NFET.

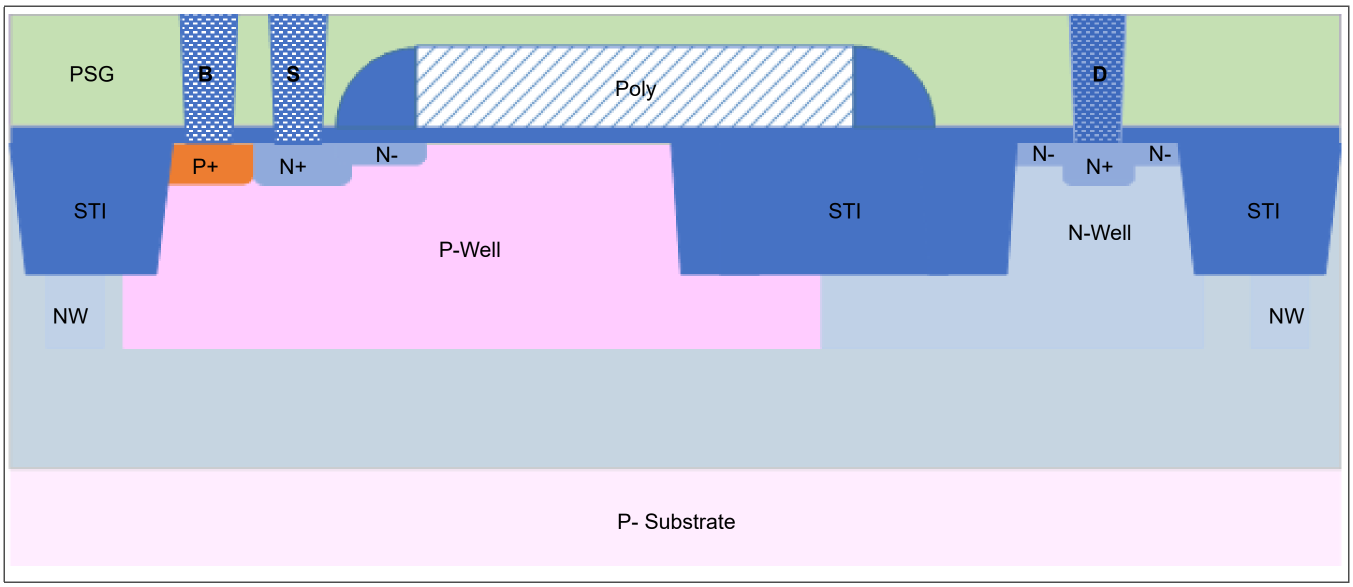


Figure 3-9. Native 20 V Drain-Extended NFET Cross Section

Figure 3-10 is a cross section of a zero-V_t 20 V drain-extended NFET.

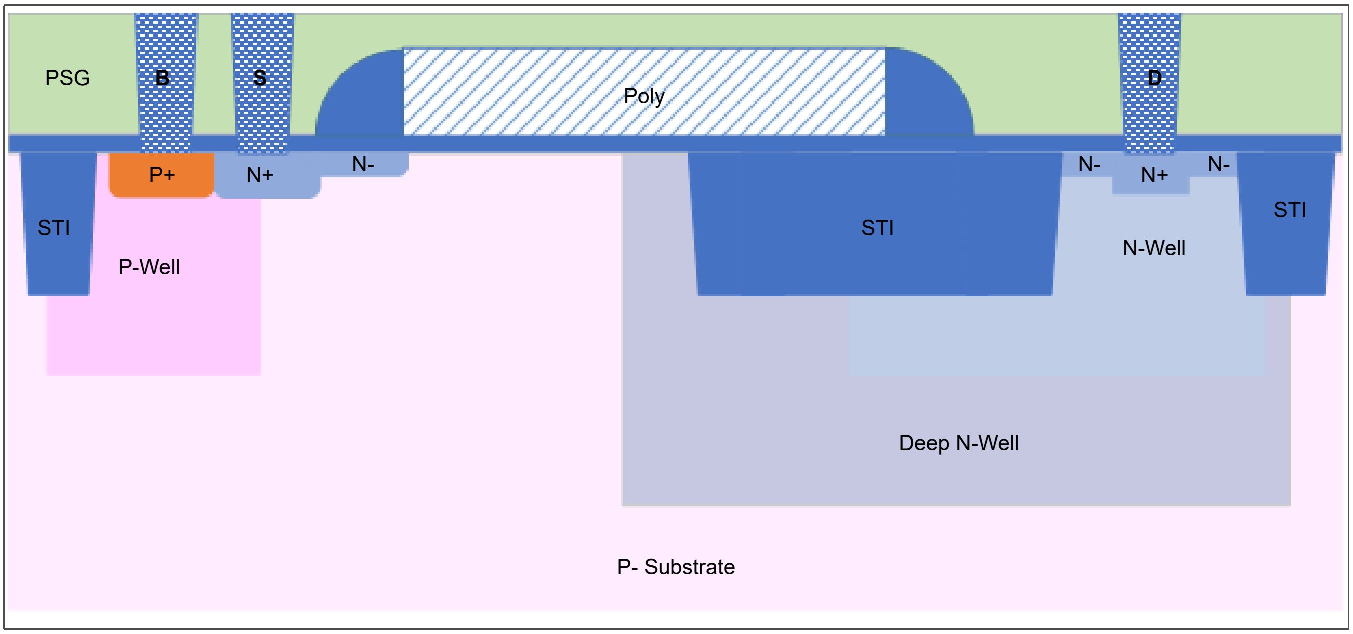


Figure 3-10. Zero-V_t 20 V Drain-Extended NFET Cross Section

3.1.2 NMOS Pcell Layouts and Parameters

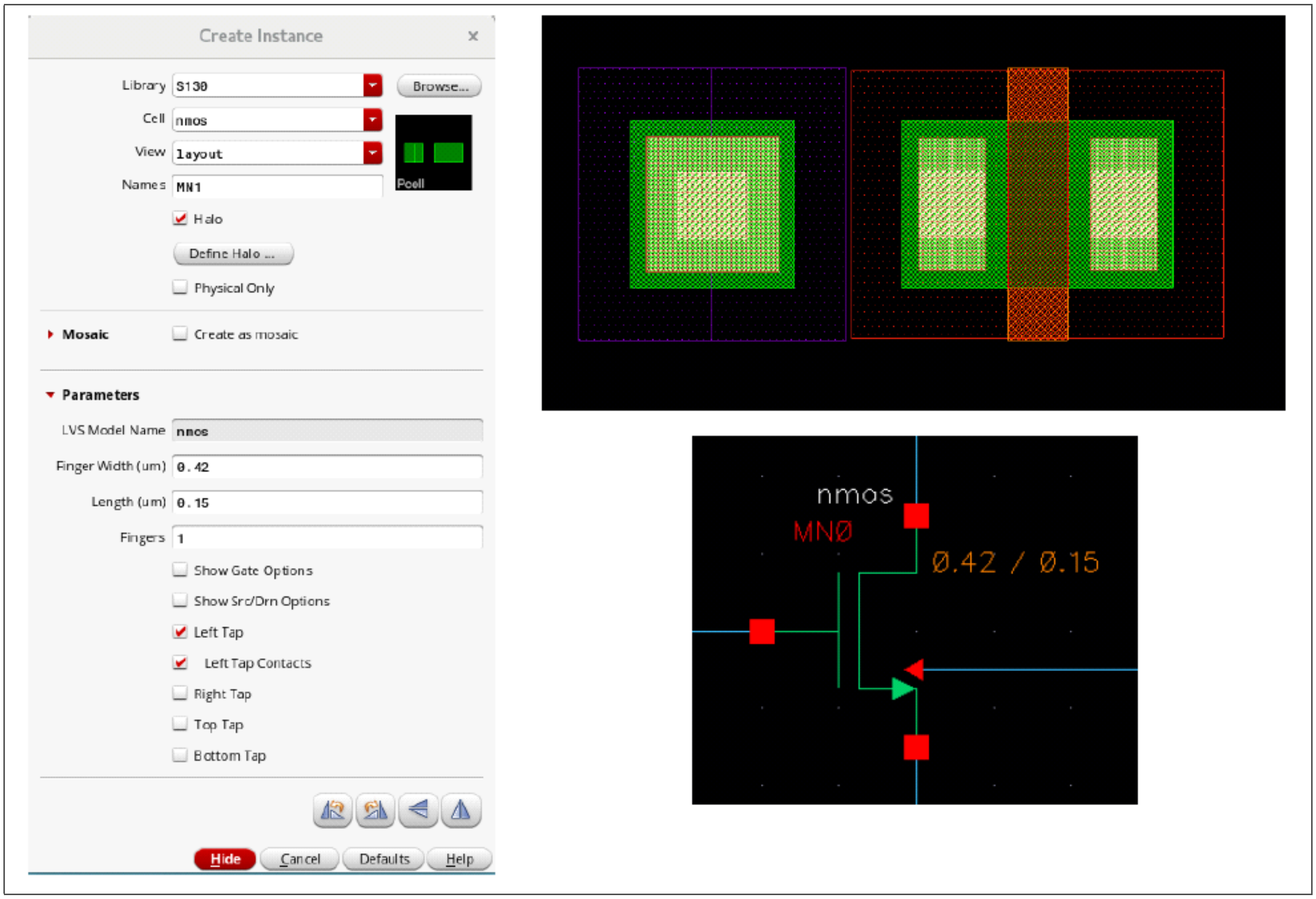


Figure 3-11. NMOS Pcell Layout

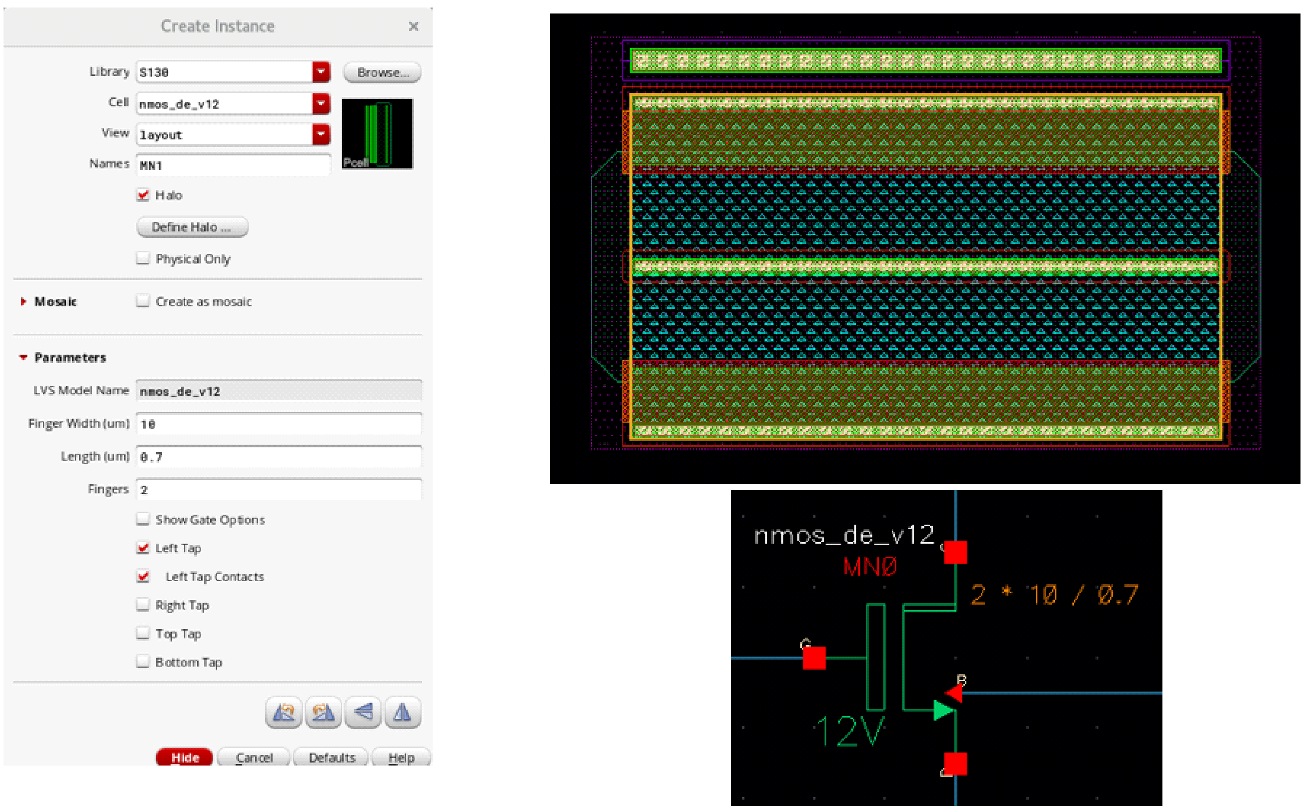


Figure 3-12. 12 V Drain-Extended NMOS Pcell Layout

Table 3-2 lists the pcell parameters available for an NMOS device. The parameter values vary depending on the type of NMOS device created.

Table 3-2. NMOS Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Finger width	0.42 µm	7.0 µm
Channel length	0.15 µm	8.0 µm
Number of fingers	1	100
Multiples[1]	1	10,000

1. The multiples value applies to the symbol only.

3.1.3 PFET Cross Sections

Figure 3-13 is a cross section of a 1.8 V PFET.

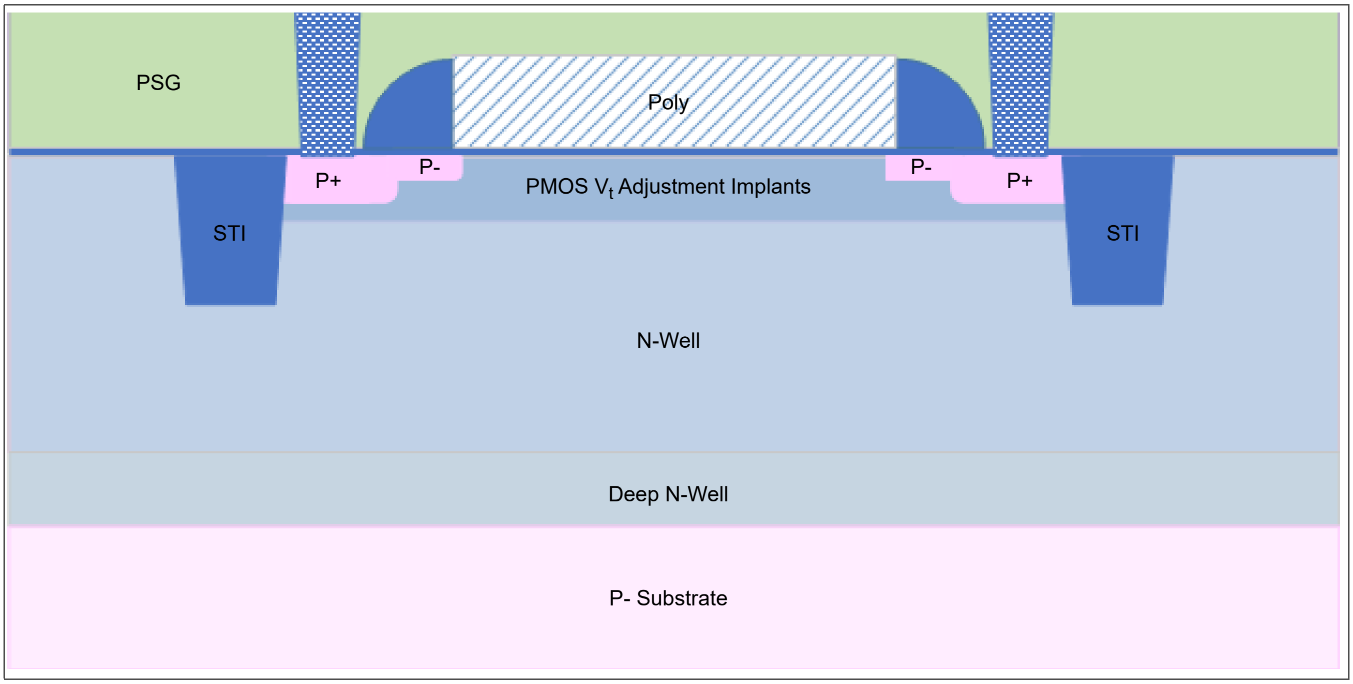


Figure 3-13. 1.8 V PFET Cross Section

The low-V_t PFET shown in *Figure 3-14* has threshold voltage implants to achieve a lower threshold voltage but is otherwise identical to the 1.8 V PFET.

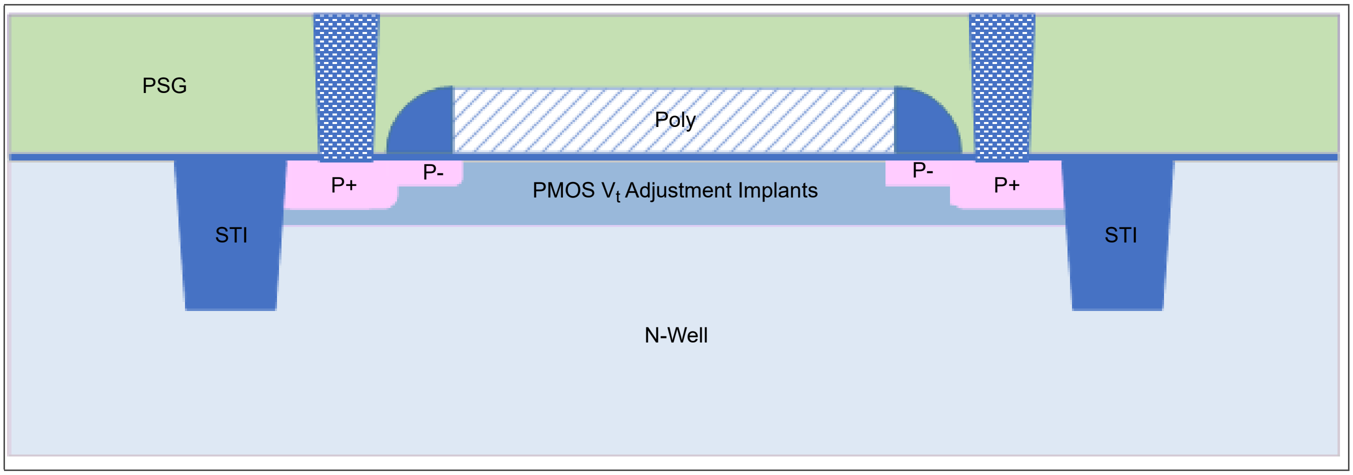


Figure 3-14. Low-V_t PFET Cross Section

The high-V_t PFET shown in *Figure 3-15* has threshold voltage implants to achieve a higher threshold voltage but is otherwise identical to the 1.8 V PFET.

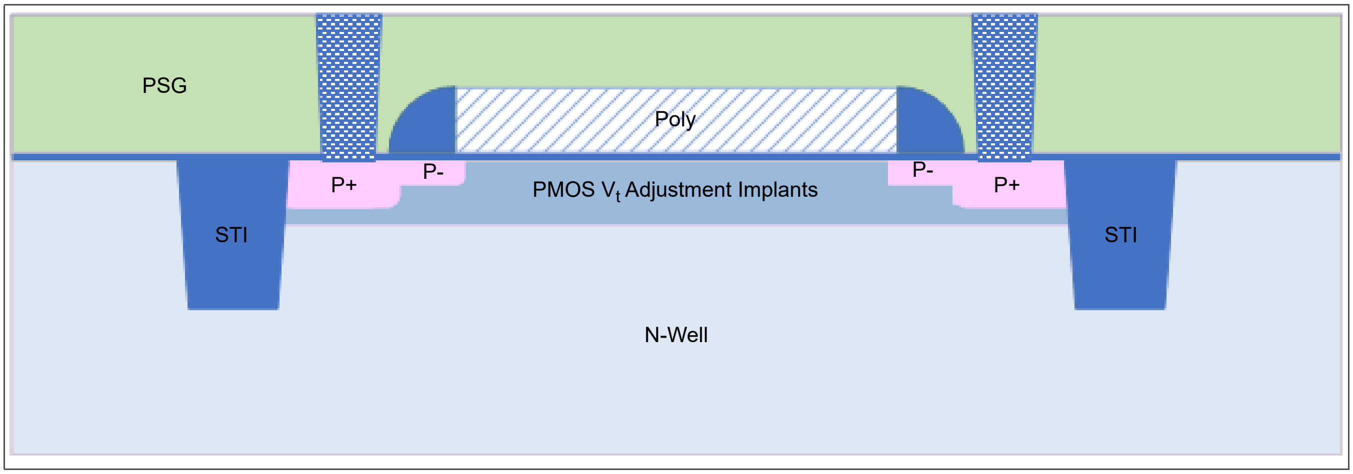


Figure 3-15. High-V_t PFET Cross Section

Figure 3-16 is a cross section of a 5 V PFET.

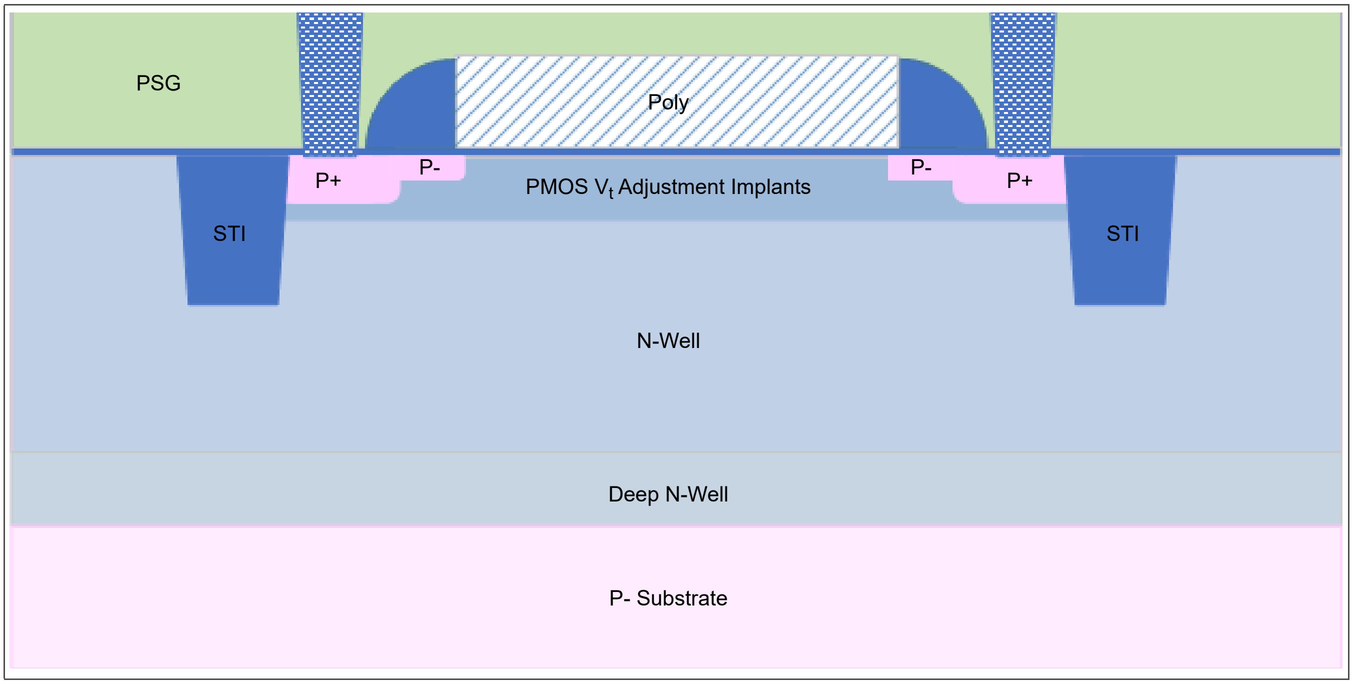


Figure 3-16. 5 V PFET Cross Section

Figure 3-17 is a cross section of a 12 V drain-extended PFET.

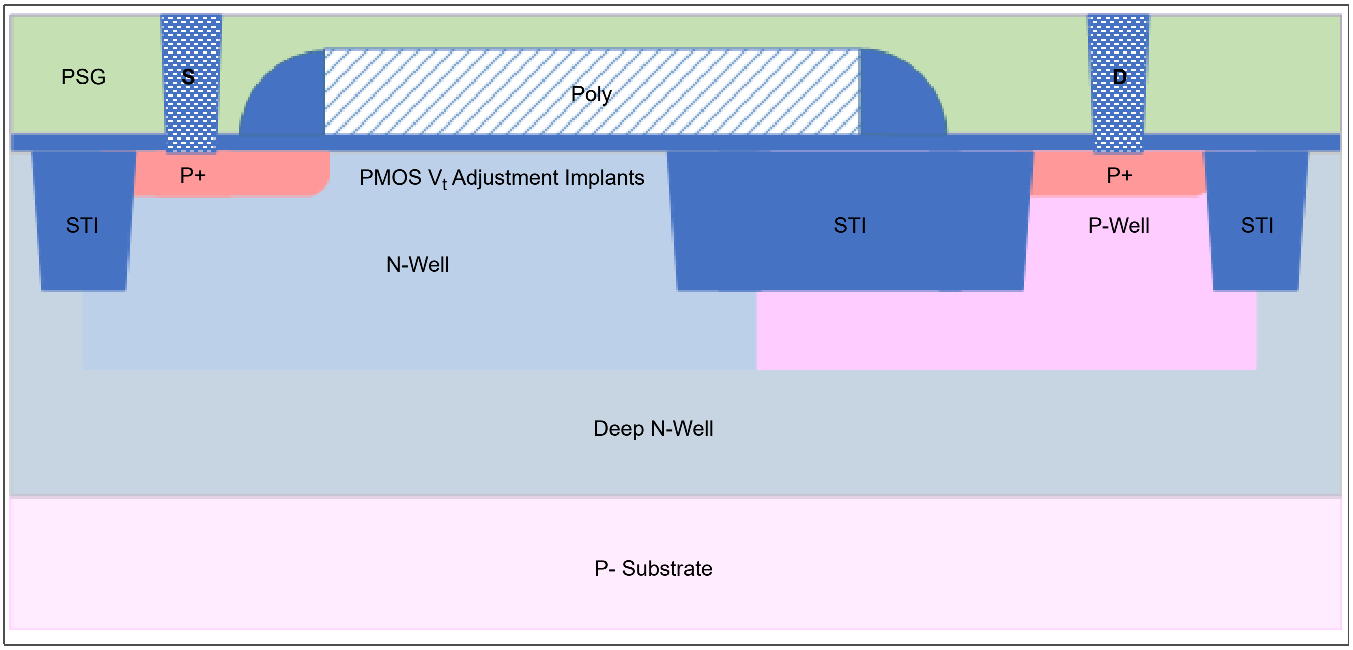


Figure 3-17. 12 V Drain-Extended PFET Cross Section

Figure 3-18 is a cross section of a 20 V drain-extended PFET.

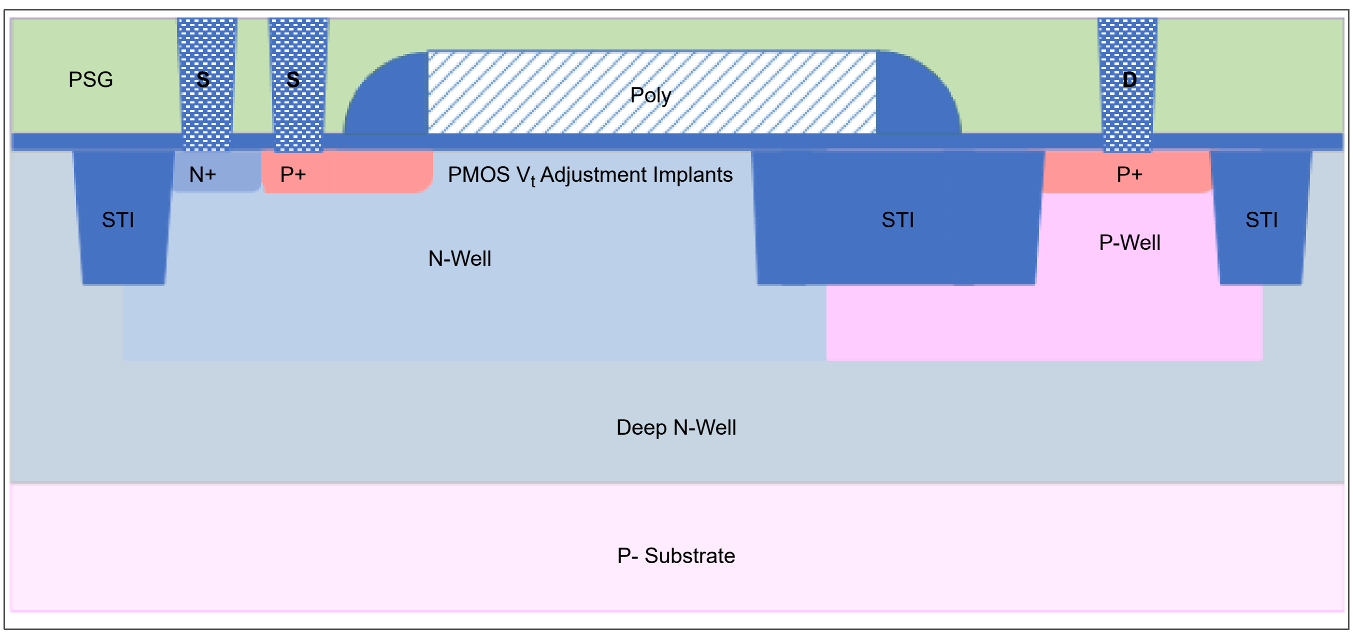


Figure 3-18. 20 V Drain-Extended PFET Cross Section

3.1.4 PMOS Pcell Layouts and Parameters

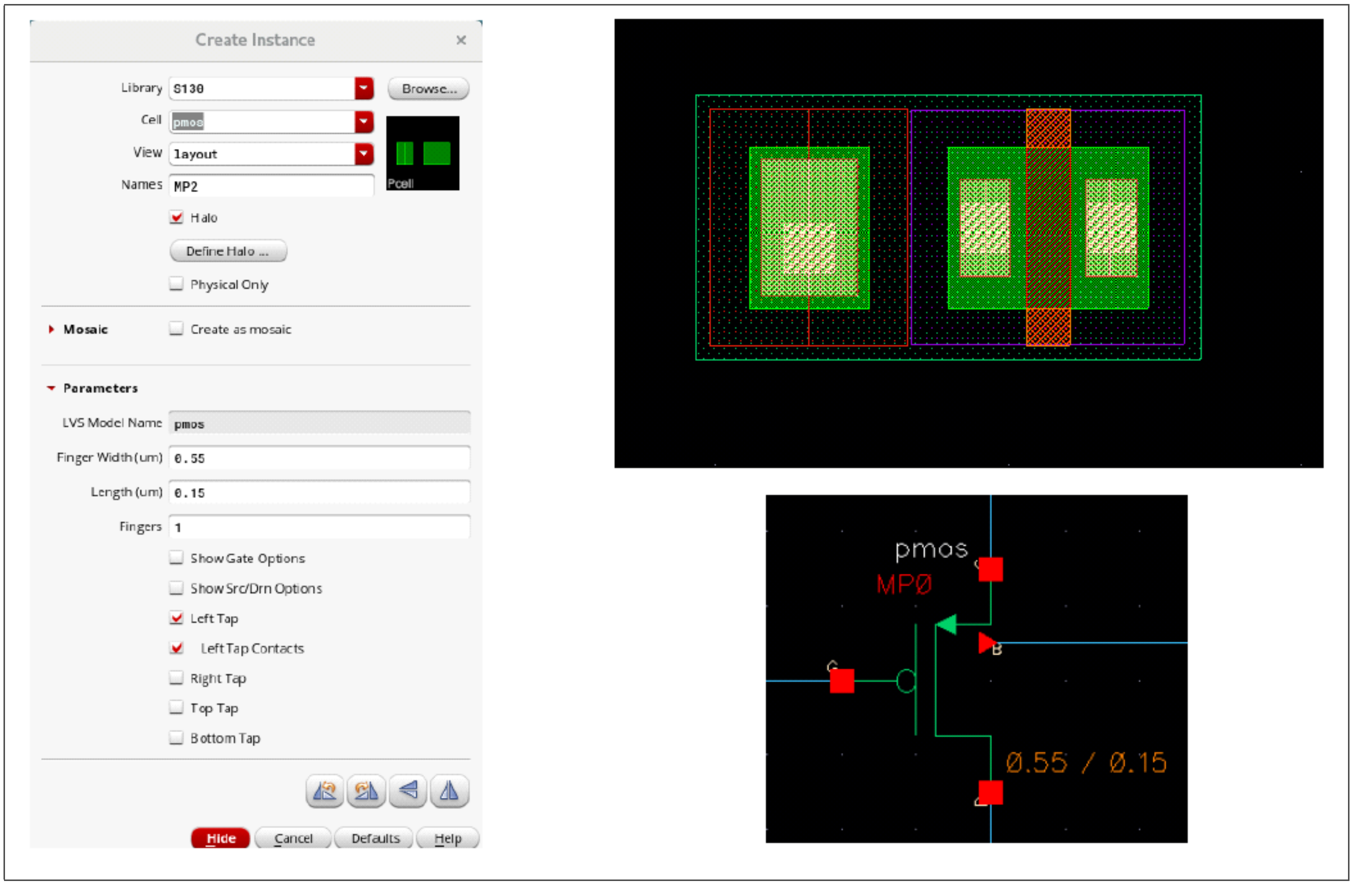


Figure 3-19. PMOS Pcell Layout

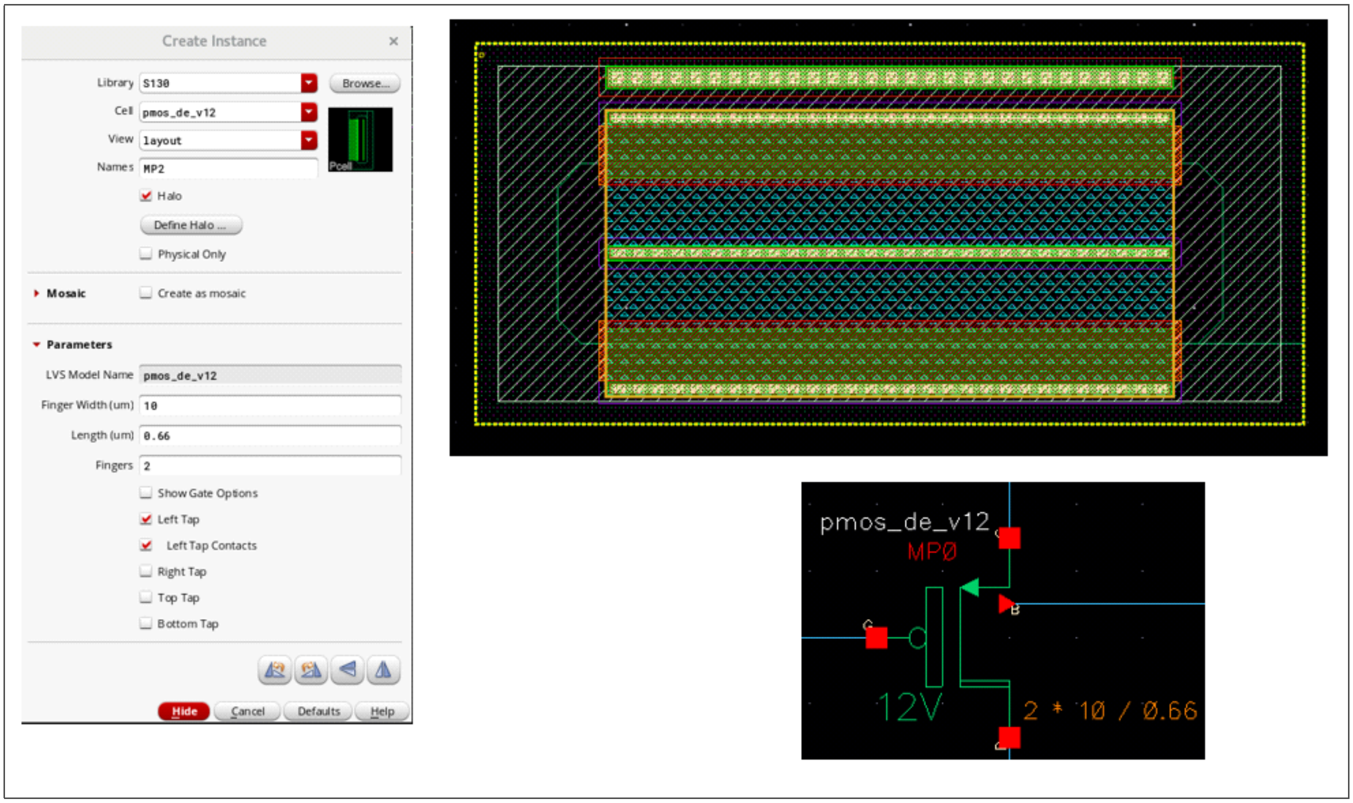


Figure 3-20. 12 V Drain-Extended PMOS Pcell Layout

Table 3-3 lists the pcell parameters available for a PMOS device. The parameter values vary depending on the type of PMOS device created.

Table 3-3. PMOS Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Finger width	0.55 µm	7.0 µm
Channel length	0.15 µm	8.0 µm
Number of fingers	1	100
Multiples[1]	1	10,000

1. The multiples value applies to the symbol only.

3.1.5 MOSFET LVS Checking Parameters

Table 3-4 lists the tolerance values associated with each of the NMOS or PMOS pcell parameters checked during LVS checking.

Table 3-4. MOSFET Pcell LVS Checking Parameters

Parameter	Tolerance Value (%)
Width	0
Length	0
Multiples	0

3.2 Bipolar Junction Transistors

Table 3-5. S130 Bipolar Devices

Device Type	Cell Name	Operating Ranges		
		$\lvert V_{CE} \rvert$ (V)	$\lvert V_{BE} \rvert$ (V)	I_{CE} (μA/μm^2)
NPN transistor	npn_1x1	0 to 5	0 to 5	0.01 to 10
	npn_1x1_v5			
	npn_1x2			
PNP transistor	pnp			
	pnp_v5			

3.2.1 NPN Transistor Cross Sections

Figure 3-21 is a cross section of a 1 × 1 or 1 × 2 npn transistor.

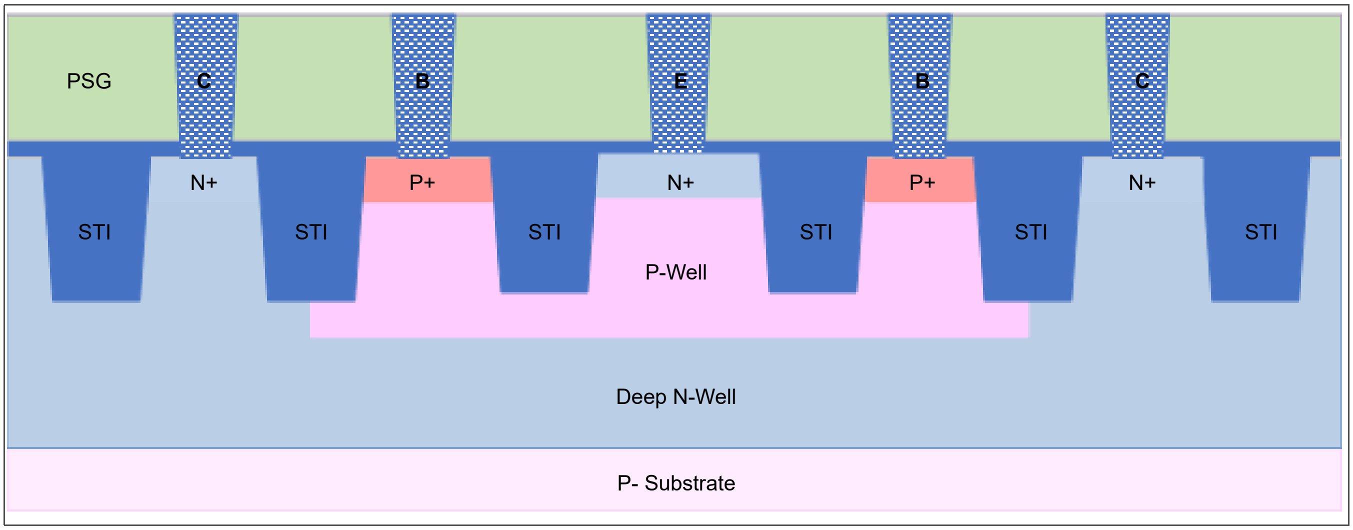


Figure 3-21. 1 × 1 or 1 × 2 NPN Transistor Cross Section

Figure 3-22 is a cross section of a 5 V 1 × 1 npn transistor. The poly gate tied to the emitter prevents the parasitic MOSFET from turning on.

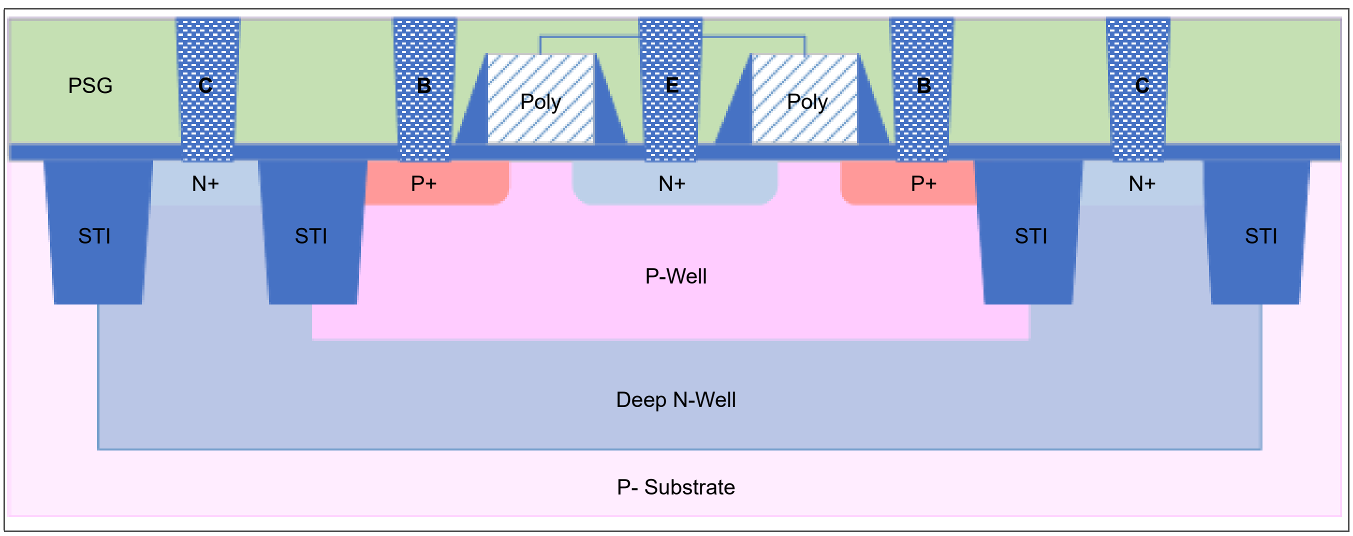


Figure 3-22. 5 V 1 × 1 NPN Transistor Cross Section

3.2.2 NPN Cell Layout

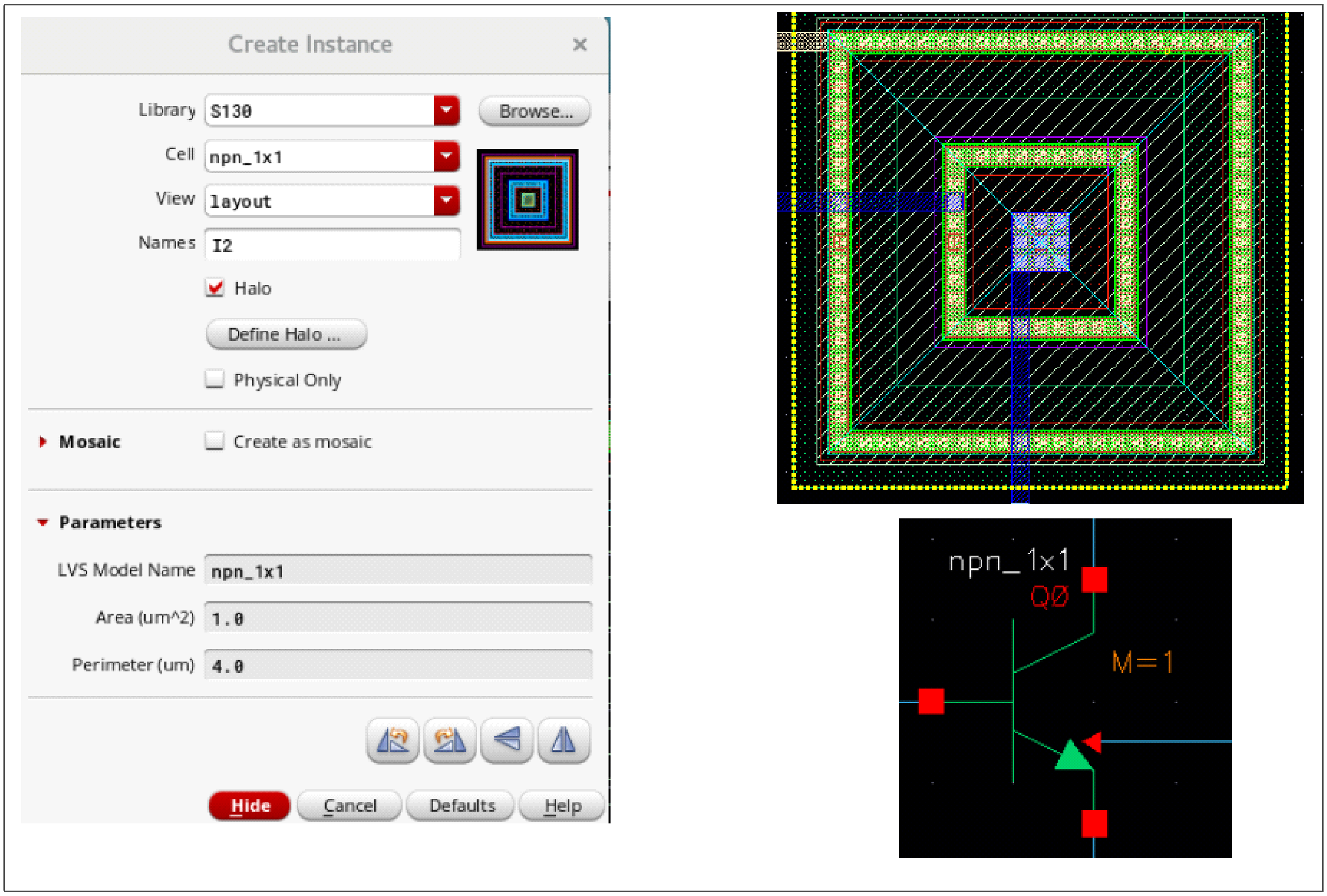


Figure 3-23. NPN Cell (npn_1x1) Layout

3.2.3 PNP Transistor Cross Section

Figure 3-24 is a cross section of a pnp transistor. This device does not contain a deep n-well; the collector is the substrate.

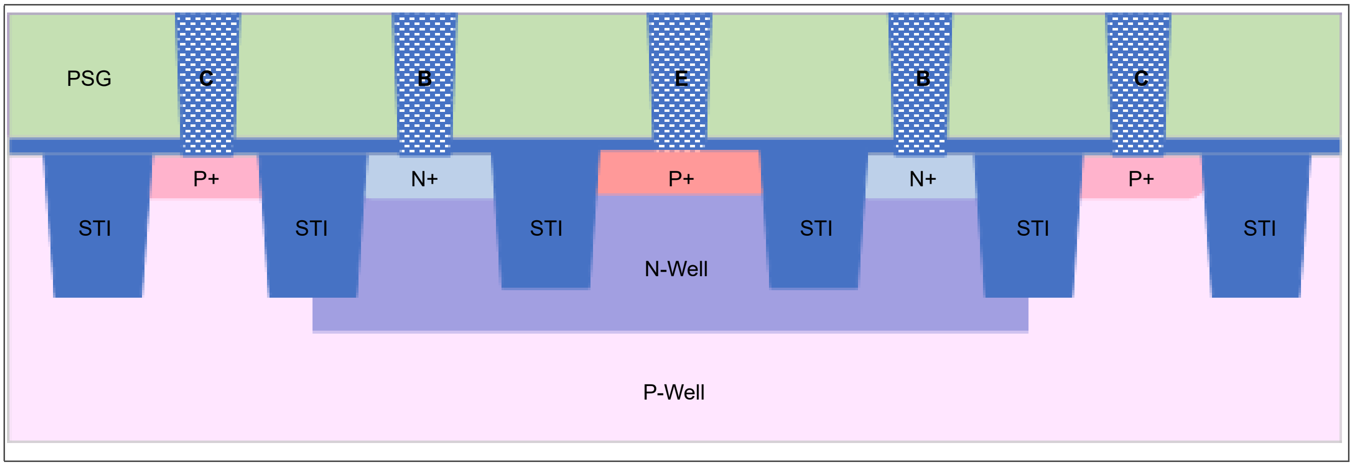


Figure 3-24. PNP Transistor Cross Section

3.2.4 PNP Cell Layout

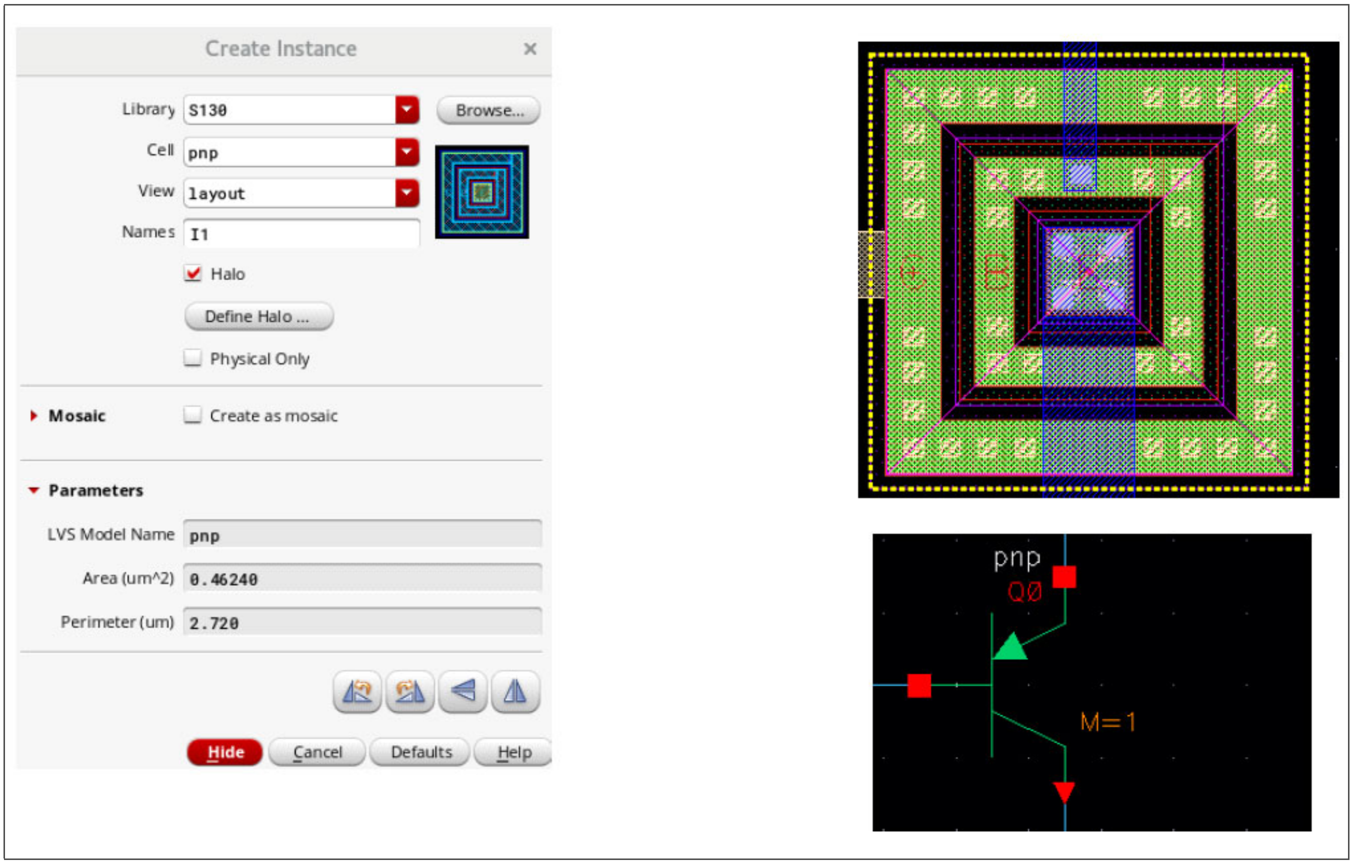


Figure 3-25. PNP Cell Layout

3.2.5 Bipolar Device Pcell Parameters

The bipolar device cells have fixed layouts, that is, they do not have configurable parameters other than the multiples value when placing schematic symbols.

3.2.6 Bipolar Device Cell LVS Checking Parameters

Table 3-6 lists the tolerance values associated with each of the bipolar device cell parameters checked during LVS checking.

Table 3-6. Bipolar Device Cell LVS Checking Parameters

Parameter	Tolerance Value (%)
Emitter area	0
Emitter perimeter	0
Multiples	0

3.3 Capacitors

Table 3-7. S130 Capacitors

Device Type	Cell Name	Operating Voltage (V)
MIM capacitor between met3 and met4	cm3m4	$\lvert V_{c0} - V_{c1}\rvert$ = 0 to 5
MIM capacitor between met4 and met5	cm4m5	

3.3.1 Capacitor Cross Section

Both MIM capacitors are constructed the same way using a thin interlayer dielectric (ILD) over metal followed by a thin conductor layer on top of the ILD. The MIM capacitors can be stacked, as shown in *Figure 3-26*, to maximize the total capacitance value.

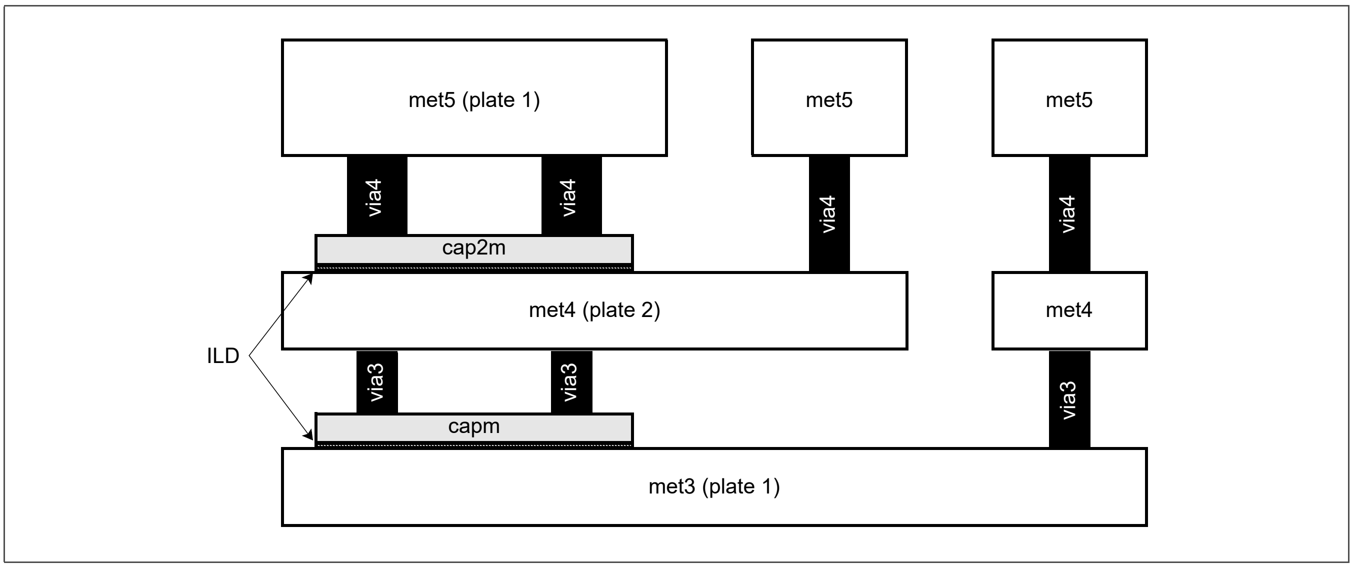


Figure 3-26. Stacked MIM Capacitor Cross Section

3.3.2 Capacitor Pcell Layouts and Parameters

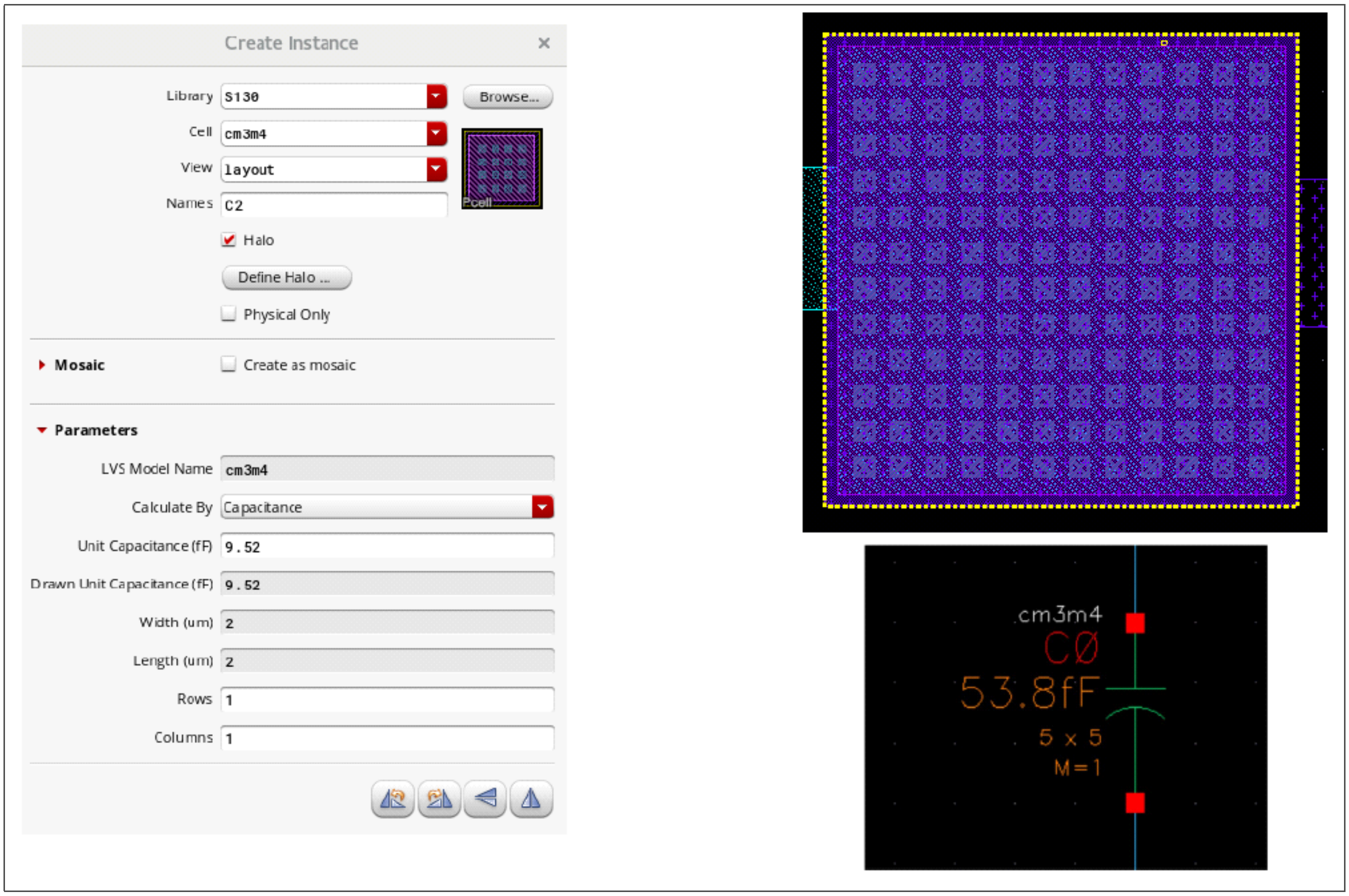


Figure 3-27. MIM Capacitor Pcell (cm3m4) Layout

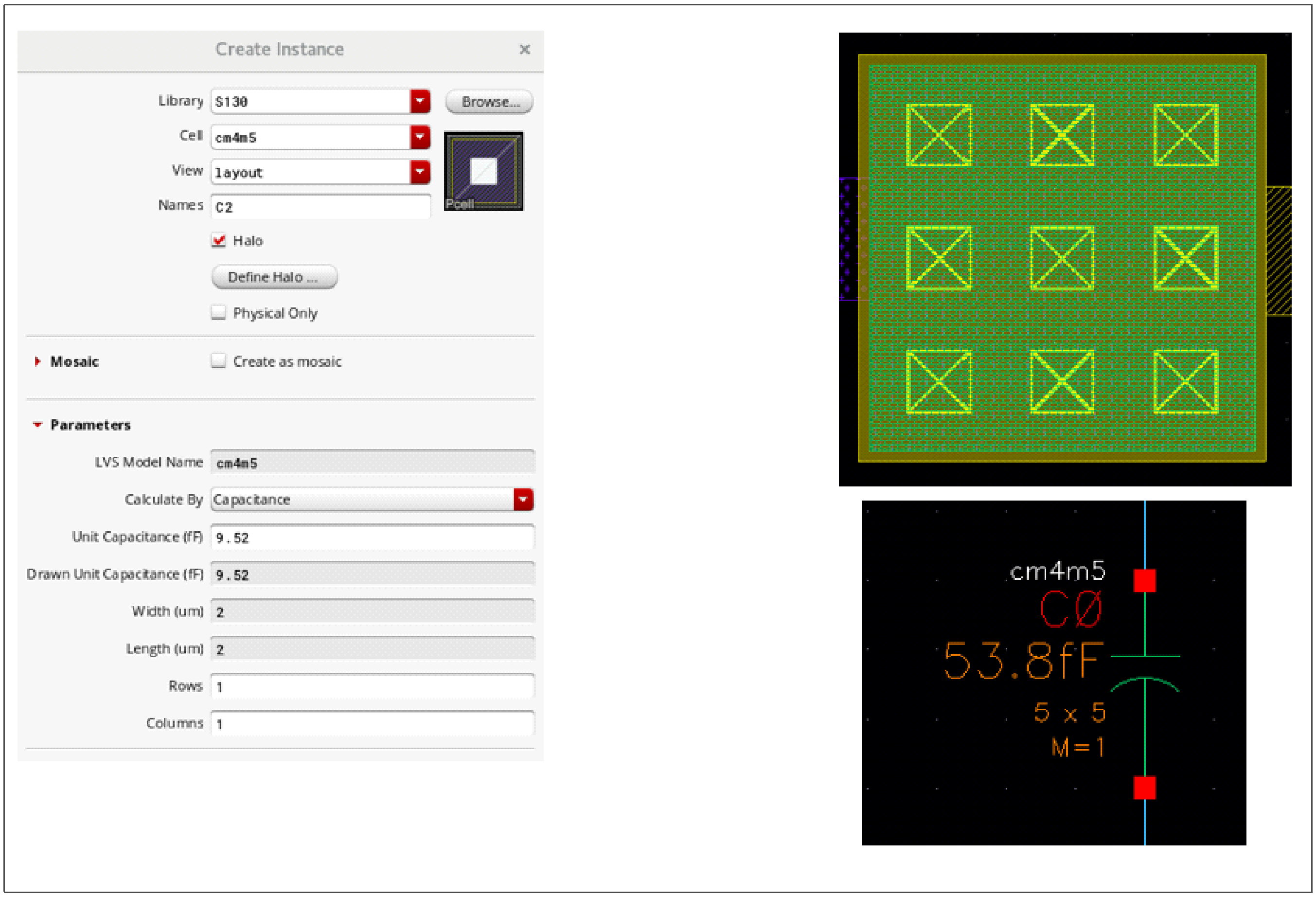


Figure 3-28. MIM Capacitor Pcell (cm4m5) Layout

Table 3-8 lists the pcell parameters available for the MIM capacitors.

Table 3-8. Capacitor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Width[1]	2 µm	100 µm
Length[1]	2 µm	100 µm
Unit capacitance	9.52 fF	1,000,000 fF
Multiples[2]	1	10,000

1. The maximum width/length ratio is 20.
2. The multiples value applies to the symbol only.

3.3.3 Capacitor Pcell LVS Checking Parameters

Table 3-9 lists the tolerance values associated with each of the MIM capacitor pcell parameters checked during LVS checking.

Table 3-9. Capacitor Pcell LVS Checking Parameters

Parameter	Tolerance Value (%)
Area	0
Perimeter	0
Multiples	0

3.4 Pads

Table 3-10. S130 Pads

Pad Type	Cell Name
Standard bond pad	pad_bond
Probe pad	pad_probe
Microprobe pad	pad_microprobe

3.4.1 Bond Pad Pcell Layout and Parameters

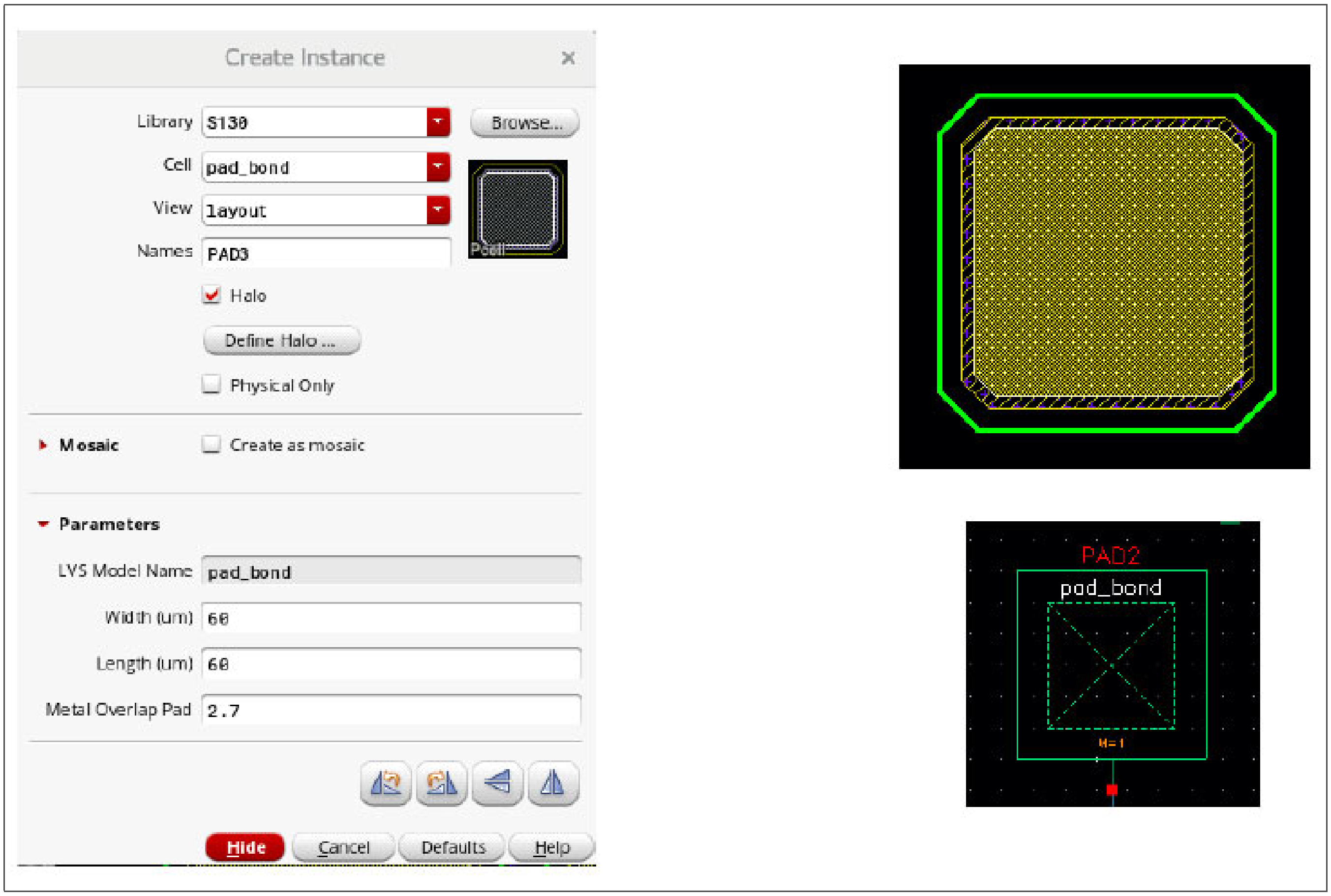


Figure 3-29. Bond Pad Pcell Layout

Table 3-11 lists the pcell parameters available for bond pads.

Table 3-11. Bond Pad Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Width	60 µm	150 µm
Length	60 µm	150 µm
Metal overlap of pad	2.7 µm	13.5 µm
Multiples[1]	1	10,000

1. The multiples value applies to the symbol only.

3.4.2 Probe Pad Pcell Layout and Parameters

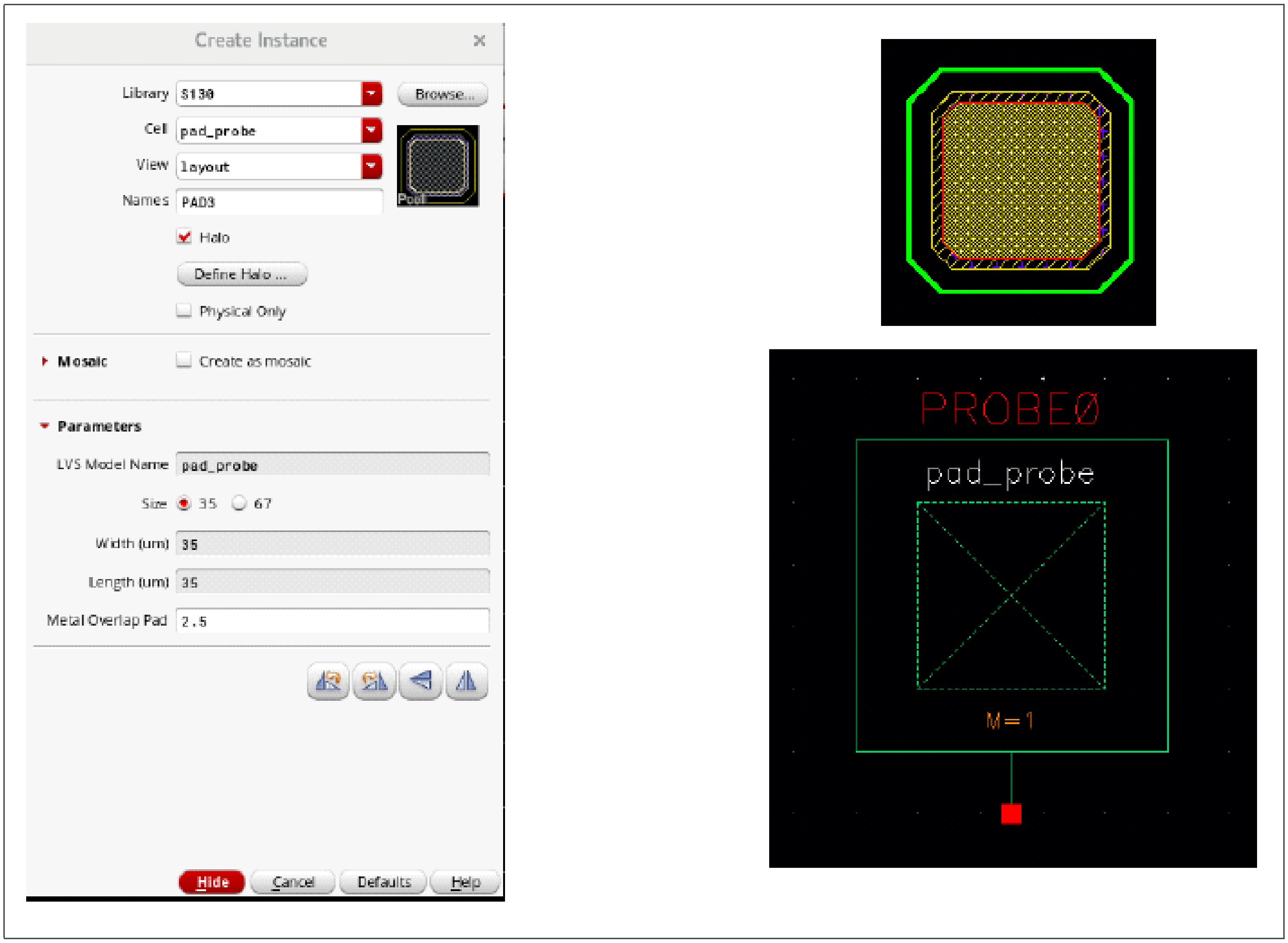


Figure 3-30. Probe Pad Pcell Layout

Table 3-12 lists the pcell parameters available for probe pads.

Table 3-12. Probe Pad Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Width	35 µm or 67 µm only	–
Length	35 µm or 67 µm only	–
Metal overlap of pad	2.5 µm	12.5 µm
Multiples[1]	1	10,000

1. The multiples value applies to the symbol only.

3.4.3 Microprobe Pad Pcell Layout and Parameters

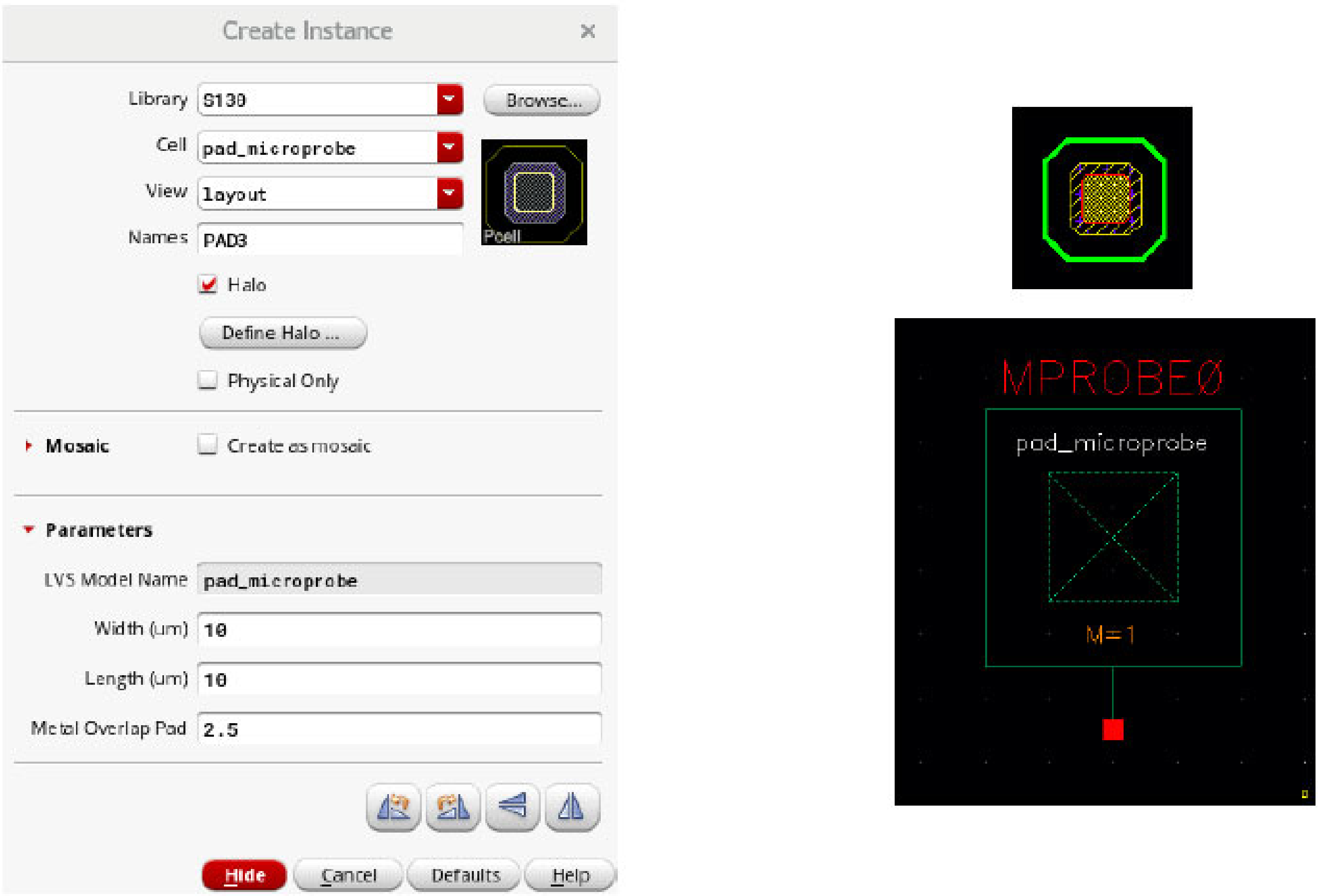


Figure 3-31. Microprobe Pad Pcell Layout

Table 3-13 lists the pcell parameters available for microprobe pads.

Table 3-13. Microprobe Pad Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Width	10 µm	35 µm
Length	10 µm	35 µm
Metal overlap of pad	2.5 µm	12.5 µm
Multiples[1]	1	10,000

1. The multiples value applies to the symbol only.

3.4.4 Pad Pcell LVS Checking Parameters

Table 3-14 lists the tolerance values associated with each of the pad pcell parameters checked during LVS checking.

Table 3-14. Pad Pcell LVS Checking Parameters

Parameter	Tolerance Value (%)
Width	0
Length	0
Multiples	0

3.5 Resistors

Table 3-15. S130 Resistors

Device Type	Cell Name	Sheet Rho (Ω/Sq.)
Diffusion resistor	rndiff	120
	rndiff_v5	114
	rpdiff	197
	rpdiff_v5	191
Poly resistor	rpoly	48.2
	rpoly_hp	317.389
	rpolyhp2K	2000
P-Well resistor	rpwell	3330
Metal resistor	li	12.8
	met1	0.125
	met2	0.125
	met3	0.047
	met4	0.047
	met5	0.0285

3.5.1 N+ Diffusion Resistor Pcell Layout and Parameters

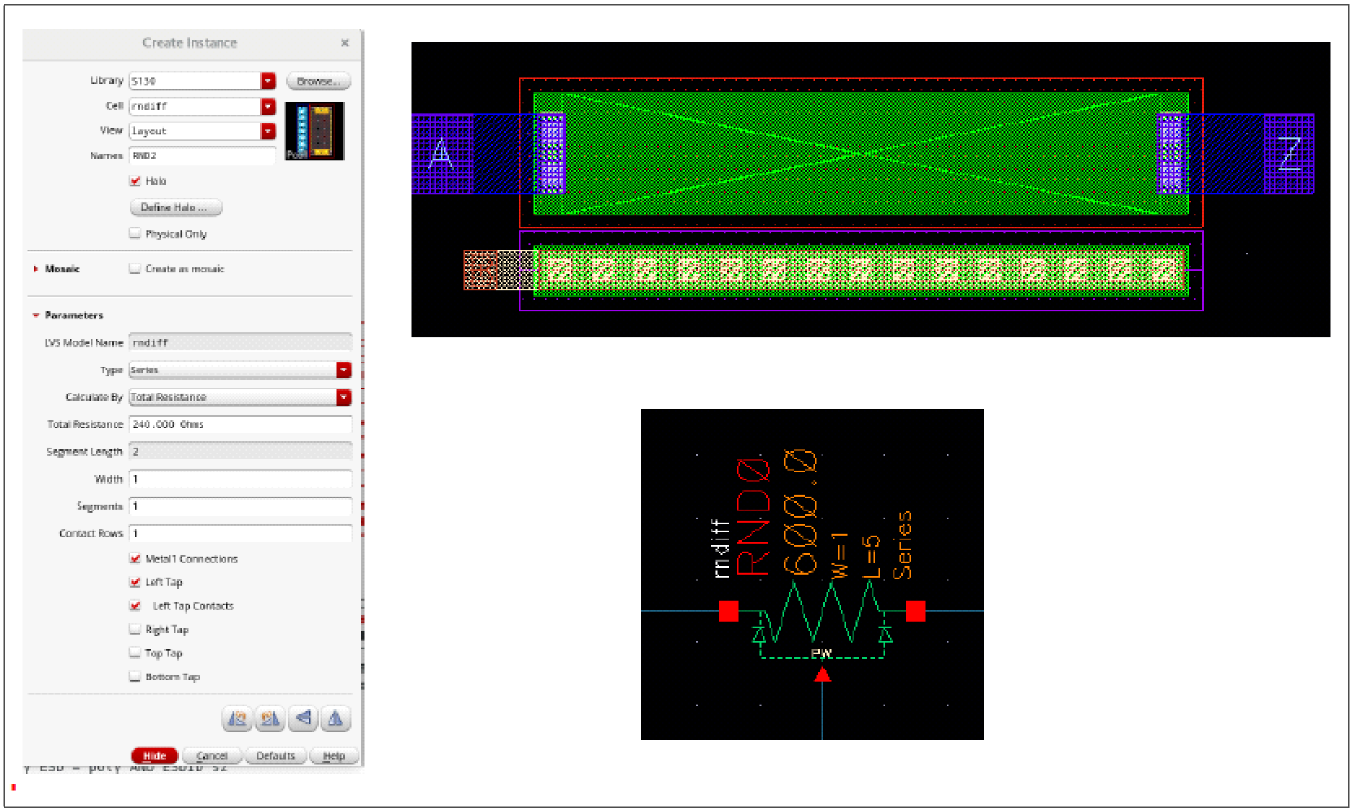


Figure 3-32. N+ Diffusion Resistor Pcell Layout

Table 3-16 lists the pcell parameters available for n+ diffusion resistors.

Table 3-16. N+ Diffusion Resistor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Type	Series, parallel	–
Width	1 µm	1000 µm
Segments	1	50
Contact rows	1	10
Calculate by Total Resistance		
Total resistance	240 Ω	60 MΩ
Calculate by Segment Length		
Segment length	2 µm	500 µm

3.5.2 5 V N+ Diffusion Resistor Pcell Layout and Parameters

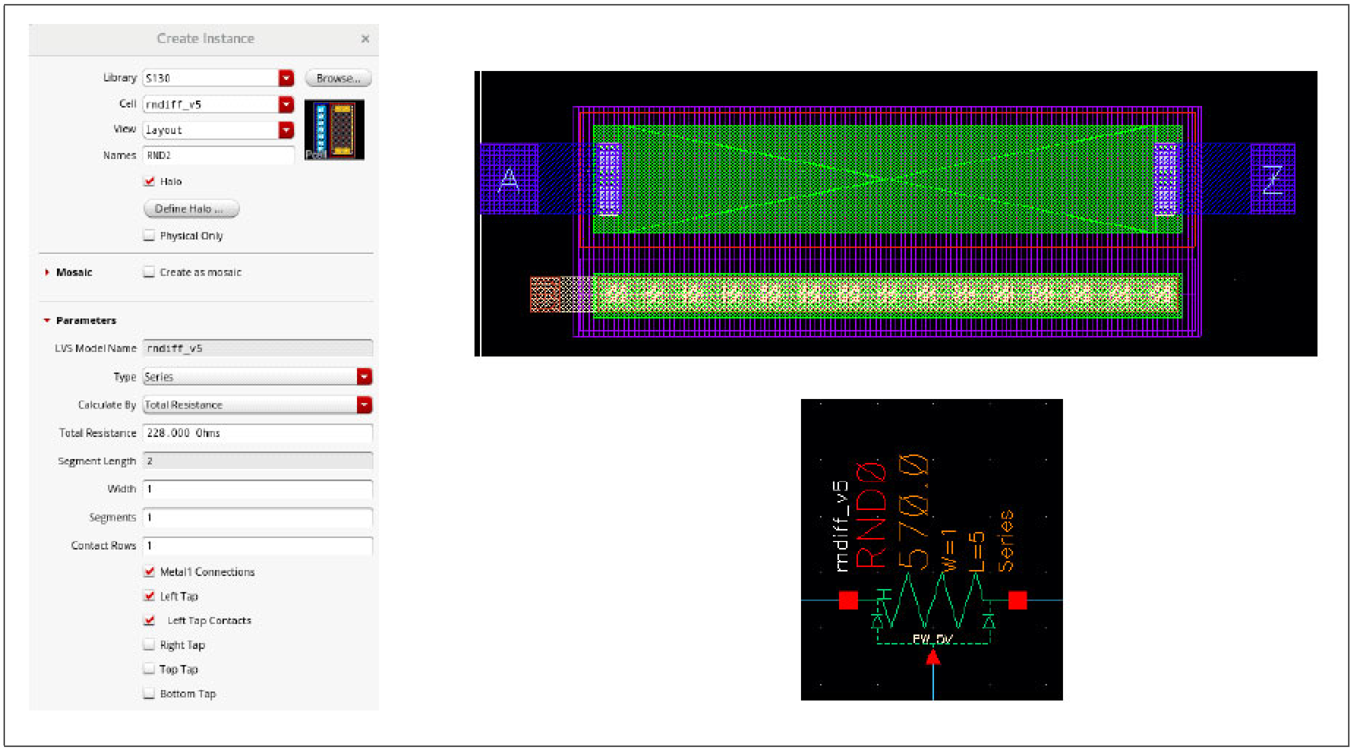


Figure 3-33. 5 V N+ Diffusion Resistor Pcell Layout

Table 3-17 lists the pcell parameters available for 5 V n+ diffusion resistors.

Table 3-17. 5 V N+ Diffusion Resistor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Type	Series, parallel	–
Width	1 µm	1000 µm
Segments	1	50
Contact rows	1	10
Calculate by Total Resistance		
Total resistance	228 Ω	10 MΩ
Calculate by Segment Length		
Segment length	2 µm	500 µm

3.5.3 P+ Diffusion Resistor Pcell Layout and Parameters

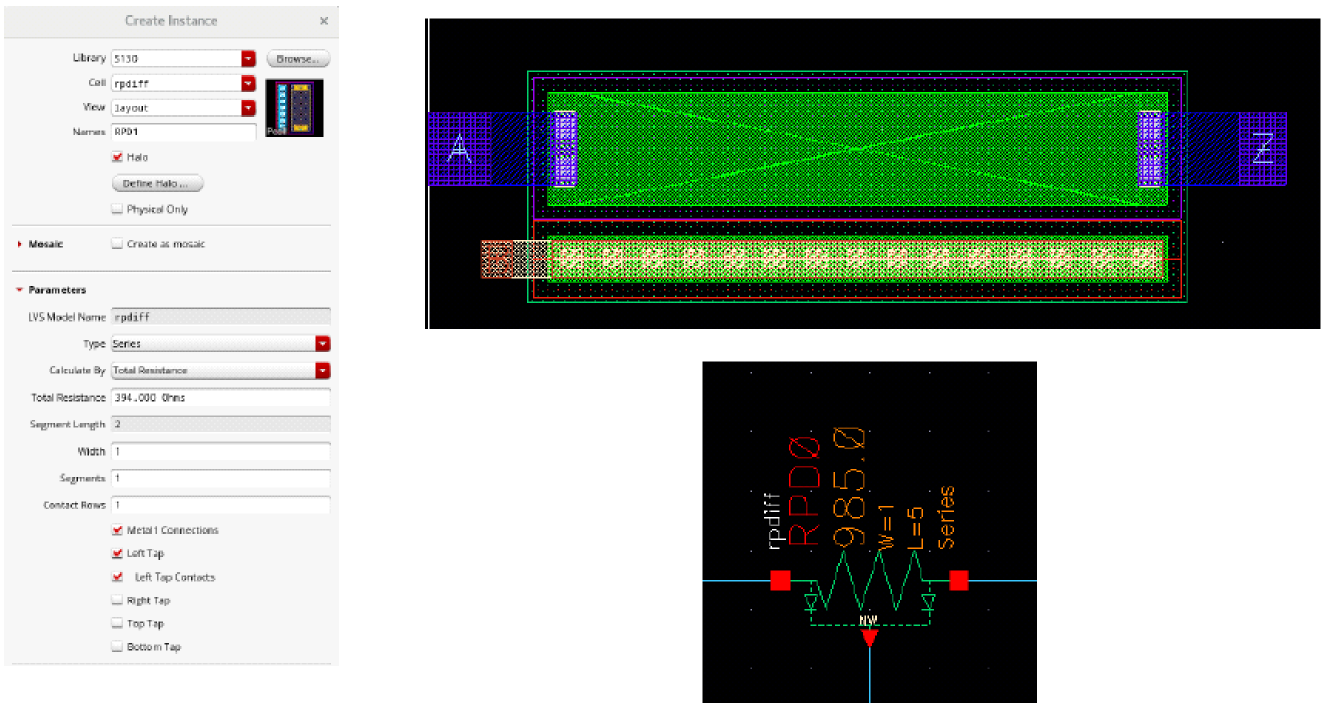


Figure 3-34. P+ Diffusion Resistor Pcell Layout

Table 3-18 lists the pcell parameters available for p+ diffusion resistors.

Table 3-18. P+ Diffusion Resistor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Type	Series, parallel	–
Width	1 µm	1000 µm
Segments	1	50
Contact rows	1	10
Calculate by Total Resistance		
Total resistance	240 Ω	10 MΩ
Calculate by Segment Length		
Segment length	2 µm	500 µm

3.5.4 5 V P+ Diffusion Resistor Pcell Layout and Parameters

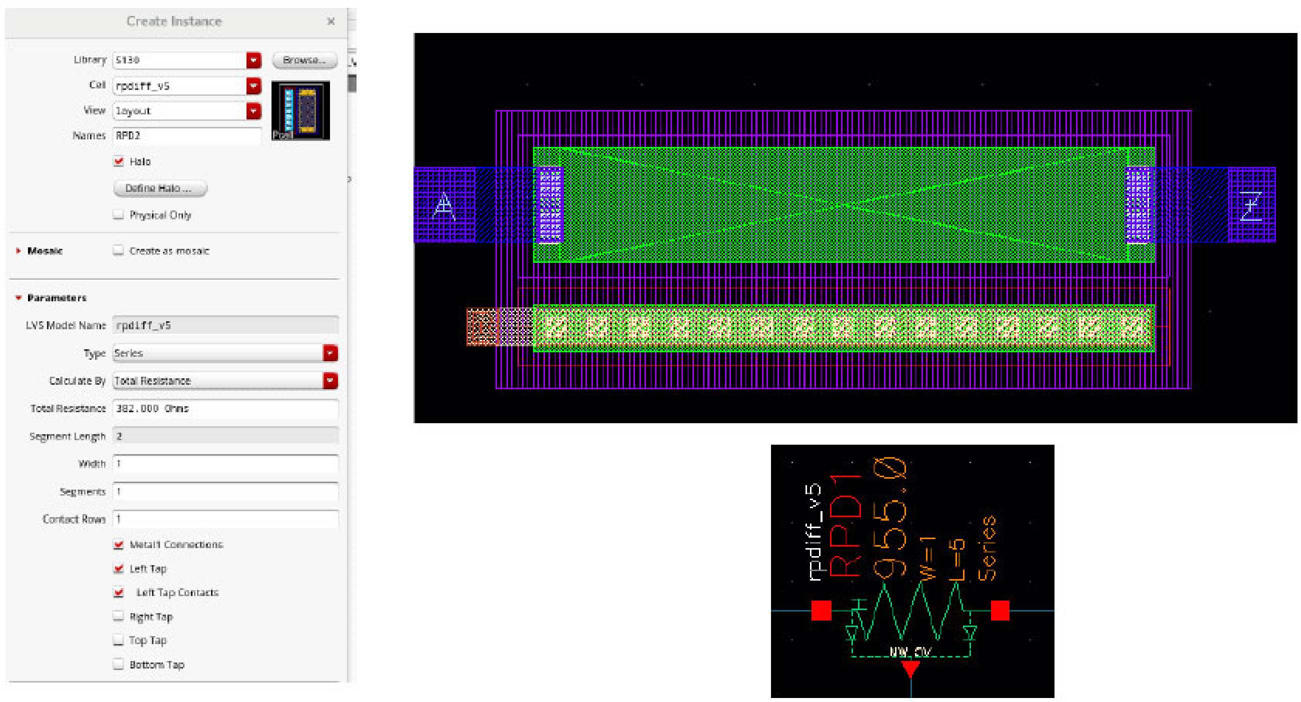


Figure 3-35. 5 V P+ Diffusion Resistor Pcell Layout

Table 3-19 lists the pcell parameters available for 5 V p+ diffusion resistors.

Table 3-19. 5 V P+ Diffusion Resistor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Type	Series, parallel	–
Width	1 µm	1000 µm
Segments	1	50
Contact rows	1	10
Calculate by Total Resistance		
Total resistance	228 Ω	10 MΩ
Calculate by Segment Length		
Segment length	2 µm	500 µm

3.5.5 Poly Resistor Pcell Layout and Parameters

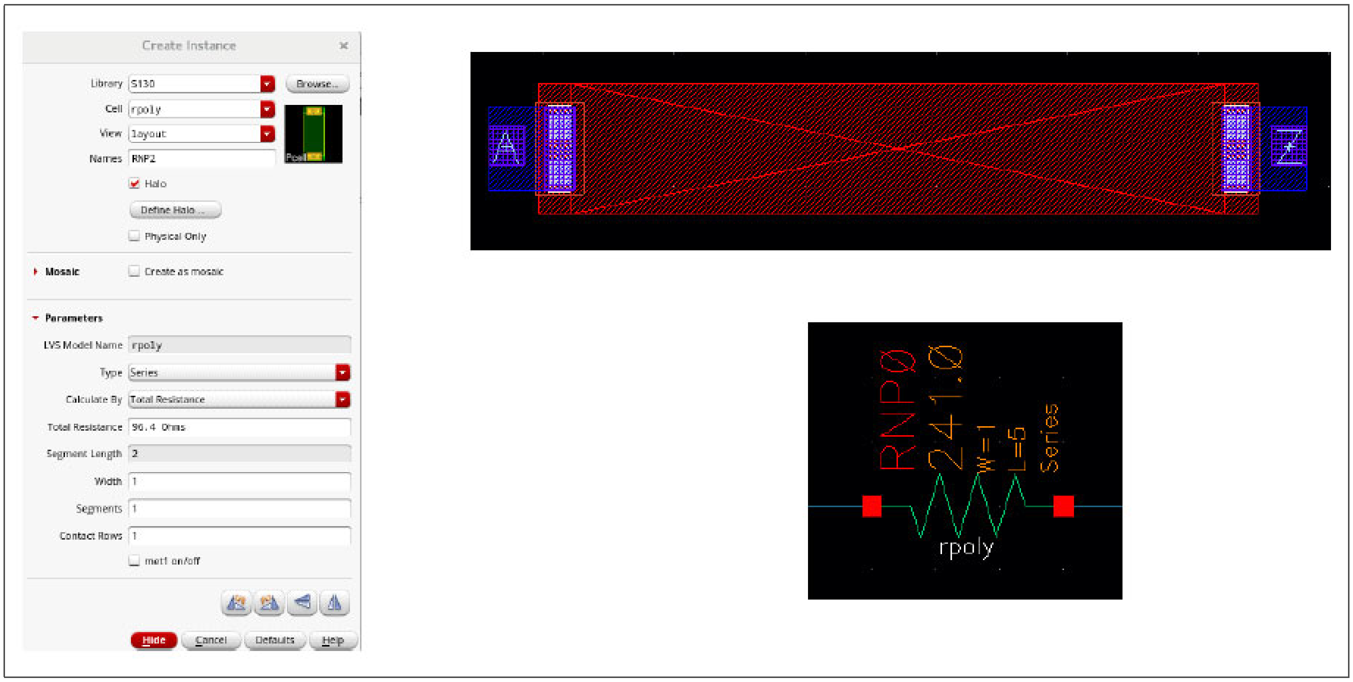


Figure 3-36. Poly Resistor Pcell Layout

Table 3-20 lists the pcell parameters available for standard poly resistors.

Table 3-20. Poly Resistor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Type	Series, parallel	–
Width	1 µm	1000 µm
Segments	1	50
Contact rows	1	10
met1 on/off	On	–
Calculate by Total Resistance		
Total resistance	96.4 Ω	10 MΩ
Calculate by Segment Length		
Segment length	2 µm	500 µm

3.5.6 High Precision Poly Resistor Pcell Layout and Parameters

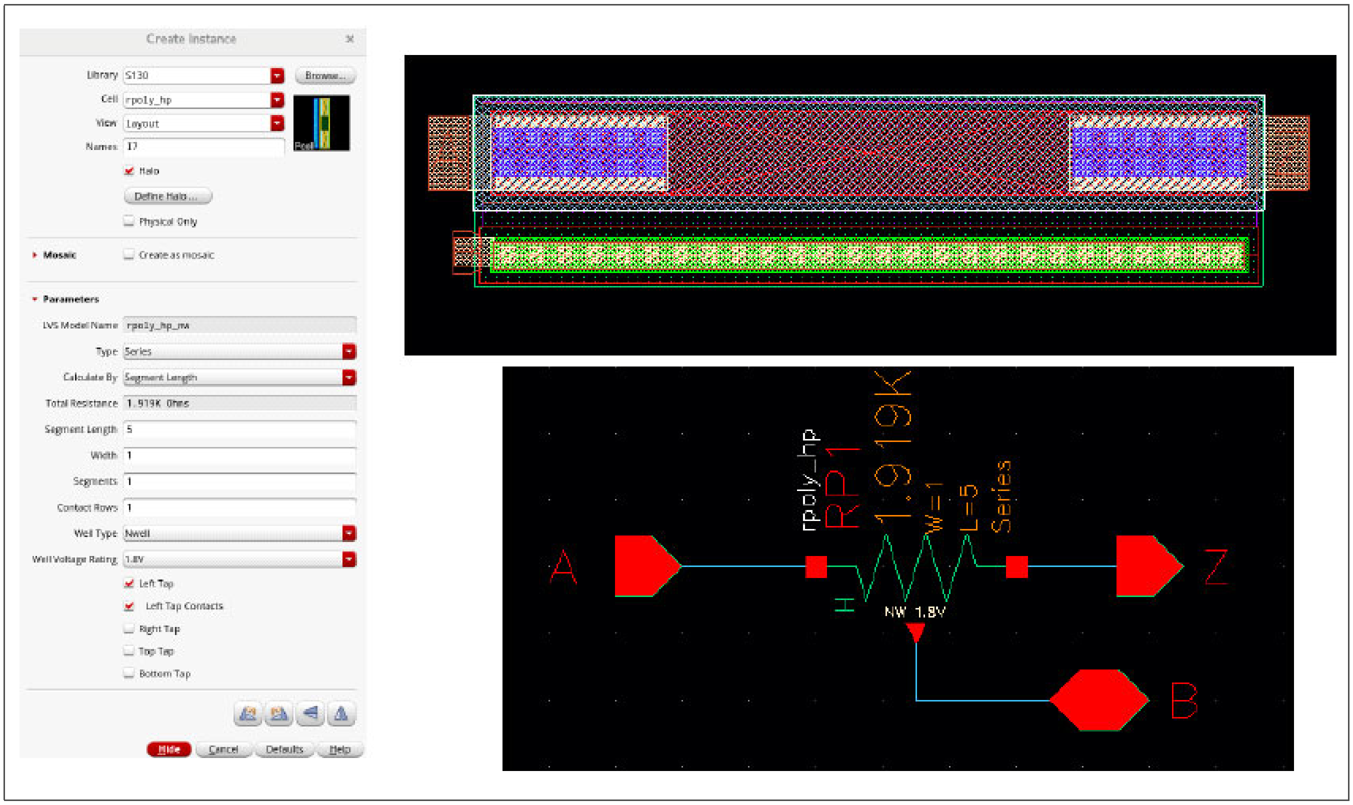


Figure 3-37. High Precision Poly Resistor Pcell Layout

Table 3-21 lists the pcell parameters available for high precision poly resistors.

Table 3-21. High Precision Poly Resistor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Type	Series, parallel, serpentine	–
Width	1 µm	1000 µm
Segments	1	50
Contact rows	1	5
Well type	N-well, p-well	–
Calculate by Total Resistance		
Total resistance	96.4 Ω	10 MΩ
Calculate by Segment Length		
Segment length	2 µm	500 µm

3.5.7 High Sheet Rho Poly Resistor Pcell Layout and Parameters

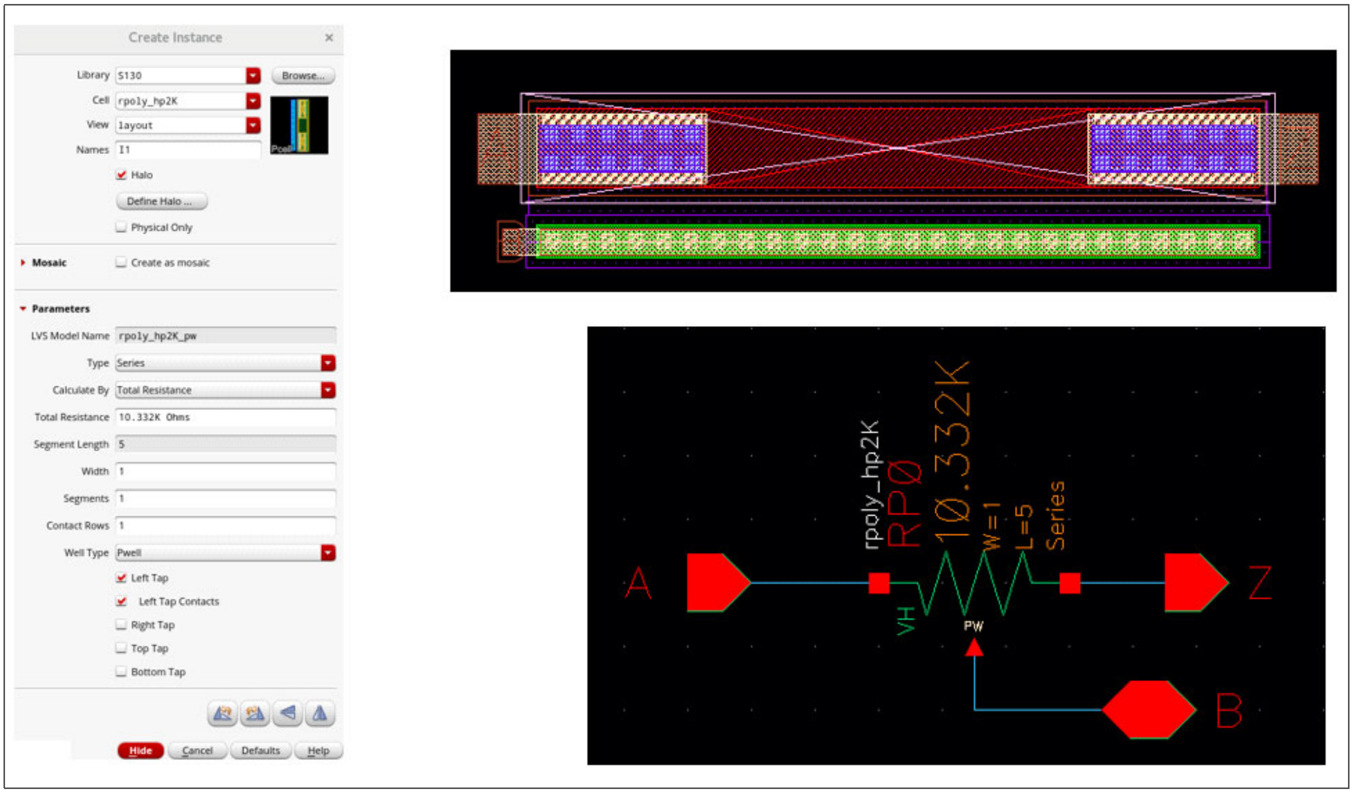


Figure 3-38. High Sheet Rho Poly Resistor Pcell Layout

Table 3-22 lists the pcell parameters available for high sheet rho poly resistors.

Table 3-22. High Sheet Rho Poly Resistor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Type	Series, parallel, serpentine	–
Width	1 µm	1000 µm
Segments	1	50
Contact rows	1	5
Well type	N-well, p-well	–
Calculate by Total Resistance		
Total resistance	4.332 kΩ	1 GΩ
Calculate by Segment Length		
Segment length	2 µm	500 µm

3.5.8 P-Well Resistor Pcell Layout and Parameters

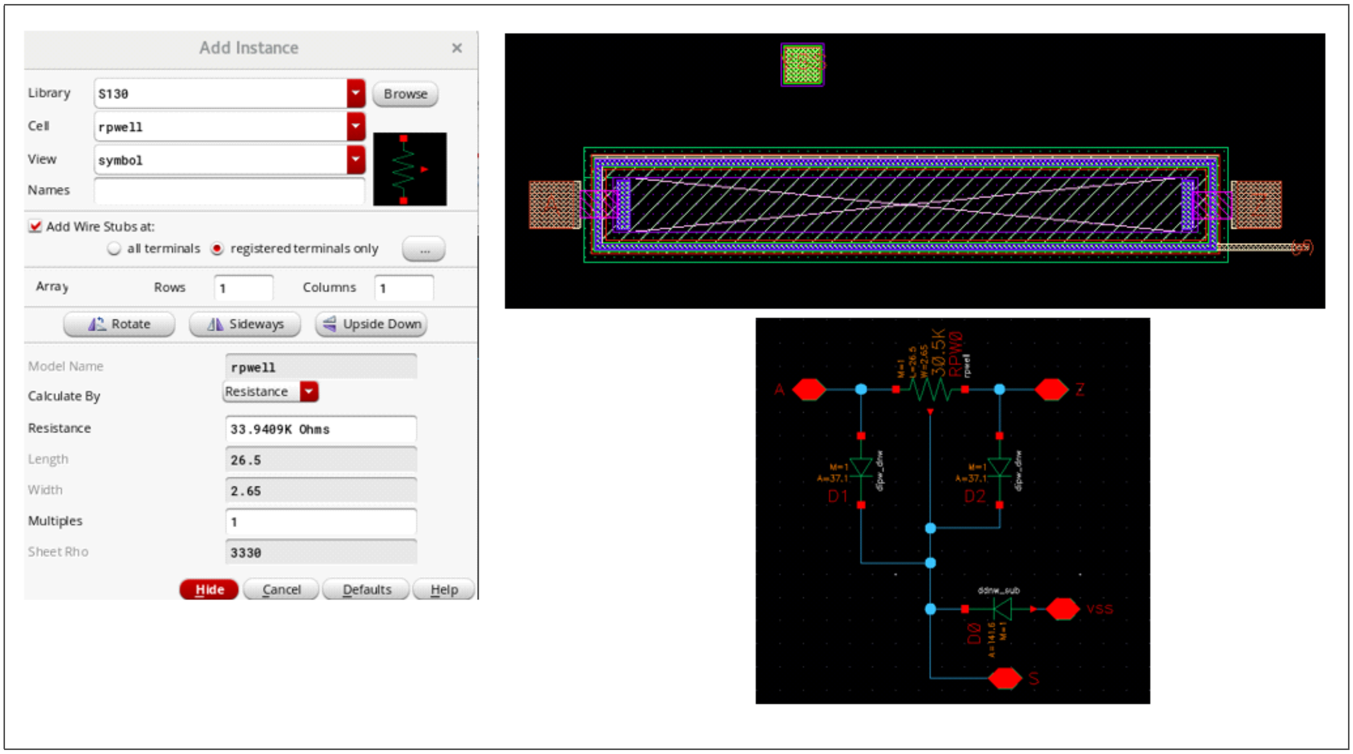


Figure 3-39. P-Well Resistor Pcell Layout

Table 3-23 lists the pcell parameters available for p-well resistors.

Table 3-23. P-Well Resistor Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Width	2.65 µm	–
Calculate by Resistance		
Resistance	33.9409 kΩ	333 kΩ
Calculate by Length		
Length	26.5 µm	265 µm

3.5.9 Metal Resistor Pcell Layouts and Parameters

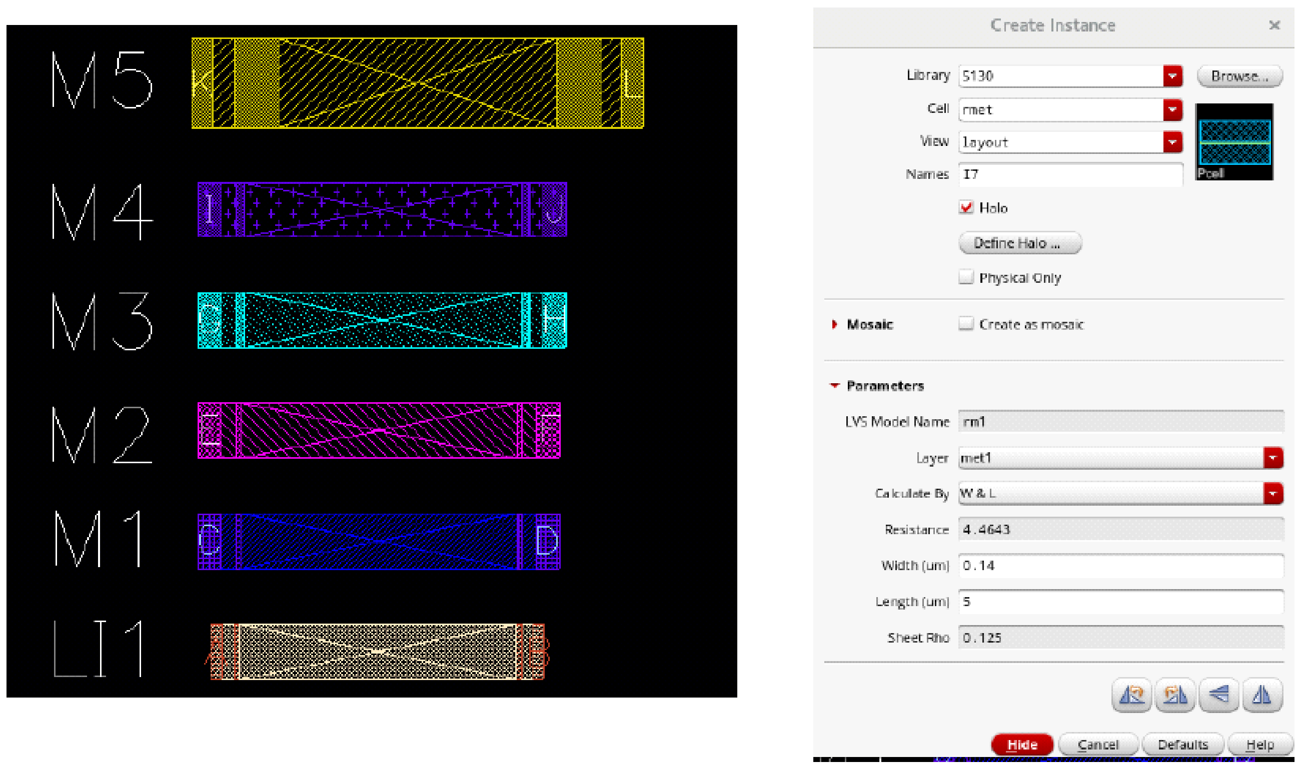


Figure 3-40. Metal Resistor Pcell Layouts

Table 3-24 lists the pcell parameters available for metal resistors.

Table 3-24. Metal Resistor Pcell Parameters (Page 1 of 2)

Parameter	Minimum (Default) Value	Maximum Value
Metal layer	li, met1–met5	–
Sheet rho	See *Table 3-15* on page 55	–
Multiples[1]	1	10,000
Calculate by Width and Length (W & L)		
Width: li met1, met2 met3,met4 met5	 0.29 µm 0.14 µm 0.3 µm 1.6 µm	100 µm
Length	0.01 µm	1,000,000 µm

1. The multiples value applies to the symbol only.

Table 3-24. Metal Resistor Pcell Parameters (Page 2 of 2)

Parameter	Minimum (Default) Value	Maximum Value
Calculate by Resistance and Width (R & W)		
Width: li met1, met2 met3,met4 met5	 0.29 µm 0.14 µm 0.3 µm 1.6 µm	100 µm
Resistance: li met1, met2 met3,met4 met5	 0.4414 Ω 4.3 mΩ 1.6 mΩ 200 µΩ	 44.138 MΩ 431.034 kΩ 156.667 kΩ 17.812 kΩ
1. The multiples value applies to the symbol only.		

3.5.10 Resistor Pcell LVS Checking Parameters

Table 3-25 lists the tolerance values associated with each of the resistor pcell parameters checked during LVS checking.

Table 3-25. Resistor Pcell LVS Checking Parameters

Parameter	Tolerance Value (%)
Width	0, 5 (serpentine only)
Length	0, 1 (serpentine only)
Multiples	0

3.6 Diodes

Table 3-26. S130 Standard Pcell Diodes

Device Type	Cell Name	Cell Description
N+ p-well diode	dnsd_pw	N+ diff in a p+ well
	dnsd_pw_v5	N+ diff AND thkox AND v5 in a p+ well
	dnsd_pw_lvt	N+ diff AND lvtn in a p+ well
	dnsd_pw_nat	N+ diff AND lvtn AND areaid:lvNative in a p+ well
	dnsd_pw_esd	N+ diff AND areaid:esd in a p+ well
	dnsd_pw_esd_v5	N+ diff AND areaid:esd AND thkox AND v5 in a p+ well
P+ n-well diode	dpsd_nw	P+ diff in nwell
	dpsd_nw_v5	P+ diff AND thkox AND v5 in nwell
	dpsd_nw_lvt	P+ diff AND lvtn in nwell
	dpsd_nw_hvt	P+ diff AND hvtp in nwell
	dpsd_nw_esd	P+ diff AND areaid:esd in nwell
	dpsd_nw_esd_v5	P+ diff AND areaid:esd AND thkox AND v5 in nwell

The diodes listed in *Table 3-27* are extracted from a deep n-well, where they acquire the highest voltage level contained within the deep n-well. For example, if the deep n-well contains a 5 V and a 12 V device, a 12 V diode is extracted.

Table 3-27. S130 Deep N-Well (Symbol Only) Diodes

Device Type	Cell Name	Cell Description
Deep n-well to substrate diode	ddnw_sub	Layout device extracted from a deep n-well; no corresponding layout pcell
	ddnw_sub_v5	
	ddnw_sub_v12	
	ddnw_sub_v20	
P-well to deep n-well diode	dipw_pw	Layout device extracted from a deep n-well; no corresponding layout pcell
	dipw_pw_v5	
	dipw_pw_v12	
	dipw_pw_v20	

3.6.1 Standard Diode Pcell Parameters

Table 3-28 lists the pcell parameters available for standard diodes.

Table 3-28. Standard Diode Pcell Parameters

Parameter	N+ P-Well Diodes		P+ N-Well Diodes	
	Minimum (Default) Value	Maximum Value	Minimum (Default) Value	Maximum Value
Contacts on diode	On/off	–	On/off	–
Multiples[1]	1	500	1	500
Calculate by Resistance				
Width	0.33 µm	1,000,000 µm	0.33 µm	250 µm
Length	0.33 µm	1,000,000 µm	0.33 µm	250 µm
Calculate by Segment Length				
Area	0.1089 $µm^2$	33,000 $µm^2$	0.1089 $µm^2$	6250 $µm^2$
Width	0.33 µm	1,000,000 µm	0.33 µm	250 µm

1. The multiples value applies to the symbol only.

3.6.2 Deep N-Well Diode Schematic Elements and Pcell Parameters

During LVS checking, one or two diodes are formed wherever a deep n-well is used. The diodes formed depend on whether a hole is cut in the deep n-well that isolates the p+ tub. These diodes must exist in the schematic for default LVS checking to run clean.

For NMOS in a deep n-well inside an n-well ring, the ddnw_sub and dipw_dnw diodes shown in *Figure 3-41* are formed.

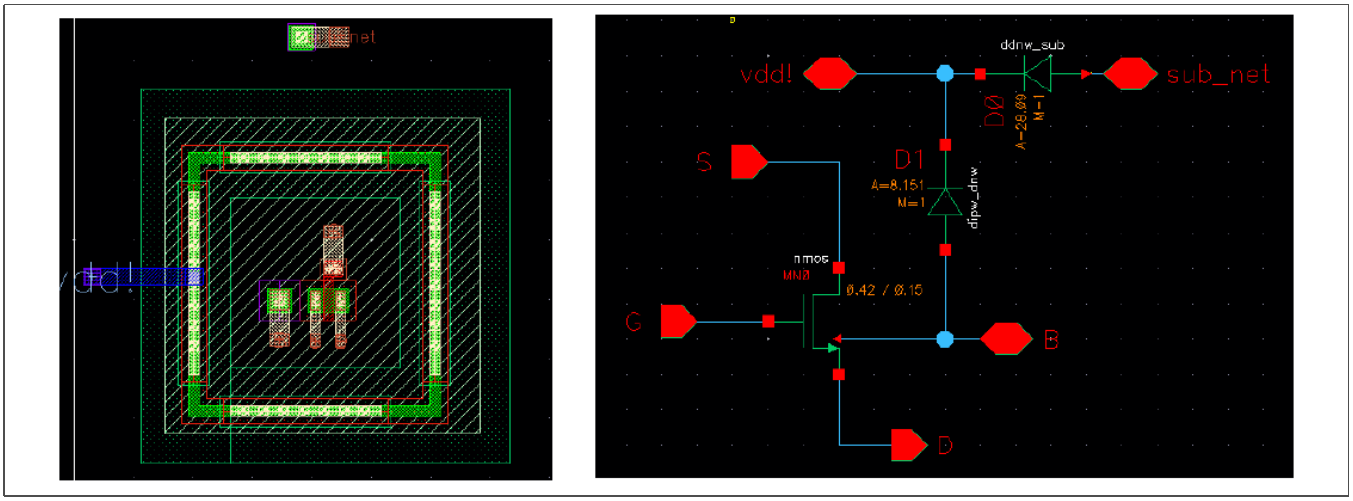


Figure 3-41. NMOS in an N-Well Ringed Deep N-Well Diode Formation

For PMOS in an n-well enclosed deep n-well, the single ddnw_sub diode shown in *Figure 3-42* is sufficient.

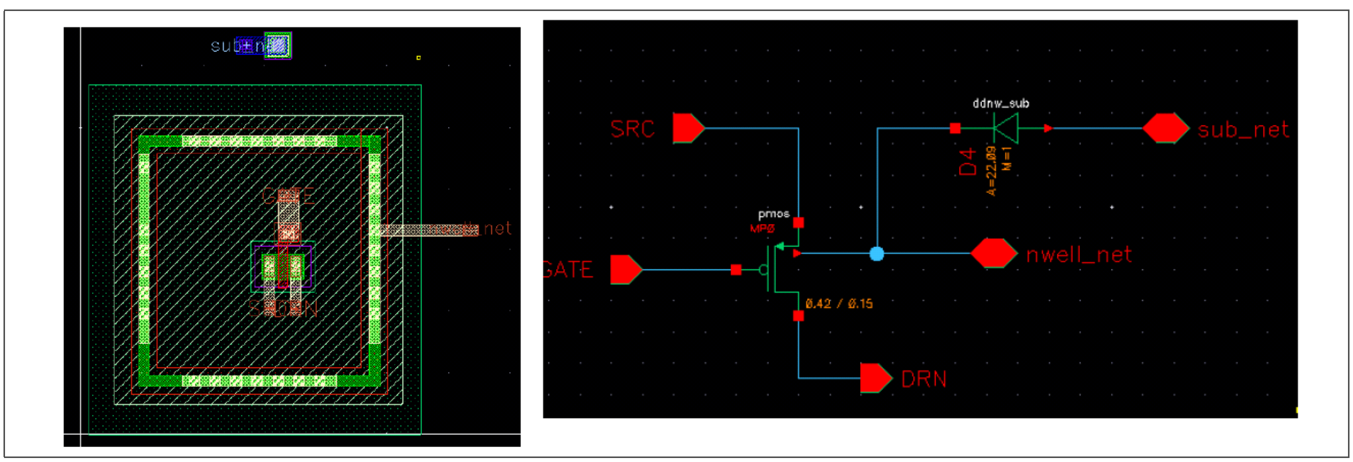


Figure 3-42. PMOS in an N-Well Enclosed Deep N-Well Diode Formation

Figure 3-43 shows an example of the diodes formed in an inverter with a shared deep n-well.

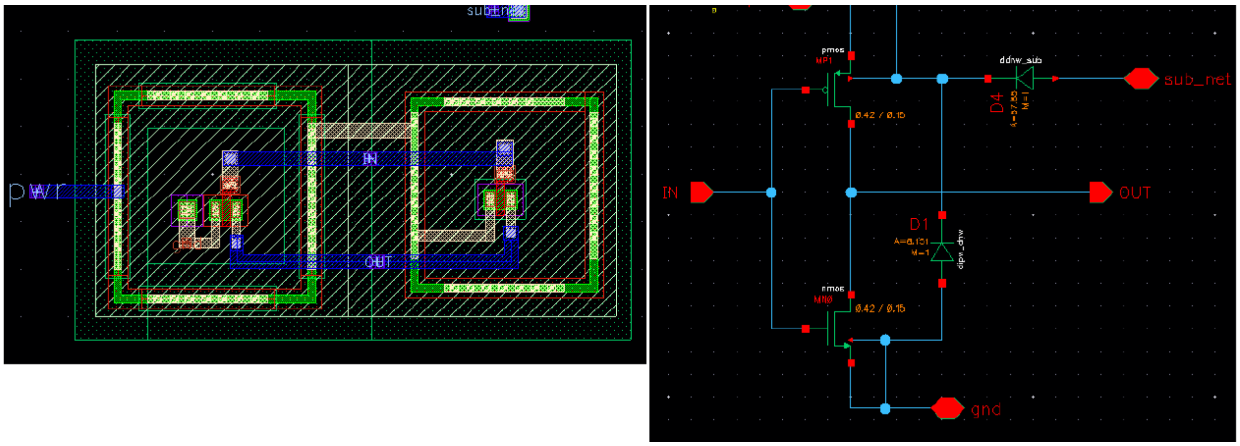


Figure 3-43. Inverter with Shared Deep N-Well Diode Formation

Table 3-29 lists the pcell parameters available for deep n-well diodes.

Table 3-29. Deep N-Well Diode Pcell Parameters

Parameter	Deep N-Well to Substrate Diodes		P-Well to Deep N-Well Diodes	
	Minimum (Default) Value	Maximum Value	Minimum (Default) Value	Maximum Value
Add wire stubs	All terminals or registered terminals only	–	All terminals or registered terminals only	–
Area	9 µm^2	10,000,000,000 µm^2	1.6129 µm^2	10,000,000,000 µm^2
Perimeter	12 µm	400,000 µm	5.08 µm	400,000 µm
Multiples[1]	1	500	1	500

1. The multiples value applies to the symbol only.

3.6.3 Diode Pcell LVS Checking Parameters

Table 3-30 lists the tolerance values associated with each of the diode pcell parameters checked during LVS checking.

Table 3-30. Diode Pcell LVS Checking Parameters

Parameter	Tolerance Value (%)[1]
Area	0
Perimeter	0
Multiples	0

1. Deep n-well (symbol only) diodes are checked to a 0.005% tolerance value because of the irregular shapes of the n-well structures in some devices, such as drain-extended devices.

3.6.4 Deep N-Well Diode Extraction Calculations for P-Well Resistors

A p-well resistor is formed from an isolated p-well (p+ tub) in a deep n-well. It requires the deep n-well diodes show in *Figure 3-44*.

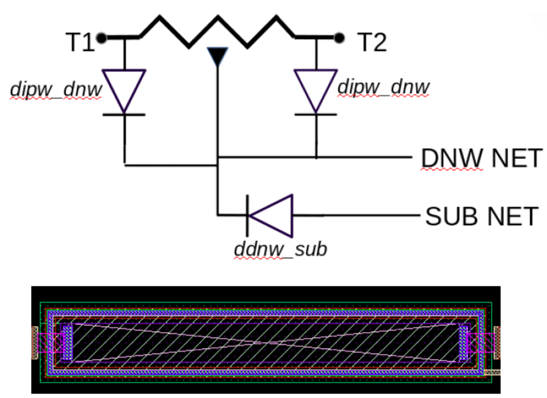
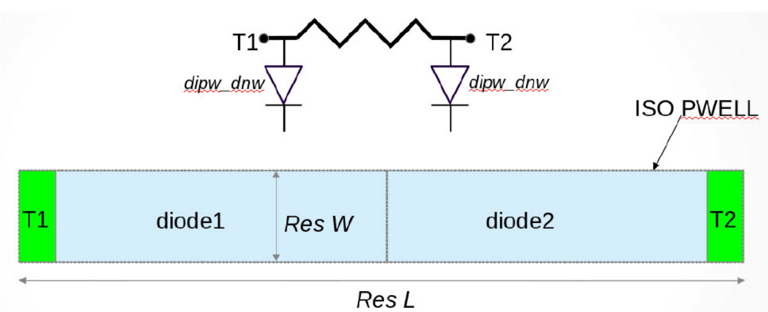


Figure 3-44. P-Well Resistor Deep N-Well Diodes

The ddnw_sub diode area and perimeter values are derived in a straightforward fashion from the area and perimeter of the deep n-well of the p-well resistor.

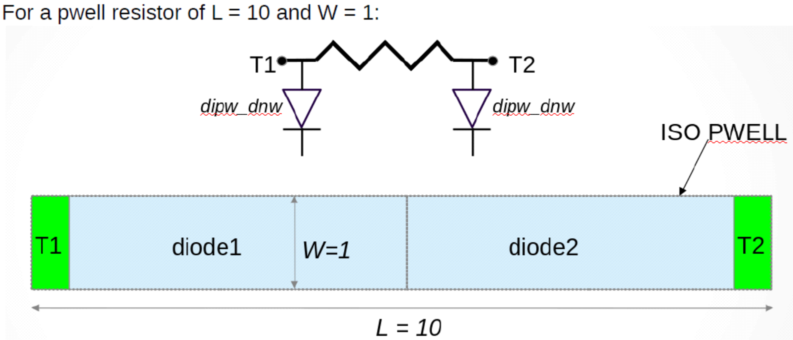
The dipw_dnw diode that normally has the same area and perimeter as the isolated p-well inside the n-well ring cannot be represented as a single diode because the isolated p-well does not have a single net. Instead, two diodes are formed, one at each resistor terminal, and each diode has an area parameter of half the isolated p-well. Because the shared perimeter is artificially induced, the perimeter of each diode is determined by the calculations shown in *Figure 3-45*.



Diode1 area = (Res(L) X Res(W)) / 2
Diode2 area = (Res(L) X Res(W)) / 2
Diode1 perimeter = Res(L) + Res(W)
Diode2 perimeter = Res(L) + Res(W)

Figure 3-45. P-Well Resistor Deep N-Well Diode Parameter Calculations (Page 1 of 2)

For a pwell resistor of L = 10 and W = 1:



Diode1 area = 10 / 2 = 5
Diode2 area = 10 / 2 = 5
Diode1 perimeter = 5 + 5 + 1 = 11
Diode2 perimeter = 5 + 5 + 1 = 11

Figure 3-45. P-Well Resistor Deep N-Well Diode Parameter Calculations (Page 2 of 2)

3.7 Miscellaneous Cells

Table 3-31. S130 Miscellaneous Cells

Device Type	Cell Name	Cell Purpose
Anchor	anchor	Used to insert anchor structures in critical side and corner areas.
Mask text	text_pcell	Used to place chip identification and other text that is visible on the chip after manufacturing.
Nikon (fixed)	nikon	Used to place additional Nikon structures other than those already integrated into the seal ring.
Parasitic device	pcapacitor, presistor	Used by parasitic extraction tools to create extracted views.
	icecap	Standard cell parasitic capacitor.
Seal	sealring	Seal ring pcell.

3.7.1 Anchor Pcell Layout

The anchor cell shown in *Figure 3-46* enables users to place anchors in the open areas of critical sides and corners.

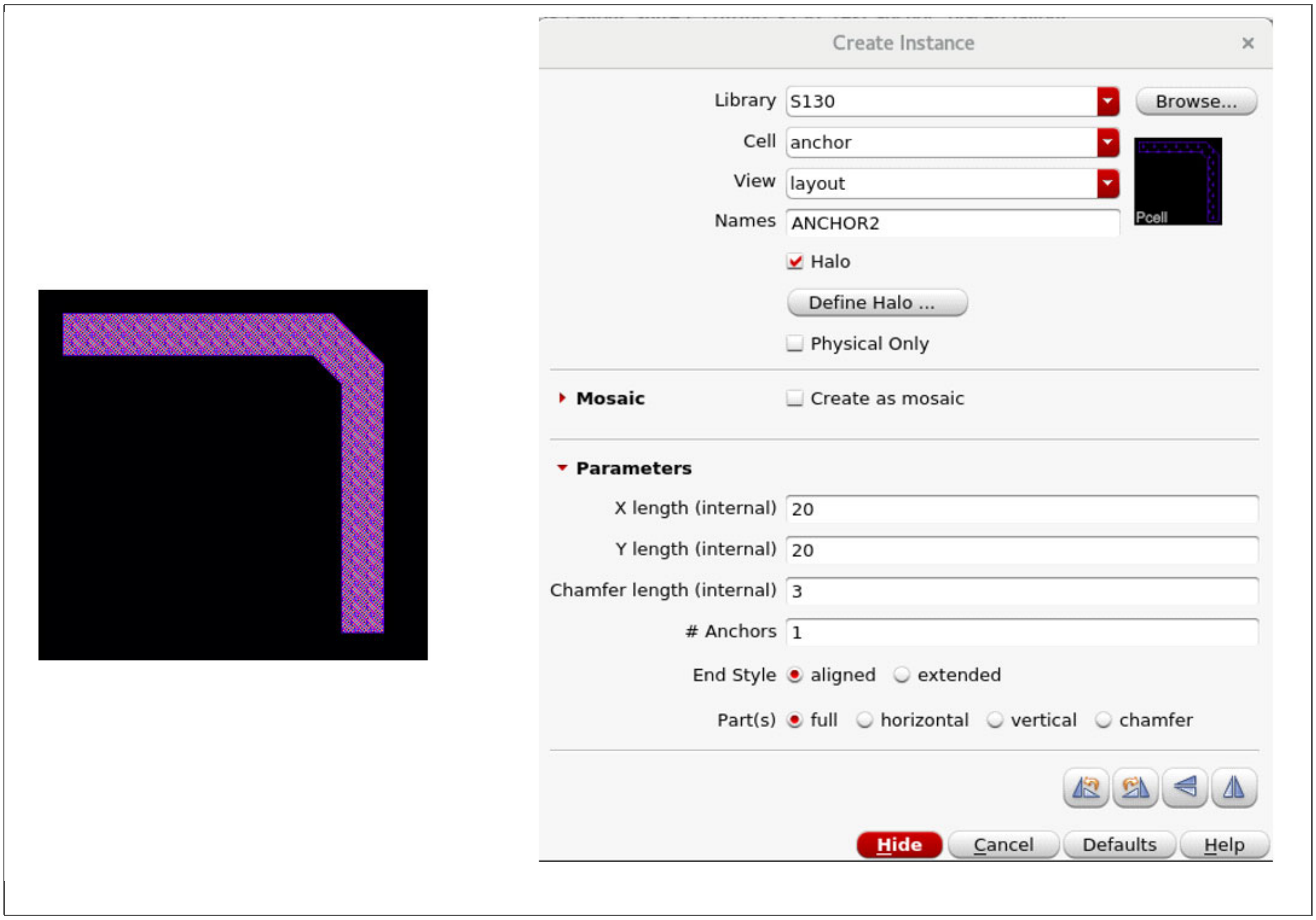


Figure 3-46. Anchor Pcell Layout

3.7.2 Mask Text Pcell Layout and Parameters

The mask text pcell shown in *Figure 3-47* enables users to add polygon shapes that resemble legible text.

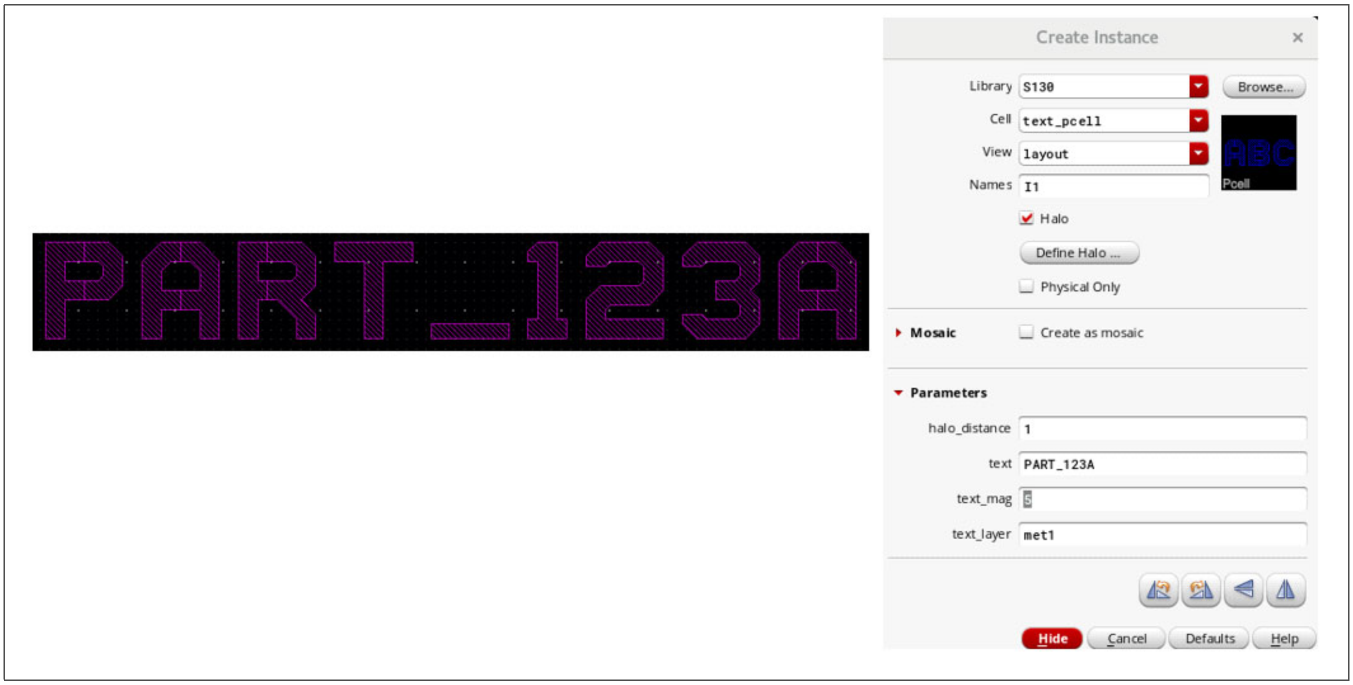


Figure 3-47. Mask Text Pcell Layout

Table 3-32 lists the pcell parameters available for mask text.

Table 3-32. Mask Text Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Halo distance[1]	1 µm	10,000,000 µm
Text	<string>	–
Text magnification	1	10,000,000
Text layer	<string>	–

1. Halo distance determines the size of the bounding box around the placed pcell.

3.7.3 Nikon Cell Layout

The Nikon cell shown in *Figure 3-48* enables users to place a fixed-size Nikon cell in addition to those already included in the seal ring pcell.

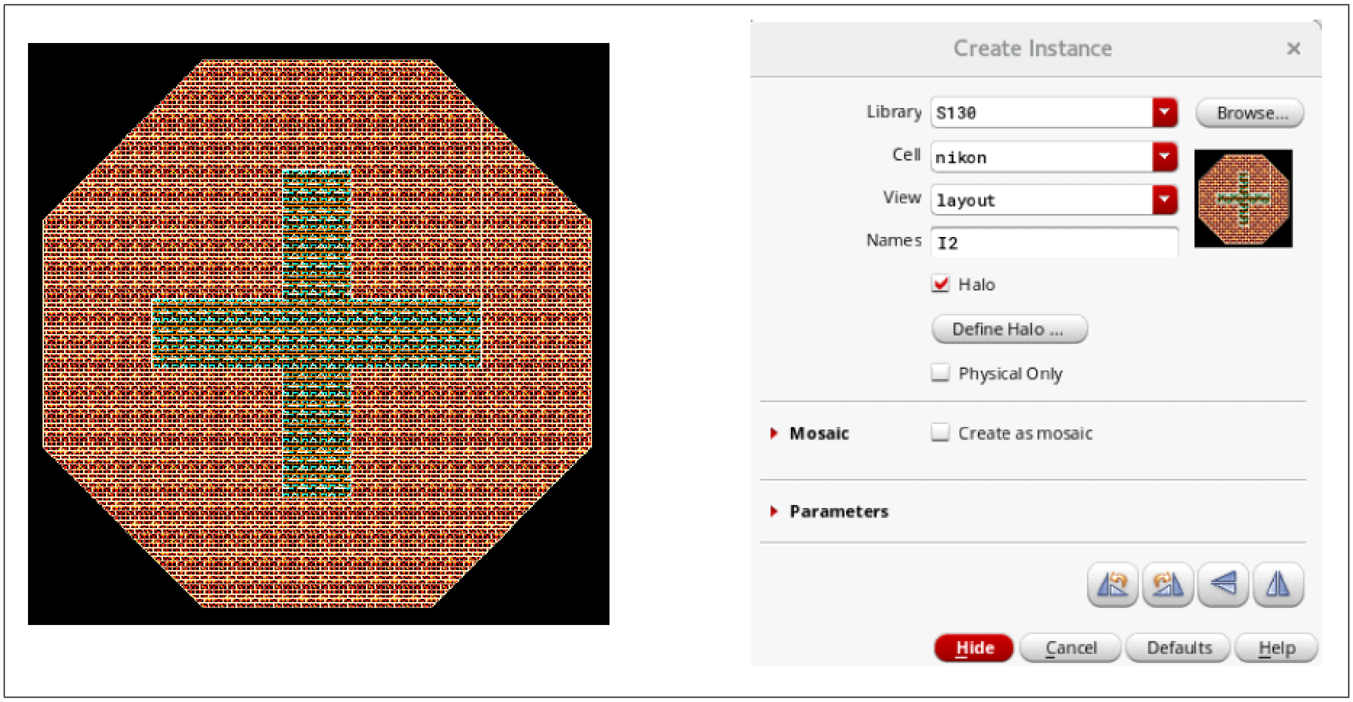


Figure 3-48. Nikon Cell Layout

3.7.4 Seal Ring Pcell Layout and Parameters

The seal ring pcell shown in *Figure 3-49* offers width and length parameters as well as an option for specifying an extra corner cut. This pcell is not checked during LVS checking.

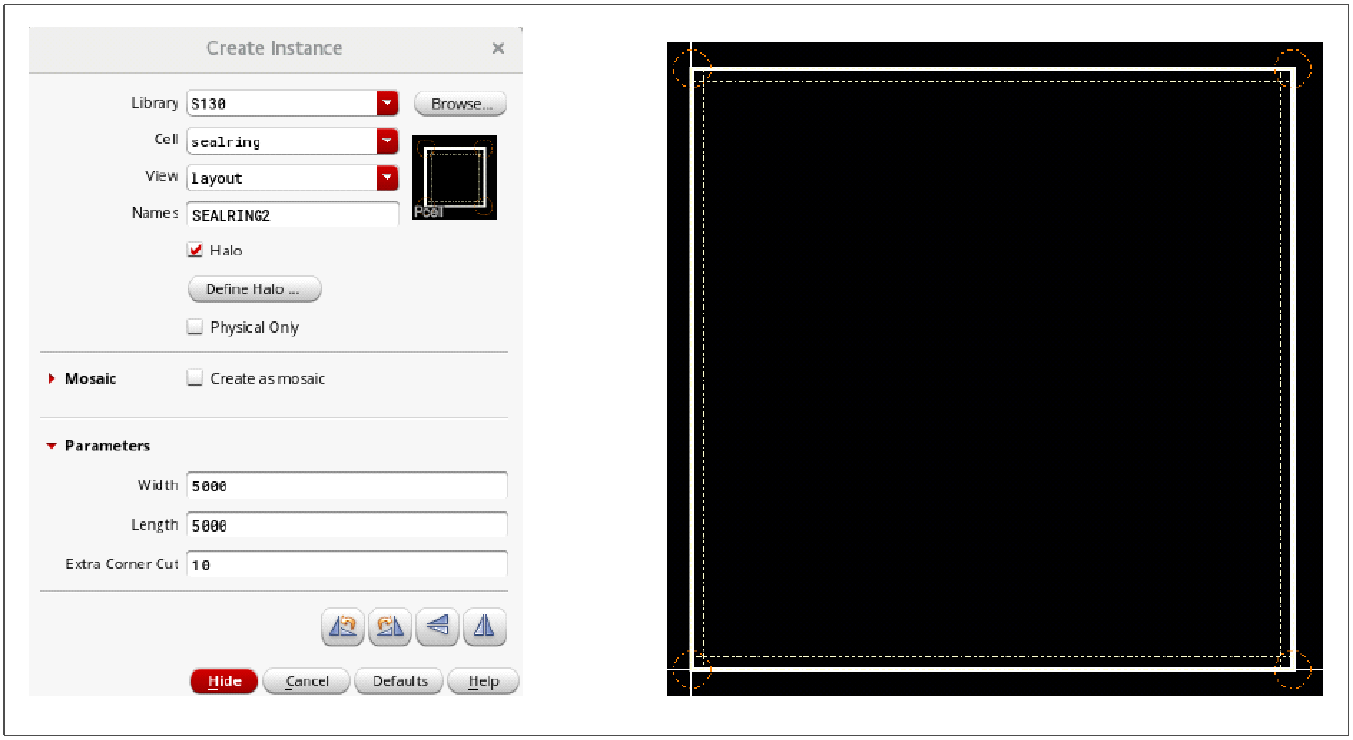


Figure 3-49. Seal Ring Pcell Layout

Table 3-33 lists the pcell parameters available for the seal ring.

Table 3-33. Seal Ring Pcell Parameters

Parameter	Minimum (Default) Value	Maximum Value
Width	1000 µm	500,000 µm
Length	1000 µm	500,000 µm
Extra corner cut	0.25 µm	250 µm

4 Layout Creation

Virtuoso XL (VXL) assists in the creation of layout from a schematic view. It can be started from either a schematic or layout view.

1. From a schematic view or open layout view, execute the menu command **Launch > Layout XL**. The Startup Option form shown in *Figure 4-1* opens when starting from a schematic without a matching layout view.

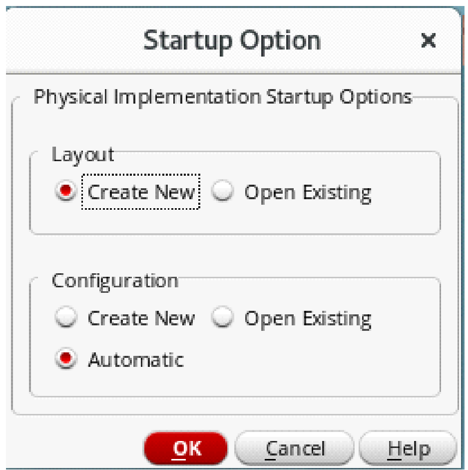


Figure 4-1. VXL Startup Option Form

2. Fill in the form with the appropriate options. Select **Create New** to open the New File form shown in *Figure 4-2* on page 77.

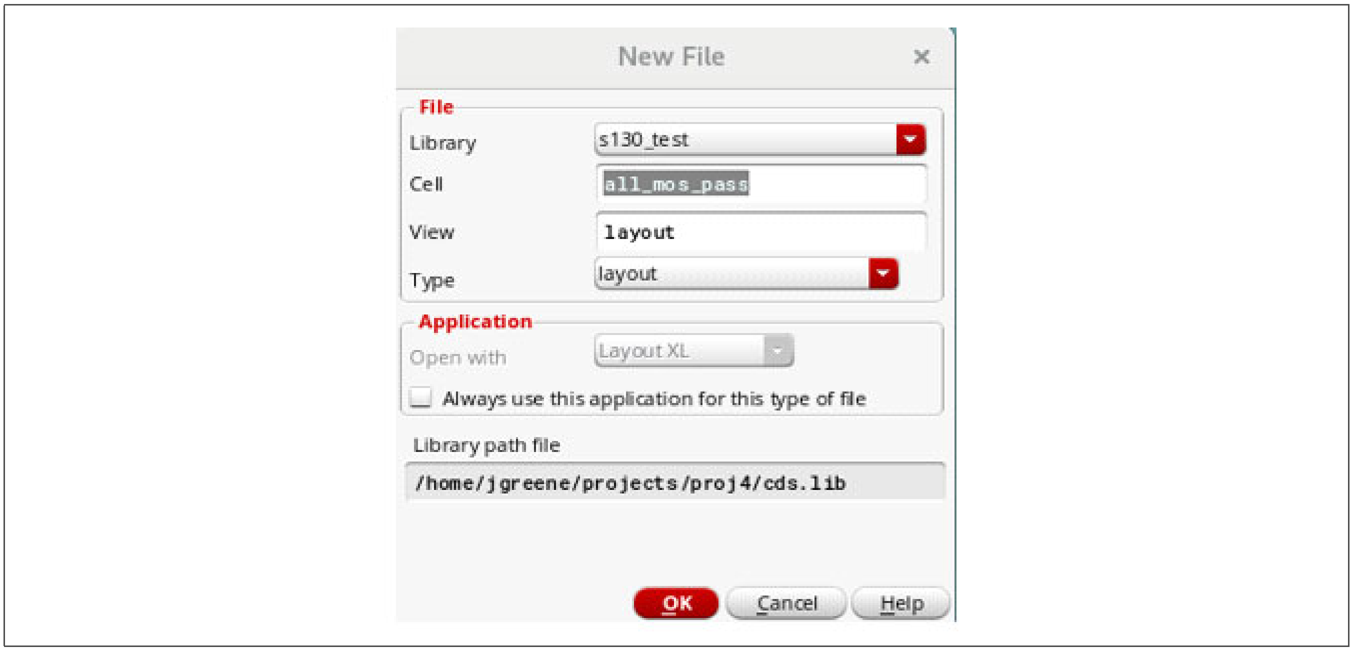


Figure 4-2. VXL New File Form

3. After confirming or editing the New File form, click **OK** to continue. The Layout Constraint Group Setting form opens and defaults to SkyWater as the constraint group.

4. Click **OK** to proceed. A layout view opens in which the layout and schematic windows are arranged side by side, as shown in *Figure 4-3*.

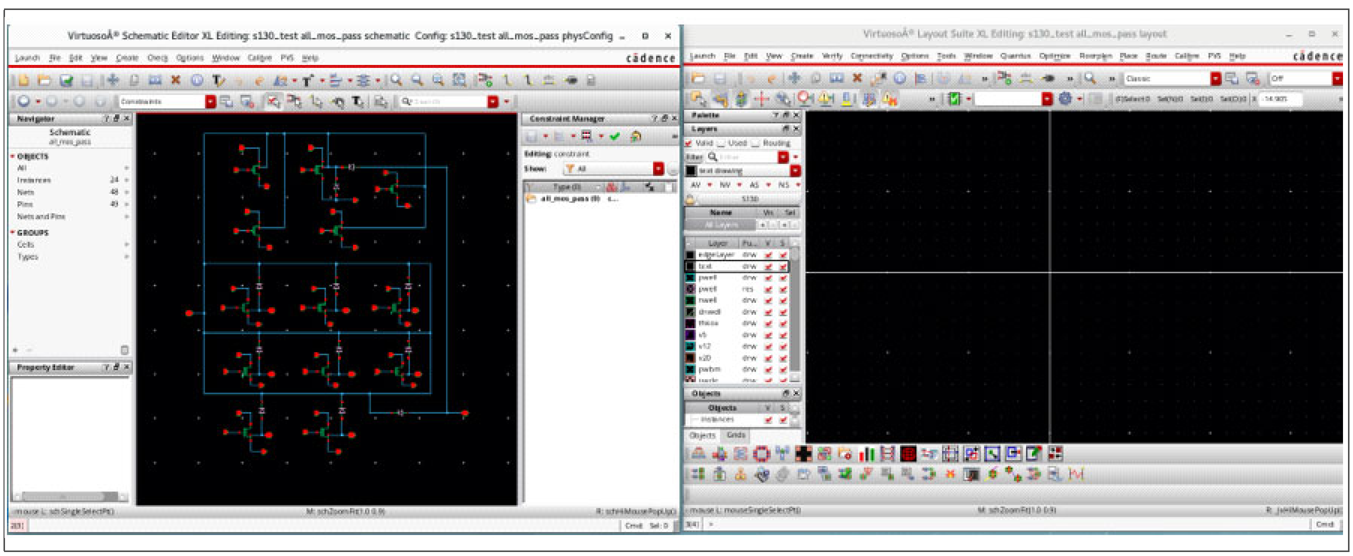


Figure 4-3. VXL Schematic Editor and Layout Suite Windows

The commands used to generate the layout are available in the menus at the top of the layout window and by using the toolbar buttons at the bottom of the layout window.

5. To initialize a layout, use the **Generate All From Source** command (lower-left toolbar button or **Connectivity > Generate > All From Source** menu option) to open the Generate Layout form shown in *Figure 4-4*.

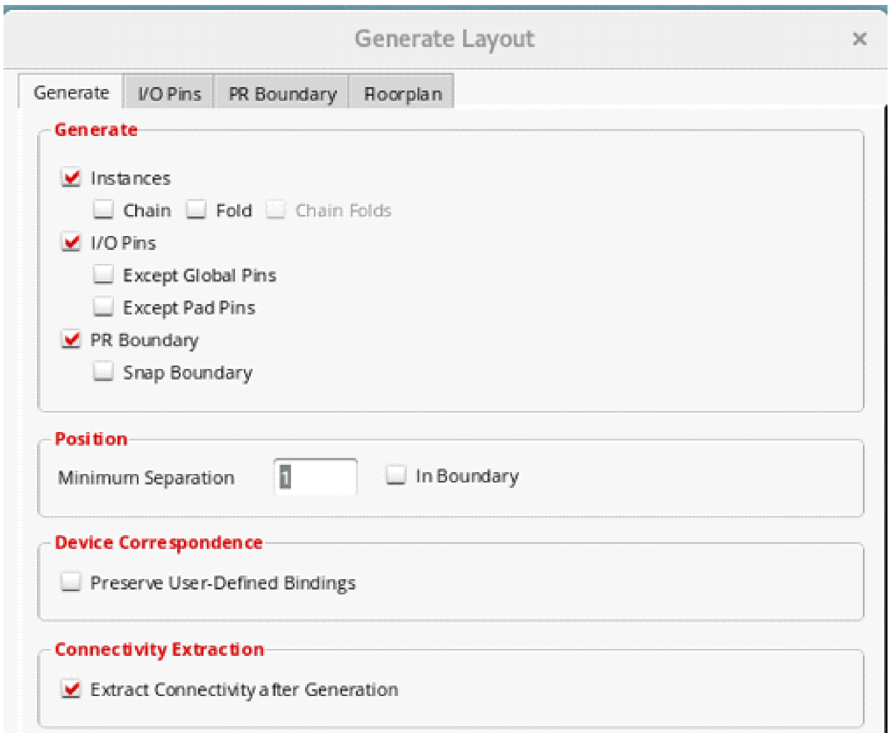


Figure 4-4. VXL Generate Layout Form

Chain and **Fold** options are supported for non-fixed-geometry MOS devices.

6. Select the **I/O Pins** tab, shown in *Figure 4-5* on page 79, to change the pin settings, as needed.

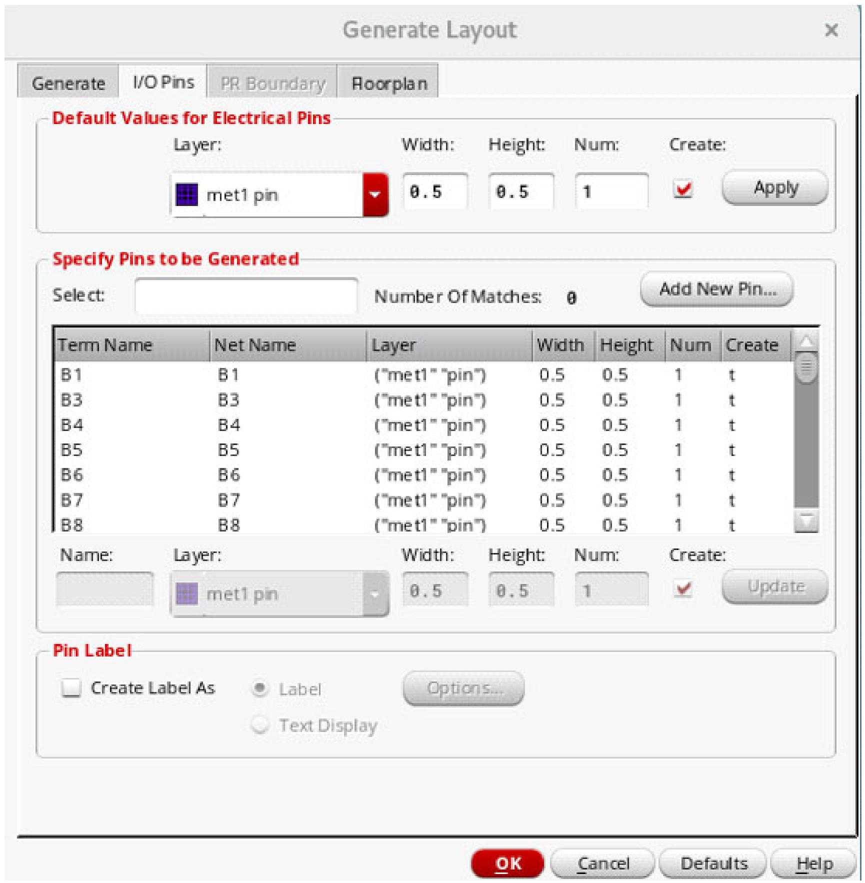


Figure 4-5. VXL Generate Layout Form I/O Pins Tab

7. Select all the other applicable settings and click **OK** to generate the layout and pin placements. The updated layout window, as shown in the example in *Figure 4-6* on page 80, includes all the pins and instances of each cell from the schematic.

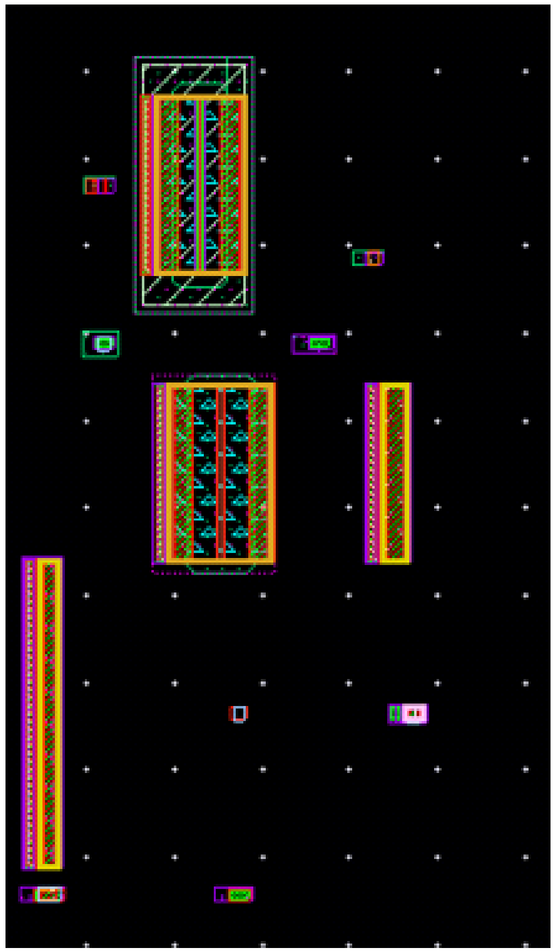


Figure 4-6. VXL Updated Layout Window Example

8. To assist with floorplanning, turn on the **Nets** option shown in *Figure 4-7* using the **Options > Display** menu command.

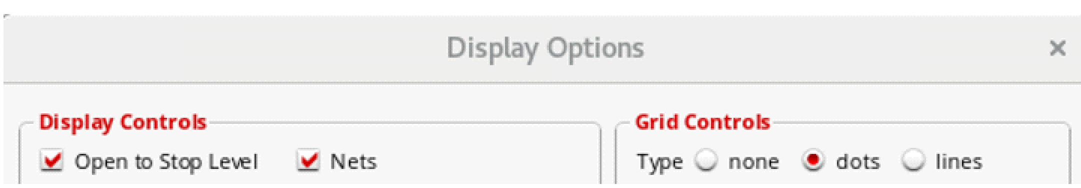


Figure 4-7. VXL Display Options Form

The **Nets** option shows the flight lines, like those shown in the example in *Figure 4-8* on page 81, that illustrate the connectivity of all the pins and devices.

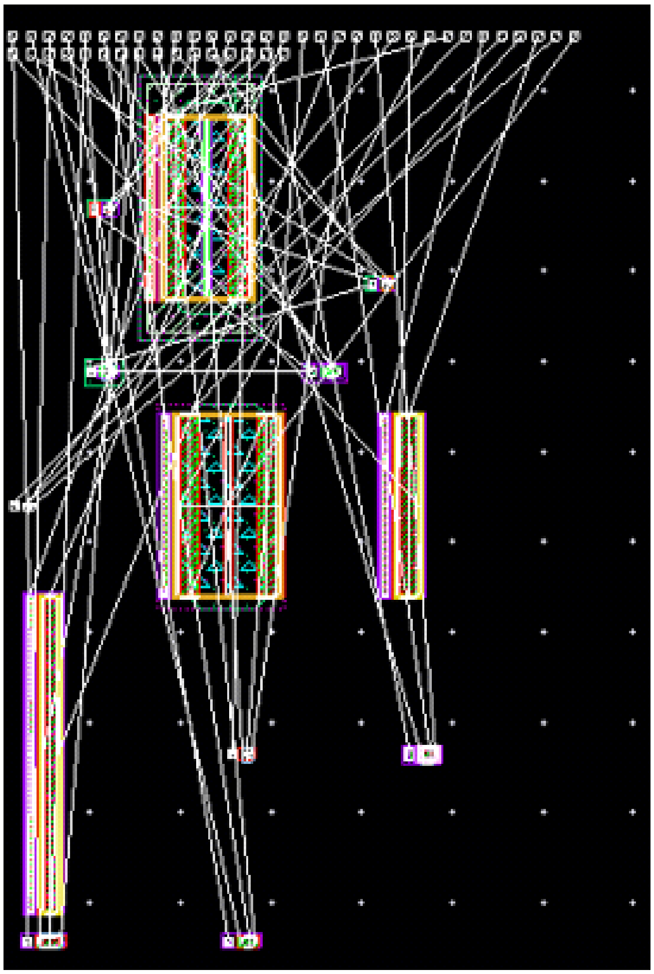


Figure 4-8. VXL Updated Layout Window Example with Net Flight Lines

Even with the Nets option turned off, connectivity lines are shown when selecting, editing, or moving a pin or device. In addition, the schematic element that matches the selected object is highlighted to confirm which object is selected.

9. While moving an object, press the F3 function key to open the Move options form, shown in *Figure 4-9* on page 82, and set the **Snap Mode**.

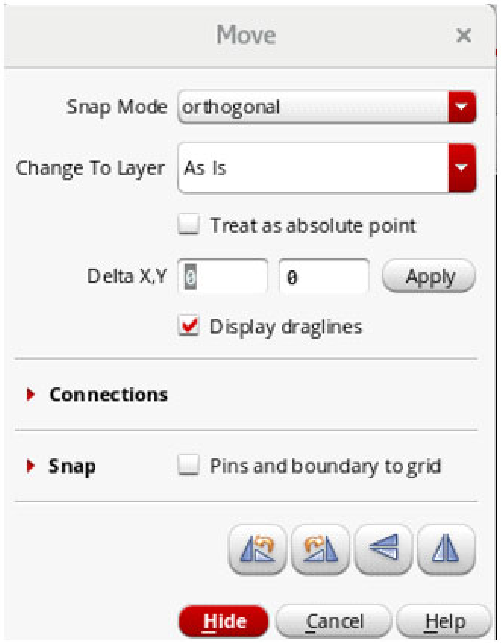


Figure 4-9. VXL Move Form

10. Use the **Create > Wire** menu command and the F3 key to open the Create Wire options form, shown in *Figure 4-10* on page 83, and expedite interconnect creation.

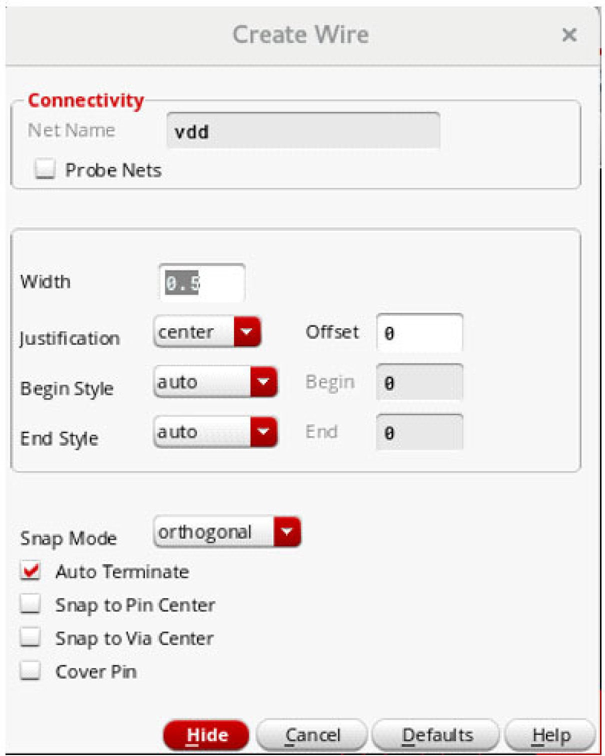


Figure 4-10. VXL Create Wire Form

While digitizing wiring, the connectivity lines are visible to assist with correct layout. A right-click while creating wires enables layers to be changed as well as automatic via placement.

11. Move the pins and instances and route wiring until the layout is complete.

12. Verify the final layout is clean using the design rule checking (DRC) and layout versus schematic (LVS) checking tools provided in the PDK. For more information, see *Section 5 Physical Verification* on page 84.

5 Physical Verification

Both Siemens Calibre and Cadence PVS are available in the S130 PDK for DRC and LVS checking.

Note: For accurate results, the PDK `.simrc` file from the `$PDK_HOME/SAMPLES` directory must be present in the working directory when running Calibre or PVS LVS.

5.1 Calibre DRC and LVS

5.1.1 Setup

Run Siemens Calibre design rule checking (DRC) or layout versus schematic (LVS) checking using the Calibre Interactive menus in either the schematic or layout view, as appropriate. These menus are automatically populated with the inputs, rule files, and run directories.

Note: Schematics must be checked and saved before running LVS or LVS fails.

5.1.2 Customization

Calibre features a DRC or LVS customization form triggered by an environment variable set in the project start script. This form opens at DRC or LVS startup or by clicking the Customization button in the Calibre Interactive form. *Figure 5-1* and *Figure 5-2* on page 85 show the DRC and LVS customization settings, respectively.

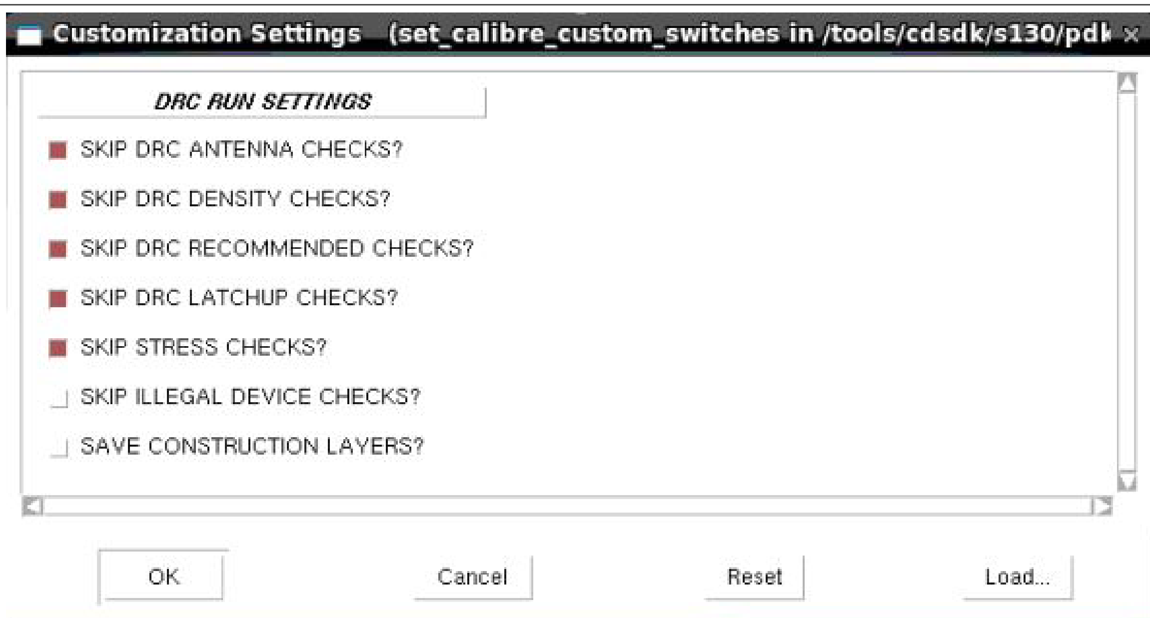


Figure 5-1. Calibre DRC Customization Form

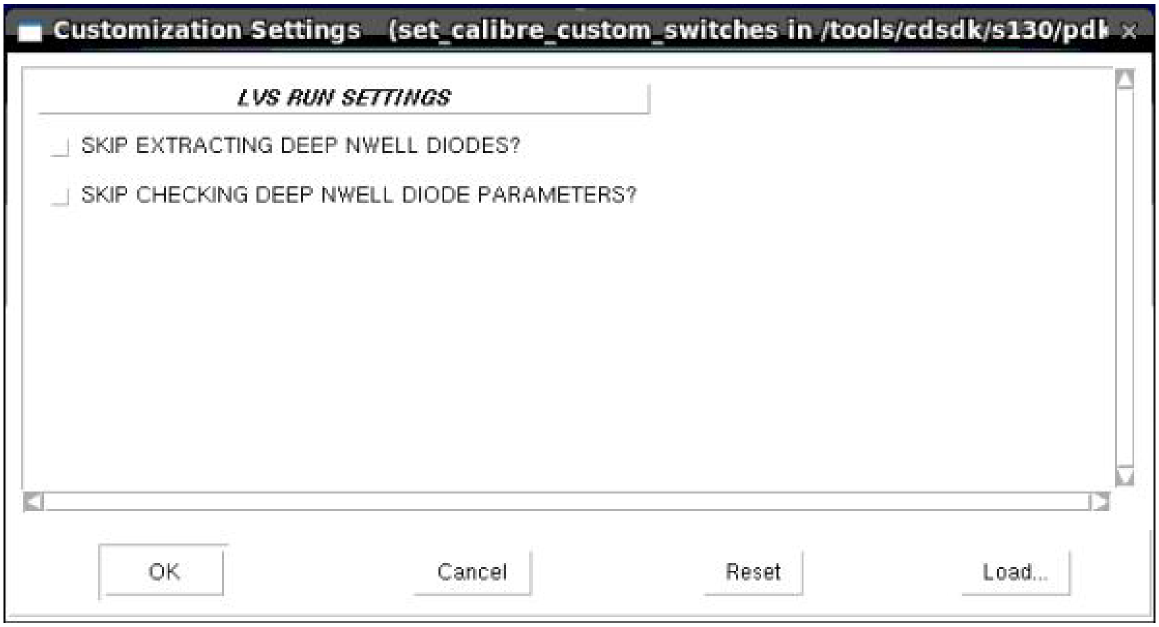


Figure 5-2. Calibre LVS Customization Form

5.1.3 LVS Soft Connection Checking

When a soft connection error occurs during LVS checking, a red “X” appears next to **Softchk Database** under **ERC** in the Calibre RVE results viewer window, as shown in *Figure 5-3*.

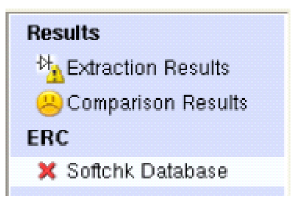


Figure 5-3. Calibre RVE Softchk Database Error Flag

Click on **Softchk Database** to view the causes of soft connection check errors, like those shown in *Figure 5-4* on page 86, in the RVE window.

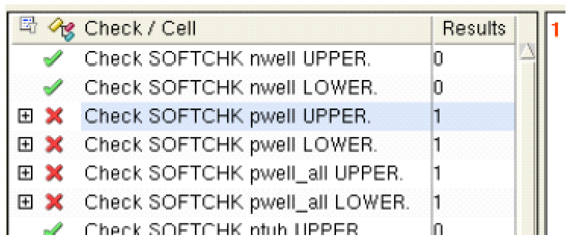


Figure 5-4. Calibre RVE Softchk Database Error Results

5.2 PVS DRC and LVS

5.2.1 Setup

Select **PVS > Run DRC** or **PVS > Run LVS** using the Cadence Physical Verification System (PVS) menu to open the DRC or LVS Run Submission form. These forms are automatically populated with the inputs, rule files, and run directories.

5.2.2 Configuration

PVS features DRC and LVS configuration option menus, shown in *Figure 5-5* on page 87 and *Figure 5-6* on page 88, respectively, that open by clicking the **Configurator** tab next to the **Tech&Rules** tab.

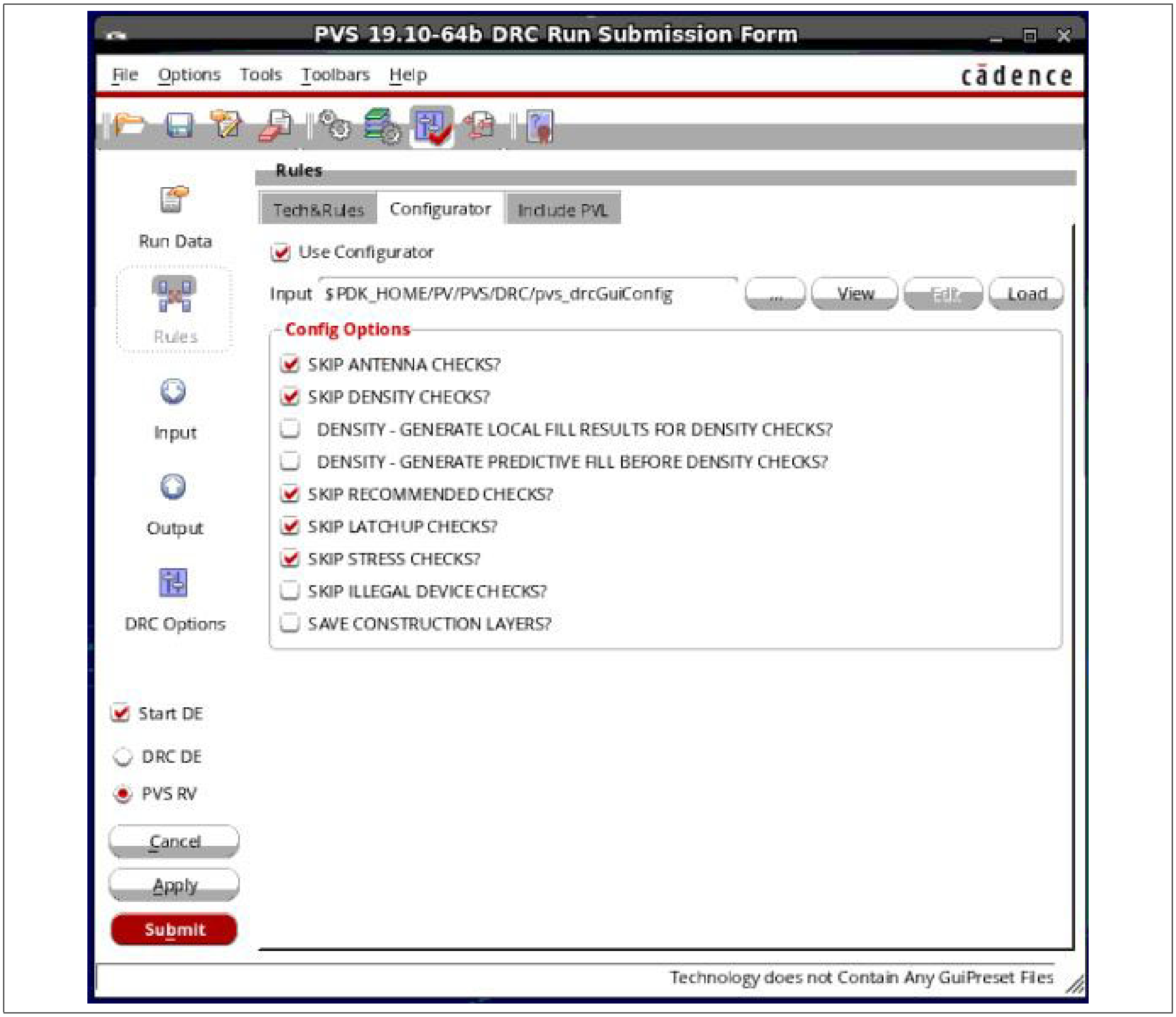


Figure 5-5. PVS DRC Configuration Options Menu

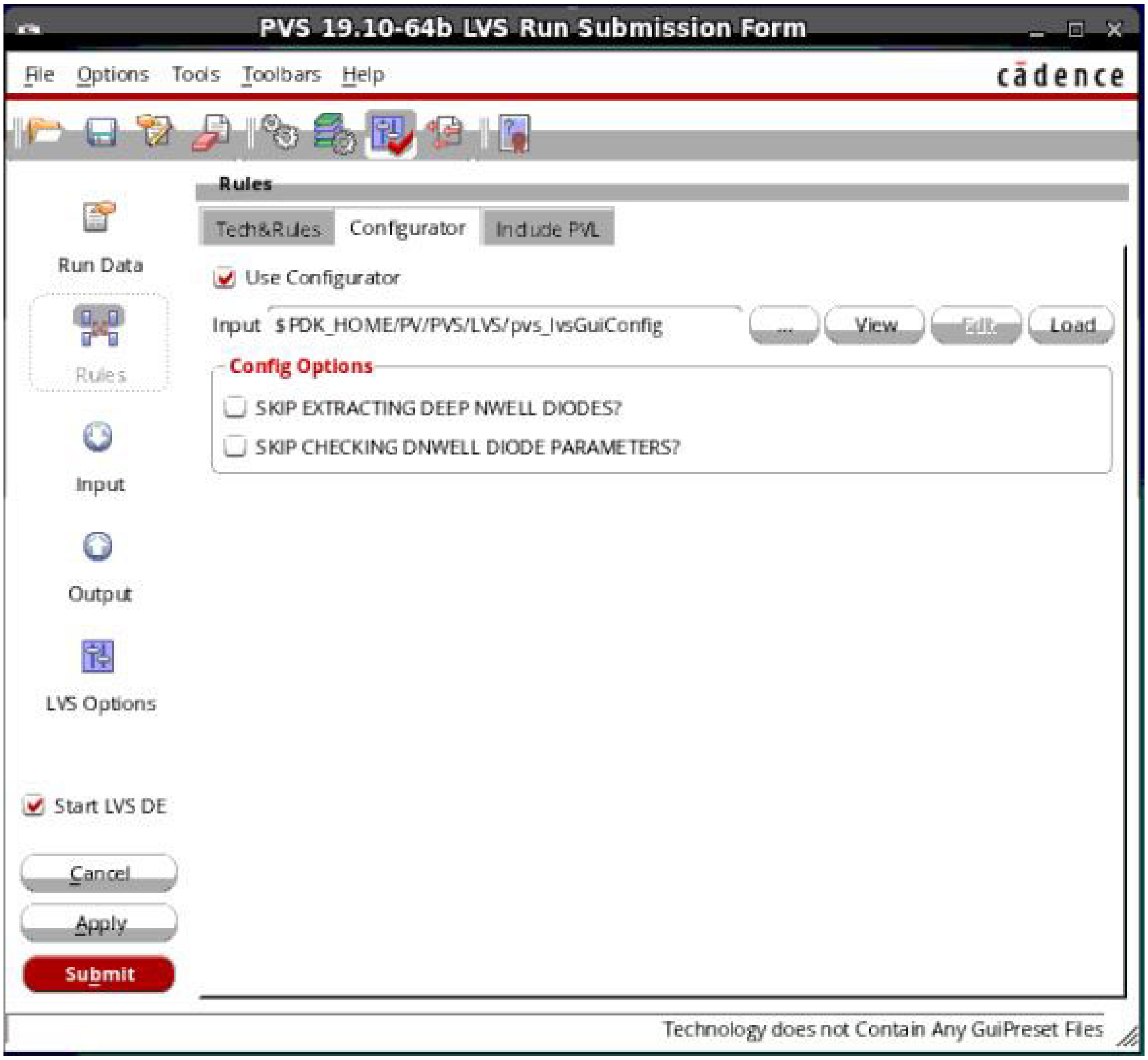


Figure 5-6. PVS LVS Configuration Options Menu

5.3 Calibre and PVS Options

5.3.1 DRC Options

The default DRC options, listed in *Table 5-1* on page 89, are typically used for cell-level designs. For larger blocks, chip-level designs, and before tapeout, set the DRC options to include all the checks. For example, antenna checks might not be relevant when running a small cell, but antennas must pass DRC before tapeout. This example is also true for recommended, latchup, stress, and illegal device checks.

Although the foundry provides post-tapeout fill, SkyWater encourages designers to check chip pattern density using the predictive fill algorithm before tapeout to prevent problems in the fab. In addition to predictive fill generation for density, a `calibre_generate_metal_fill.rul` file is available in the `$PDK_HOME/PV/Calibre/DRC` directory that can be used to generate metal fill before tapeout.

Table 5-1. Calibre and PVS DRC Options

DRC Option	Description
Skip antenna checks?	Antenna rules are skipped by default. Select this option to turn on antenna rule checking.
Skip density checks?	Density rules are skipped by default. Select this option to turn on density rule checking.
Density—Generate local fill results for density checks?	Local density (smaller window and step size than chip density) checks are skipped by default. Although they are not required by the foundry, local density check results can be used to localize foundry-required, chip-level density violations.
Density—Generate predictive fill results before density checks?	Predictive density checks are skipped by default. They generate fill shapes before density checks for more accurate results. **Note:** The fill patterns generated by this option are not the same as those performed by the foundry; however, they might help to predict pattern density issues before tapeout. See the *Appendix A Density Checking* on page 103 for more information about predictive fill.
Skip recommended checks?	Recommended checks are skipped by default. They check areas for lonely vias, floating interconnects, and other poor design practices.
Skip latchup checks?	Latchup checks are skipped by default. They ensure that well and substrate taps, for example, are placed within foundry-specified limits.
Skip stress checks?	Stress checks are skipped by default. Stress checks report errors related to chip-level assembly.
Skip illegal device checks?	Illegal device checks are skipped by default. They flag devices overlapped by layers that can cause device failure.
Save construction layers?	Construction layers are not saved by default. This option enables the creation of intermediate layers or supplemental shapes used to debug complex rules, such as slotting rules.

5.3.2 LVS Options

The LVS options that control extraction and verification of deep n-well diodes, listed in *Table 5-2*, can be skipped until the top-most cell in which deep n-wells are merged. If lower level cells have separate deep n-wells that are merged at a higher level of hierarchy, it is not necessary to check the area and perimeter parameters at the lower levels.

Table 5-2. Calibre and PVS LVS Options

LVS Option	Description
Skip extracting deep n-well diodes?	Diodes are extracted by default whenever deep n-wells are placed in the layout (see *Section 3.6 Diodes* on page 66 for more information). Select this option to skip deep n-well diode extraction.
Skip checking deep n-well diode parameters?	The area and perimeter of the deep n-well diodes extracted during LVS are calculated and compared in LVS by default. If a deep n-well diode is to be resized at a higher level of hierarchy (by abutment, for example), turn off these parameter checks at the lower level of hierarchy.

6 Parasitic Extraction

The Siemens Calibre PEX and Cadence PVS Quantus tools both extract parasitic devices from the layout that are not present in the schematic. For example, when two conductive layers overlap or run in parallel, they can act like a capacitor. Conductive layers in the layout have inherited resistance that can delay net information from one layout design device terminal to another. These parasitic devices impact the performance of a chip, and designers need to know how they affect the chip before fabrication occurs.

Note: Before running the Calibre PEX or PVS Quantus extraction tool, confirm that both the layout and corresponding schematic data have clean LVS checking results.

6.1 Calibre PEX

The Calibre PEX tool generates a “calibre” view that contains the design and parasitic devices in a layout. Users can simulate this view to get a better idea of how the chip will perform.

Follow these steps to run the Calibre PEX tool:

1. From the layout view, select **Calibre > Run PEX...**, as shown in *Figure 6-1*, to open the Calibre user interface PEX runset form shown in *Figure 6-2* on page 91 that has all the fields populated with the information needed to execute the run.

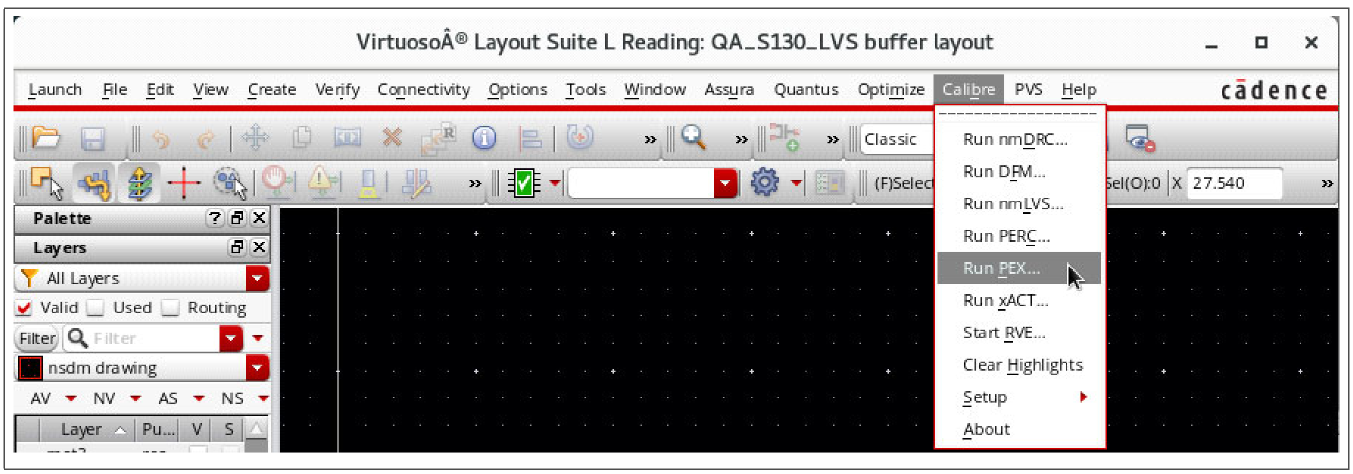


Figure 6-1. Calibre Tools Menu

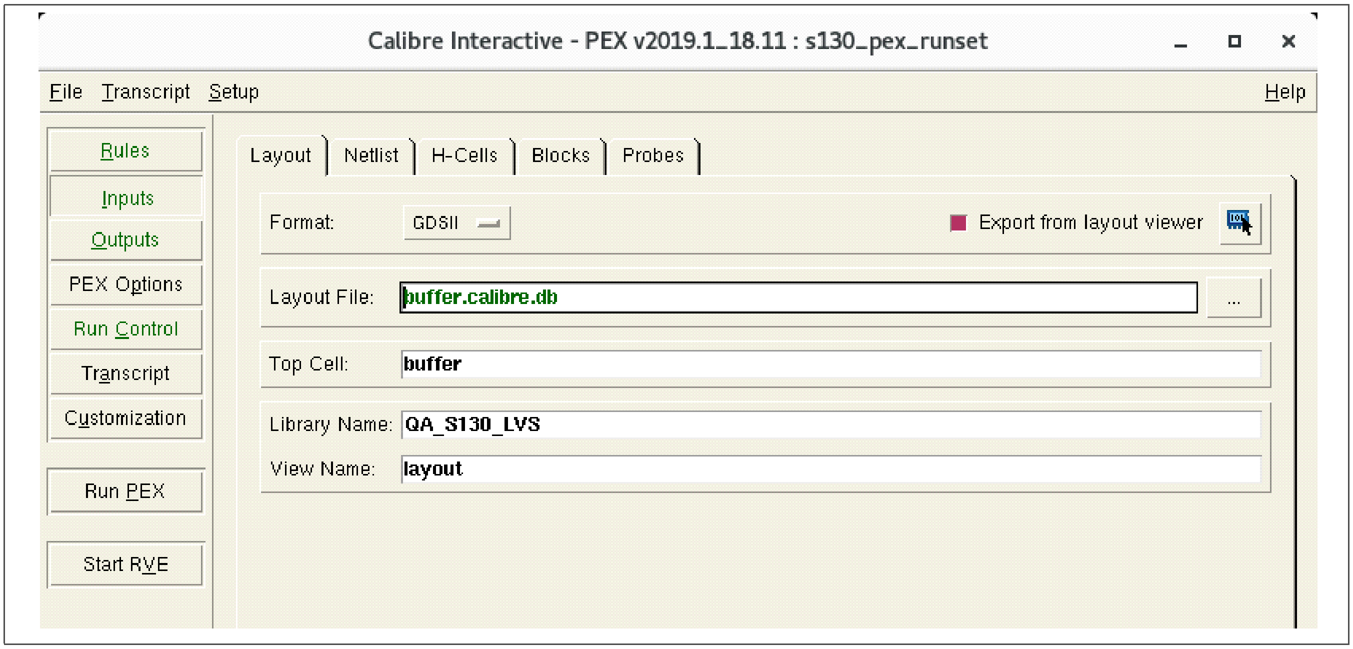


Figure 6-2. Calibre PEX Runset Form

2. Examine the information under each tab and make any necessary modifications before executing the run.

3. Click on the **Run PEX** button in the Calibre user interface, as shown in *Figure 6-3* on page 92, to begin the run.

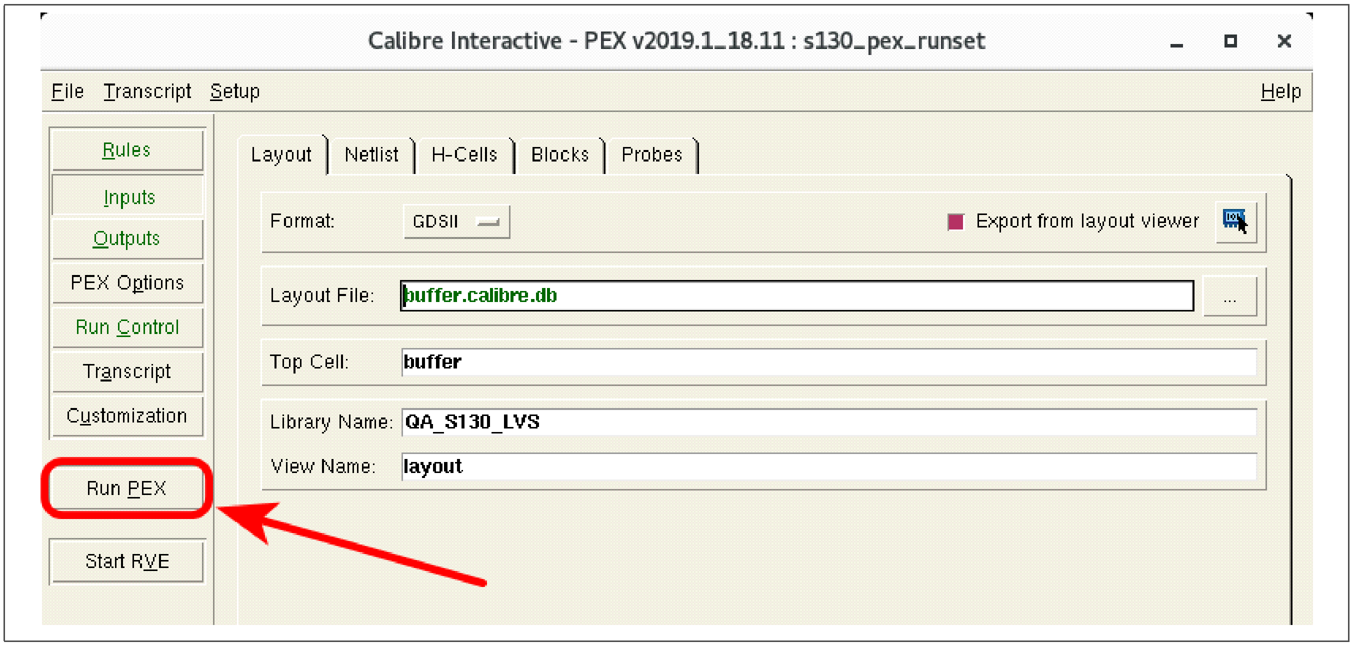


Figure 6-3. Calibre Run PEX Command

4. If this is not the first time Calibre PEX has run in the specified run directory, two “Overwrite file?” dialog boxes might open, one for the existing layout file, like the example shown in *Figure 6-4*, and one for the existing schematic file.

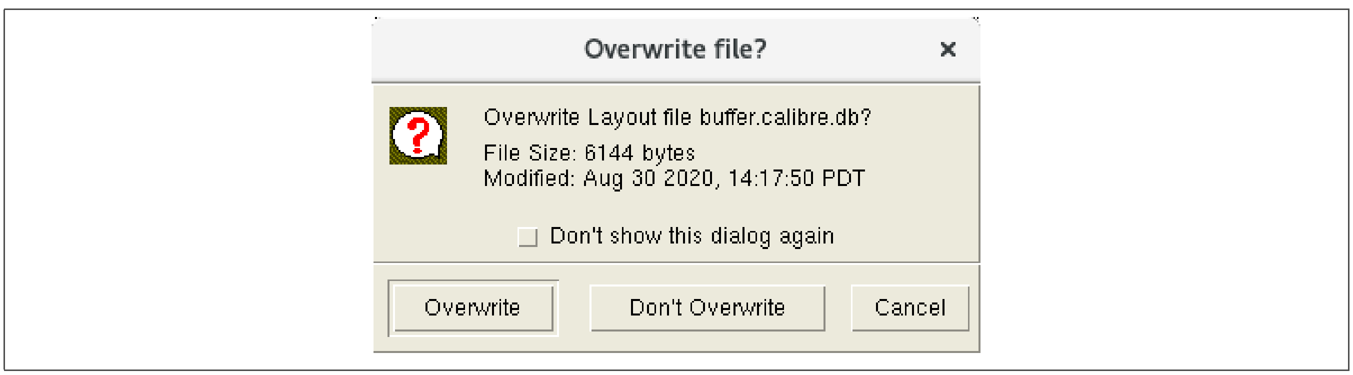


Figure 6-4. Calibre PEX Overwrite File? Window

Click on the **Overwrite** button in the first dialog box if the layout file has changed since the last PEX run, and click on the **Overwrite** button in the second dialog box if the schematic file has changed since the last PEX run. Click on the **Don’t Overwrite** button in the corresponding dialog box if the layout or schematic file has not changed since the previous run.

5. When the run completes, an ASCII netlist of all the design and parasitic devices in the layout opens in a window, like that shown in *Figure 6-5*.

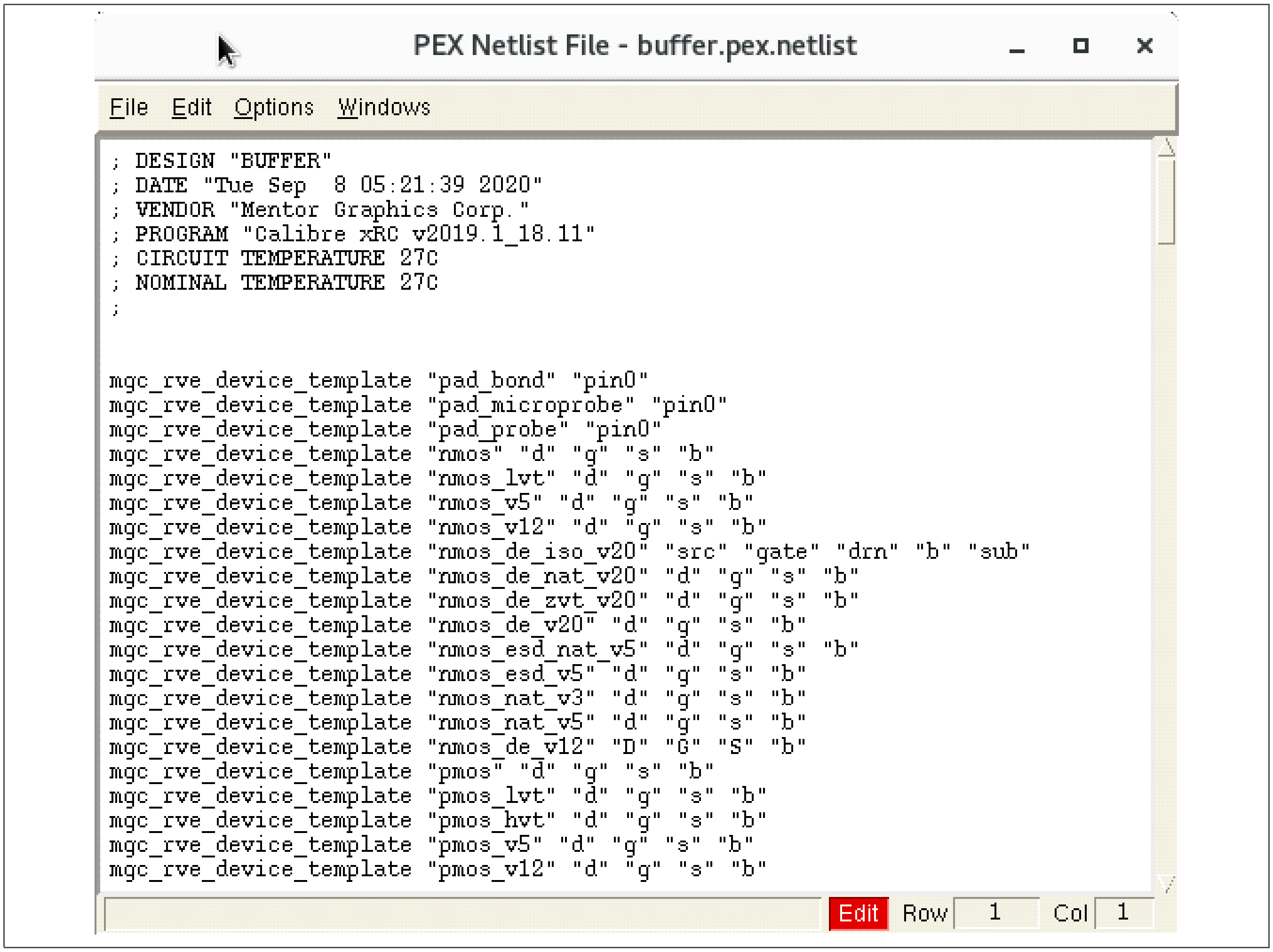


Figure 6-5. Calibre PEX Netlist File

In addition, the Calibre View Setup form, shown in *Figure 6-6* on page 94, opens. Click **OK** at the bottom of the Calibre View Setup form to generate the Calibre view.

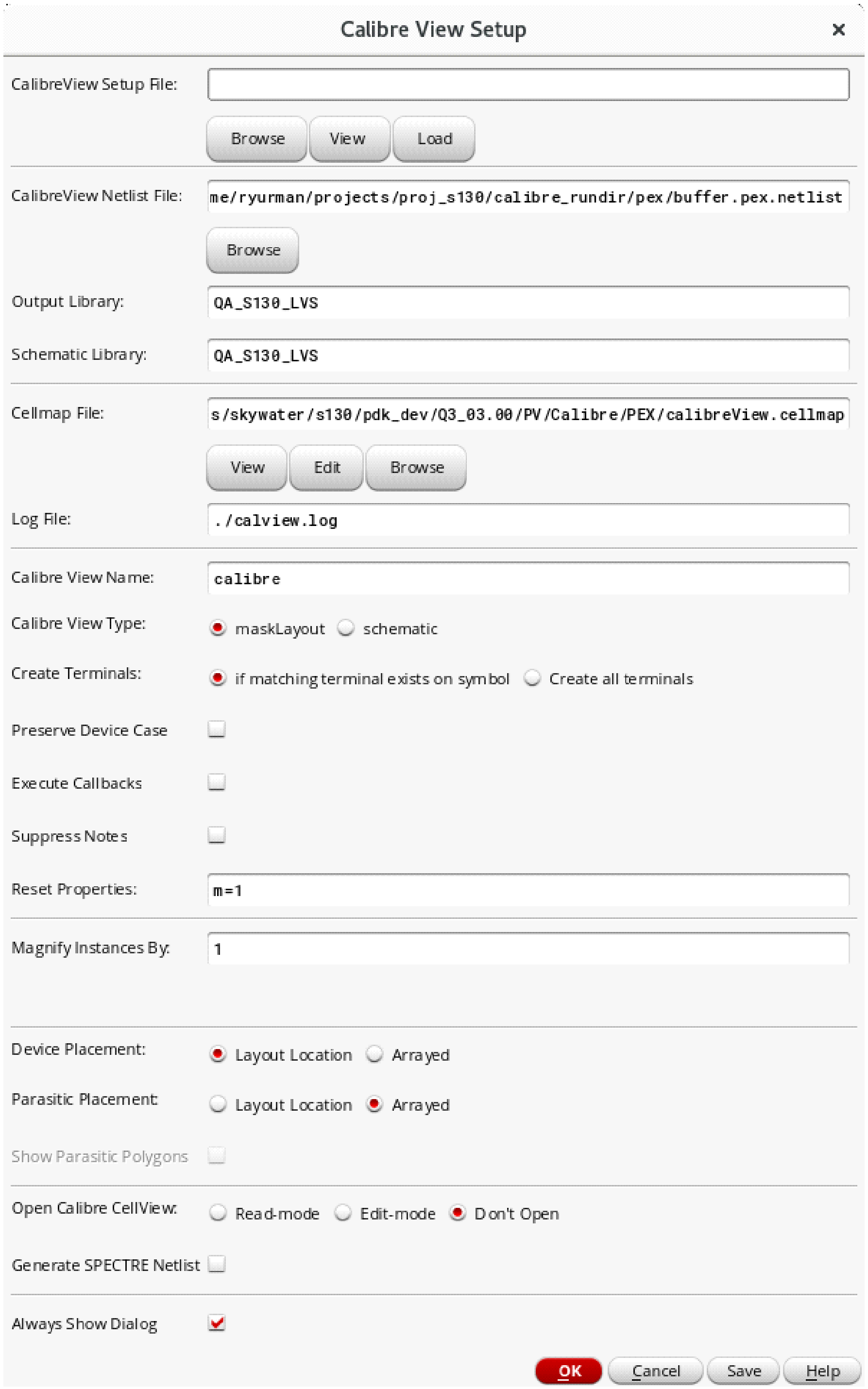


Figure 6-6. Calibre View Setup Form

6. When the Calibre view is generated, close the Calibre Info window, like the example shown in *Figure 6-7*, that indicates the number of warnings and errors found.

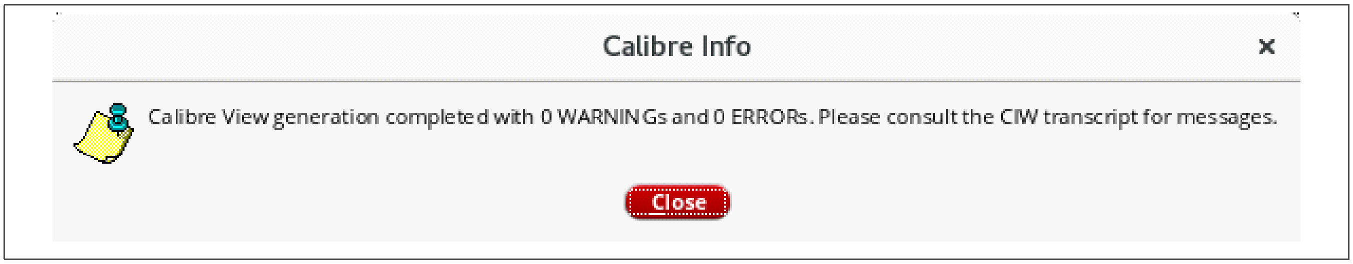


Figure 6-7. Calibre View Info Window

7. Look in the Library Manager to confirm that a Calibre view is listed for the given cell. If so, proceed with simulation, as needed.

6.2 PVS Quantus Extraction

The PVS Quantus extraction tool generates an “av_extracted” view that contains the design and parasitic devices in a layout. Users can simulate this view to get a better idea of how the chip will perform.

Follow these steps to run the PVS Quantus extraction tool:

1. From the layout view, select **Quantus > Run PVS > Quantus**, as shown in *Figure 6-8*, to open the Quantus (PVS) Interface form shown in *Figure 6-9* on page 96. Click **OK**.

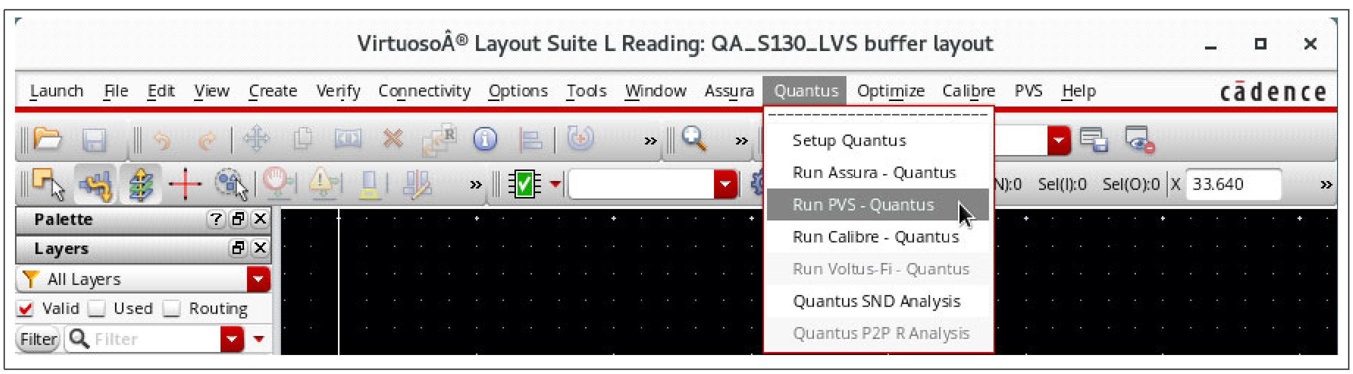


Figure 6-8. PVS Quantus Tool Option

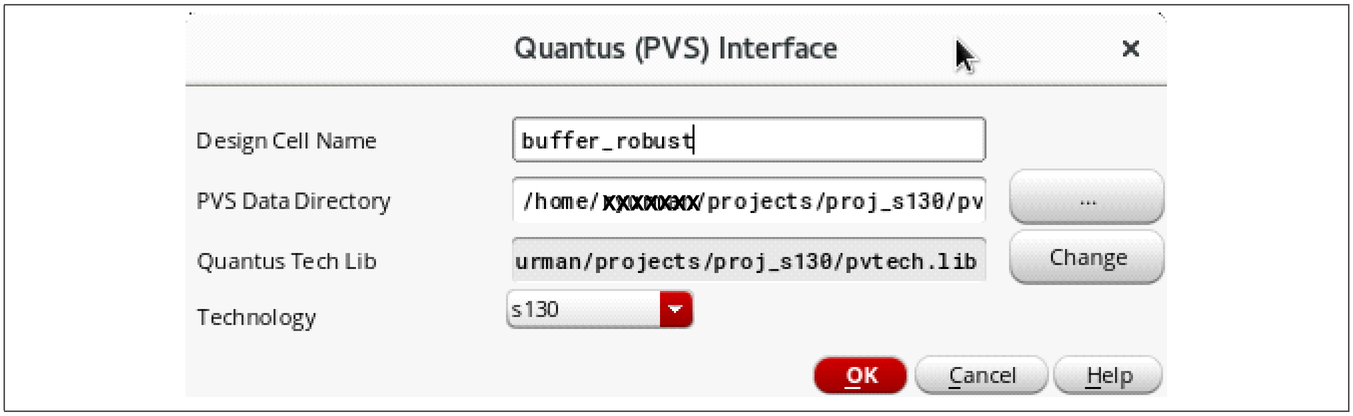


Figure 6-9. Quantus (PVS) Interface Form

2. When the Quantus (PVS) Parasitic Extraction Run Form shown in *Figure 6-10* on page 97 opens, set the **Technology** field to s130 and the **RuleSet** field to QRC.

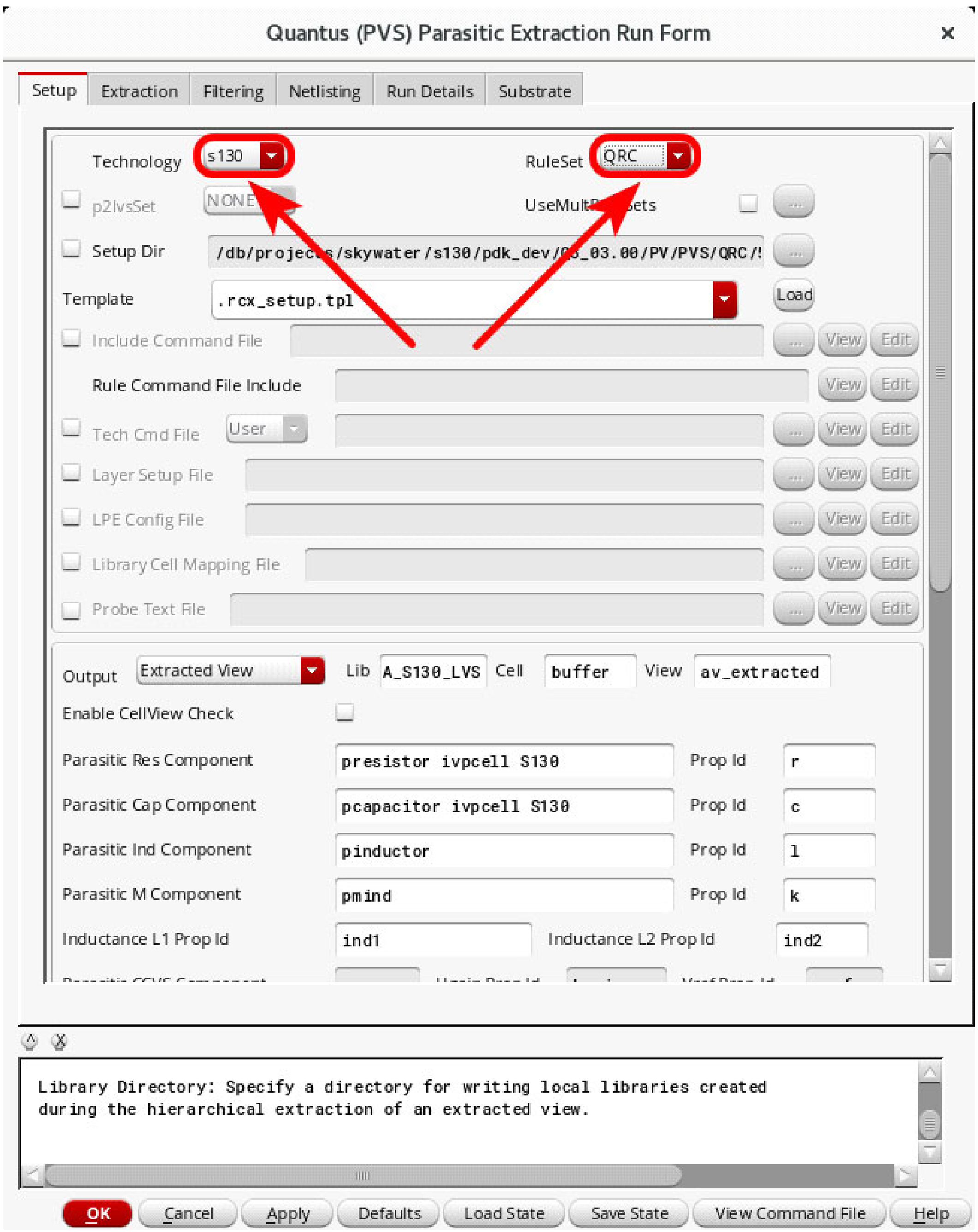


Figure 6-10. Quantus (PVS) Parasitic Extraction Run Form

3. Examine the information under each tab and make any necessary modifications before executing the run.

4. Click **OK** at the bottom left of the Quantus (PVS) Parasitic Extraction Run Form (see *Figure 6-10* on page 97) to begin the run.

5. *Figure 6-11* is an example of the Quantus Progress Form. When it opens, click the **Watch Log file...** button to view the run progress.

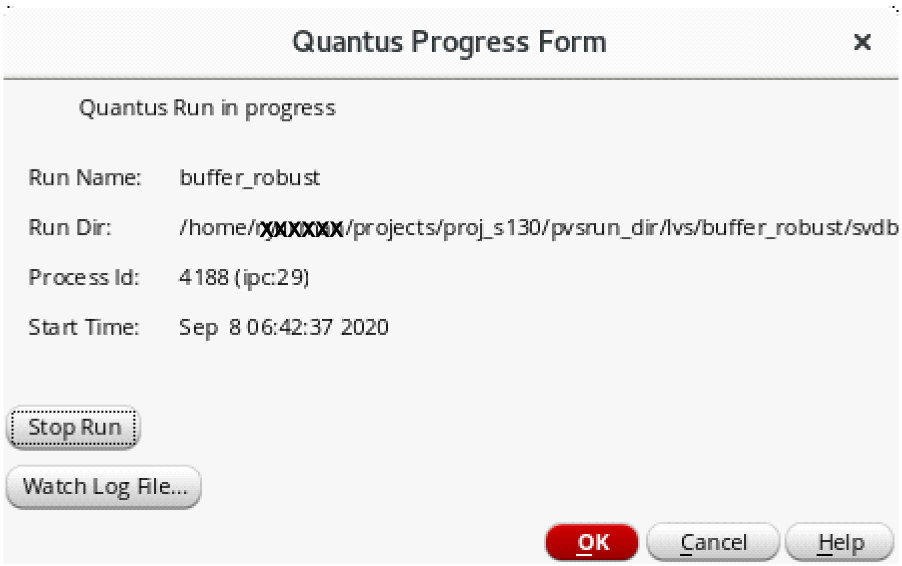


Figure 6-11. Quantus (PVS) Progress Form

6. When the extraction run completes, close the Quantus Run window that indicates the tool successfully created an av_extracted view for the given cell.

7. Look in the Library Manager to confirm that an av_extracted view is listed for the given cell. If so, proceed with simulation, as needed.

7 Simulation

This section describes circuit simulation within the context of the Cadence Virtuoso ADE Explorer. This tool enables circuit exploration of analog and RF integrated circuits at an early stage of design. It supports the following analyses:

- Process corner analysis
- Monte Carlo statistical analysis
- Sensitivity and reliability analysis (important for ISO 26262 compliance)

The GUI displays the simulation data in a coherent view, showing the circuit status under various operating conditions or over process corners.

The ADE Explorer can function in two modes:

Schematic mode	Integrates with the Cadence Spectre Simulation Platform to support real-time tuning.
GUI mode	Features a specification-driven environment that allows designers to set measurements and develop datasheets which show the pass/fail status of the measurements during simulation.

The following sections describe circuit simulation using the Spectre Simulation Platform within ADE.

7.1 Model Selection

To start ADE, select **Setup > Model Libraries**. A form similar to the example shown in *Figure 7-1* opens.

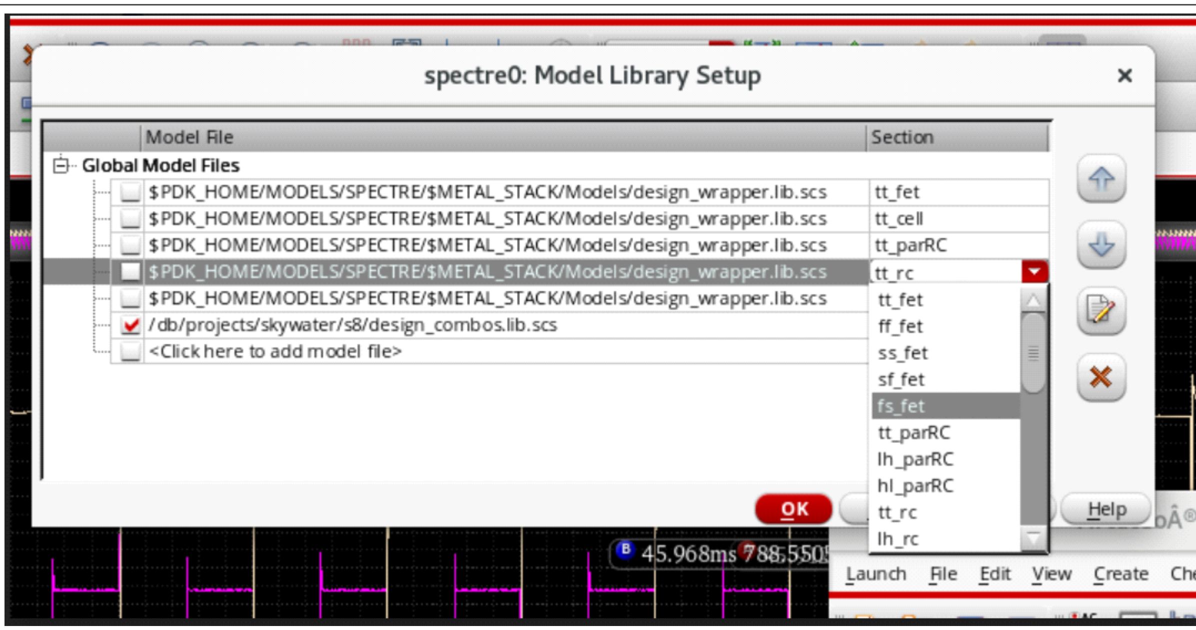


Figure 7-1. Library Model Selection Example for Simulation

Table 7-1 describes the selection of device models listed under **Section** on the right side of the setup form.

Table 7-1. Device Model Selection

Section	Description
**_fet	MOS field-effect transistor models
**_parRC	Extracted parasitic passive element model
**_rc	Resistor or capacitor model
npn_*	Bipolar junction transistor corner model

The default MOS FET model selection is tt_fet, where "tt" indicates that both the PMOS and NMOS FET values are typical. The same applies to the default resistor and capacitor model selection, tt_rc, where both the resistor (r) and capacitor (c) values are typical. See *Table 7-2 Device Process Corners* on page 102 for more information about the items listed under **Section** in *Figure 7-1* on page 99.

To change the model selection, double-click the **Section** to be changed. Then click the red drop-down arrow to the right of the **Section** field to expand the menu.

Select a model combination for each Section type (fet, cell, rc, and so on). When done, click **OK** at the bottom of the form.

Note: All model Sections are contained within a single file, that is, the file name listed under **Model File** on the Model Library Setup form is the same for all **Section** selections. Therefore, a user can select, for example, tt_rc and lh_rc on two separate lines. This practice is discouraged because the model on the second line redefines the model on the first line, causing simulation run-time errors.

7.2 Parameter Definition

To parameterize each Section, double-click the **Section** field for the .lib file shown in *Figure 7-2* on page 101 (that is, the file with the check mark on the left), and then type the VAR("<parameterName>") value in the field. After clicking **OK**, the <parameterName> is automatically added to list of **Design Variables** in the ADE L form. In the example shown in *Figure 7-2*, the <parameterName> "MODELS" in the Model Library Setup window on the left is listed under **Design Variables** in the ADE L form on the right.

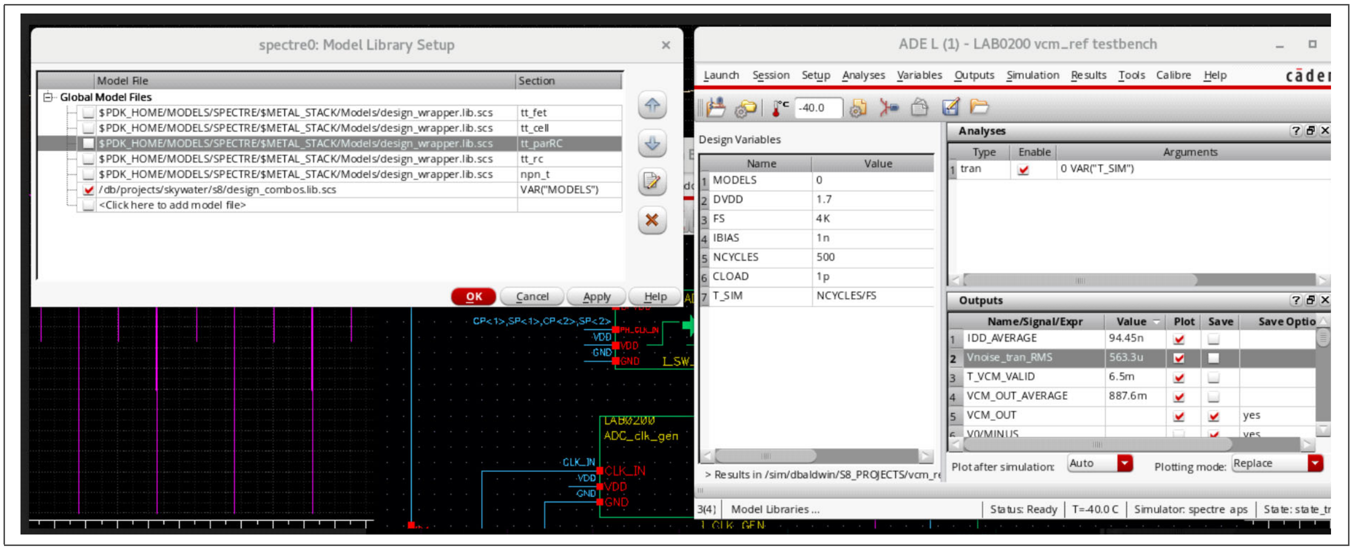


Figure 7-2. Simulation Parameter Form Example

7.3 Process Corners and Monte Carlo Simulation

Use the Model GUI ADE Simulation Setup form shown in *Figure 7-3* to change the process corners for Monte Carlo simulation.

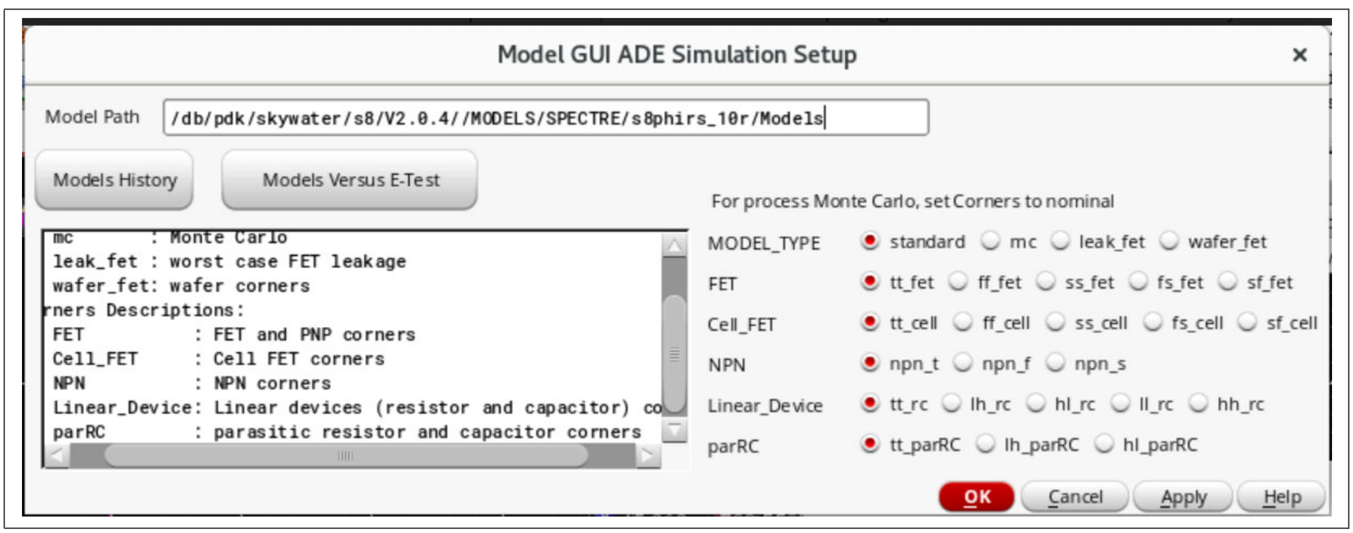


Figure 7-3. Setup Form for Changing Monte Carlo Simulation Corners

Table 7-2 defines the device process corners used during Monte Carlo simulation that correspond to the device models listed under **Section** in the Model Library Setup Form (see *Figure 7-1* on page 99).

Table 7-2. Device Process Corners

Device Corner	Devices	Section	Included File	Description
Transistor corner	FETs, PNP transistors	tt_fet	tt.cor	Typical PMOS, typical NMOS
		ff_fet	ff.cor	Fast PMOS, fast NMOS
		ss_fet	ss.cor	Slow PMOS, slow NMOS
		fs_fet	fs.cor	Fast PMOS, slow NMOS
		sf_fet	sf.cor	Slow PMOS, fast NMOS
Parasitic corner	Parasitic resistors and capacitors	tt_parRC	trtc.cor	Typical resistor, typical capacitor
		hl_parRC	hrlc.cor	High resistor, low capacitor
		lh_parRC	lrhc.cor	Low resistor, high capacitor
Linear device corner	Resistors and capacitors	tt_rc	trtclin.cor	Typical resistor, typical capacitor
		lh_rc	lrhclin.cor	Low resistor, high capacitor
		hl_rc	hrlclin.cor	High resistor, low capacitor
		ll_rc	lrlclin.cor	Low resistor, low capacitor
		hh_rc	hrhclin.cor	High resistor, high capacitor
Cell transistor corner	Cell FETs, low-leakage models	tt_cell	ttcell.cor	Typical PMOS, typical NMOS
		ff_cell	ffcell.cor	Fast PMOS, fast NMOS
		ss_cell	sscell.cor	Slow PMOS, slow NMOS
		fs_cell	fscell.cor	Fast PMOS, slow NMOS
		sf_cell	sfcell.cor	Slow PMOS, fast NMOS
Wafer	Standard FETs, bipolar junction transistors	wafer_fet	wafer.cor	FET models extracted from wafer
	Cell FETs	wafer_cell	wafer_cell.cor	
Leakage	Standard FETs, bipolar junction transistors	leak_fet	leak.cor	Average worst-case leakage models
	Cell FETs	leak_cell	leakcell.cor	
NPN corner	NPN corners	npn_t	npn_t.cor	Typical NPN
		npn_f	npn_f.cor	Fast NPN
		npn_s	npn_s.cor	Slow NPN
Monte Carlo	All devices	mc	monte.cor, critical_params.cor	Monte Carlo process and mismatch

To run a Monte Carlo simulation, all the corner sections as well as the mc section must be included.

Appendix A. Density Checking

A.1 Calibre DRC Density Checking

A.1.1 Density Checks Customization Menu

The check settings customization menu opens when Calibre starts. To change the check settings between runs, click the Customization button on the left side of the Calibre Interactive menu. In the customization menu shown in *Figure A-1*, density checks will be run and the options to check local density and generate predictive fill are selected.

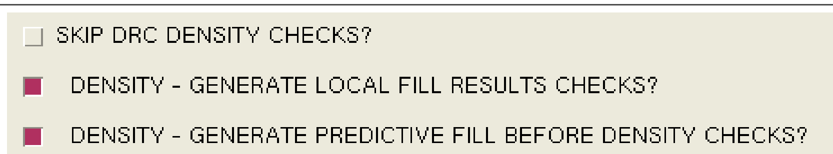


Figure A-1. Calibre DRC Density Checks Customization Form

A.1.2 Density DRC Setup

Calibre has a default limit of 1000 outputs for any given error. To override this option, click **Setup** at the top of the form shown in *Figure A-2*. Select **DRC Options**, and set the **Max. errors generated per check** to **All** before clicking the **Run DRC** button.

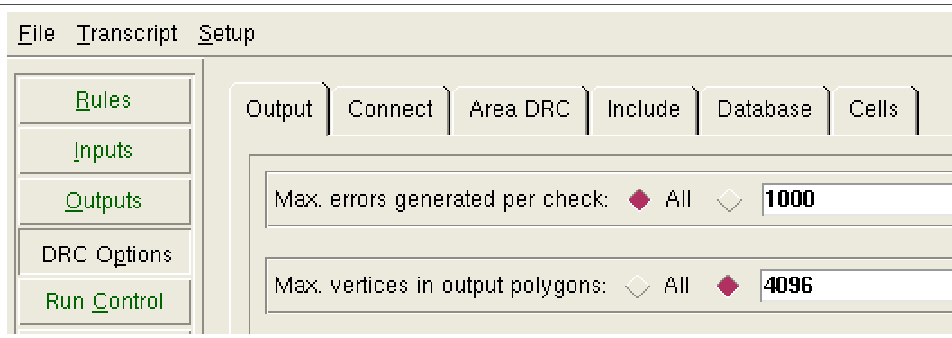


Figure A-2. Calibre DRC Options Tab

A.1.3 Density Check Results

There are two types of Calibre DRC density checks:

- Chip-level density checks run in a 700 µm × 700 µm window stepped at 70 µm. They are the foundry-required checks.
- Local density checks run in a 100 µm × 100 µm window stepped at 50 µm. They can be used to localize areas that cause chip-level density violations; otherwise, they can be ignored.

Density check results enable users to view predictive fill for each layer. A large number of results, like those shown in the example in *Figure A-3*, can take a long time to view. In this example, only local density violations are listed, and they can be ignored because there are no corresponding chip-level violations.

Check / Cell	Results	Flat
Check view_met1_predictive_fill	226517	231328
Check met1.local.high.DEN.2	3	3
Check view_met2_predictive_fill	277815	277839
Check met2.local.low.DEN.1	1	1
Check met2.local.high.DEN.2	3	3
Check view_met3_predictive_fill	242890	242890
Check met3.local.high.DEN.2	2	2
Check view_met4_predictive_fill	238623	238631
Check met4.local.high.DEN.2	2	2

Figure A-3. Calibre Met1–Met4 Density Check Results Example

In the example shown in *Figure A-4*, met5 has a chip-level violation. Because the violation is based on predictive fill, which is only an estimate for foundry-produced fill, the user must check the results and view the value of the violation in the ASCII results database.

Check view_met5_predictive_fill	19098	19098
Check met5.local.high.DEN.2	3	3
Check met5.chip.high.DEN.4	1	1

Figure A-4. Calibre Met5 Density Check Results Example

Calibre refers to the ASCII results database as an RDB file. The density results can be viewed by executing the **File > Side RDBs…** command in the RVE window. For the met5 chip-level density violation, select the `met5_chip_high.rdb` file from the list of available files. *Figure A-5* on page 105 shows the new tab for the side RDB file in the RVE pane.

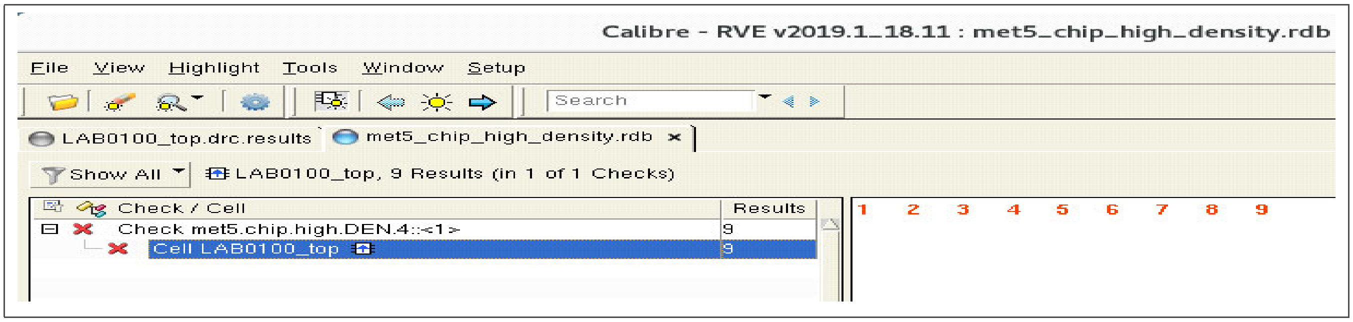


Figure A-5. Calibre RVE Side RDB File Tab

Click each listed density violation for more information. For example, *Figure A-6* shows details about the met5.chip.high.DEN.4 chip density rule violation.

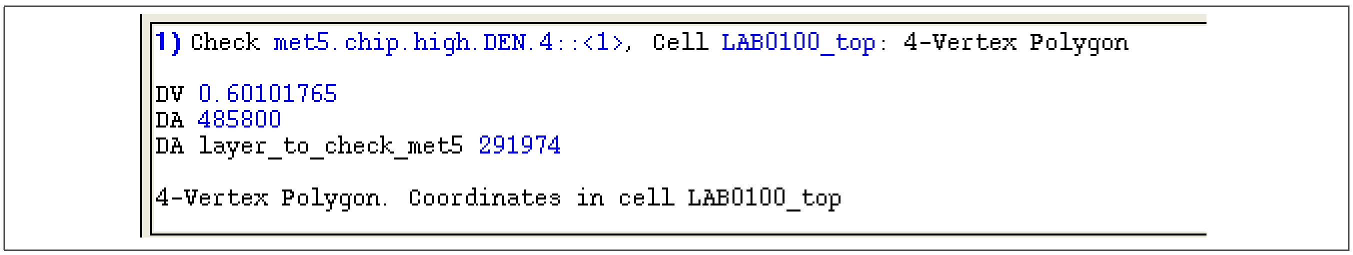


Figure A-6. Calibre Met5 Chip Density Violation Details

In this example, the DV value (times 100) indicates the violation is a chip density value of 60.101765%. The *S130 Design Rule Manual* lists a maximum density of 60% for the met5.chip.high.DEN.4 design rule. This violation can likely be ignored because the check used predictive fill, which is estimated.

View the details for each violation. Any result within 5% of the target can also likely be ignored.

A.1.4 Density Check Results for Poly and Diffusion

Simulated chip density fill favors diffusion density over poly density (the two cannot overlap). For S130, the diffusion first-pass fill size of 4.08 μm^2 tends to inhibit the production of simulated poly fill, which has a first-pass fill size of only 0.72 μm^2. Consequently, predictive fill for poly is less reliable than predictive fill for other layers. The poly density target is also a narrow range, from 30–40%.

A.1.5 Density Check Results—Histogram Examples

A histogram can be produced for any density result by right-clicking over the violation in the side RDB window and selecting **Histogram > DV**. *Figure A-7* on page 106 is an example of a histogram for a low poly chip density result. This histogram shows a poly density range of 27.8% to 29.75%, which is within the 5% error margin. Therefore, this low poly chip density error can probably be ignored.

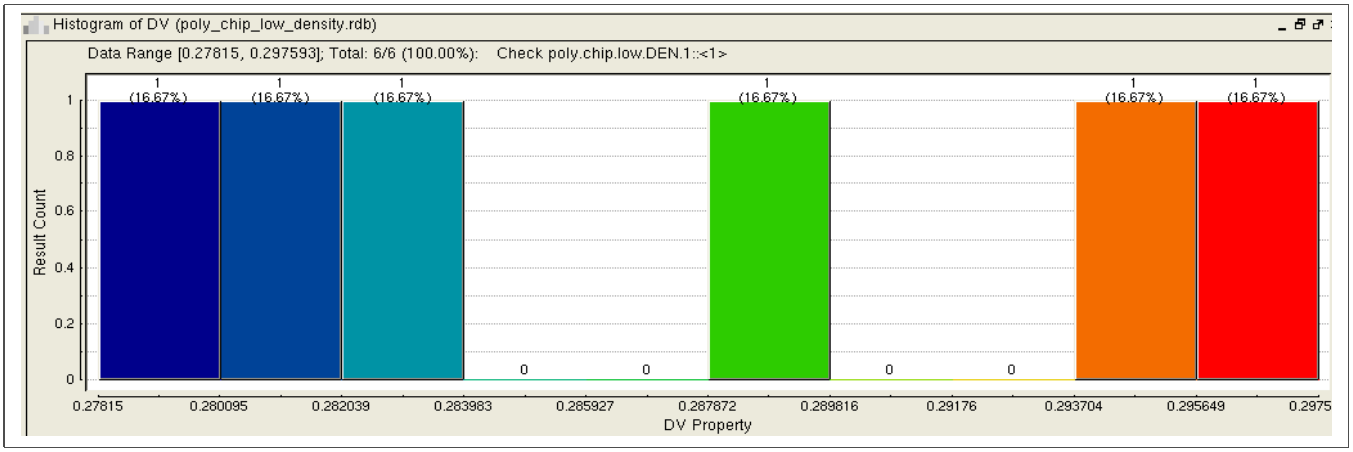


Figure A-7. Calibre Low Poly Chip Density Histogram Example

The histogram example in *Figure A-8* shows a low diffusion chip density in the 20% range, well below the 28% diff.chip.low.DEN.1 design rule minimum.

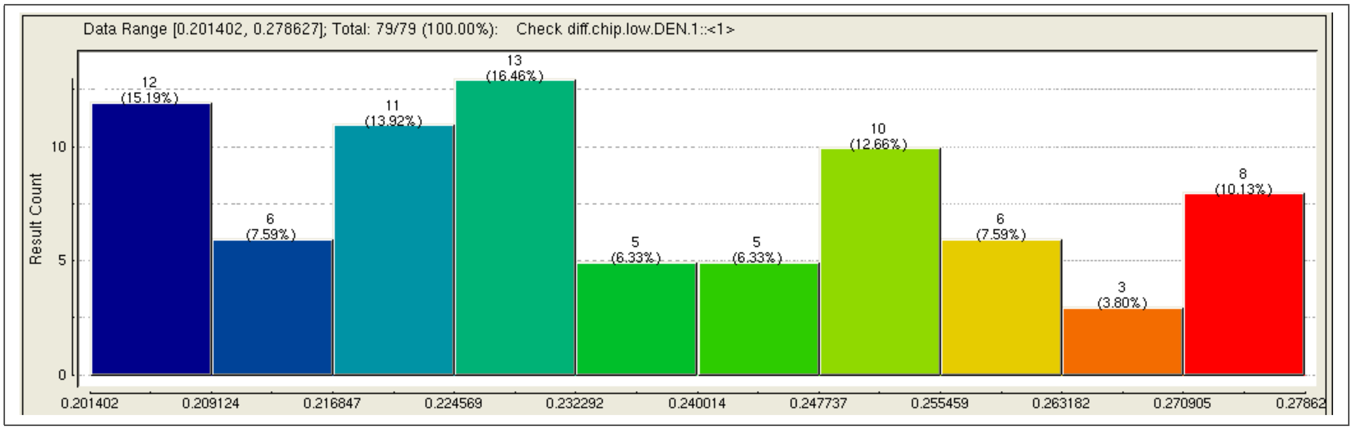


Figure A-8. Calibre Low Diffusion Chip Density Histogram Example

Highlighting the 20% histogram bar in the layout, as shown in *Figure A-9* on page 107, does little to localize the chip density error.

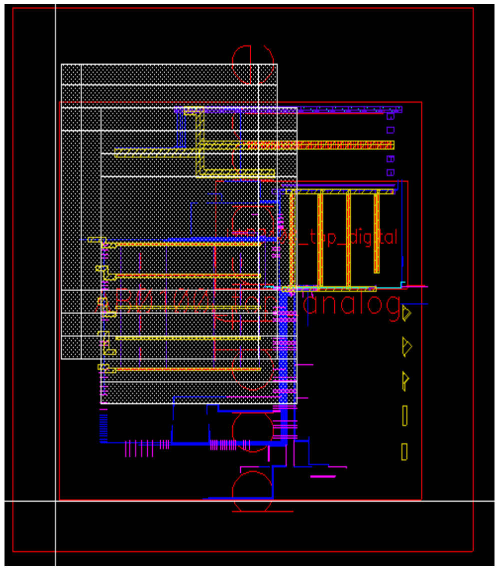


Figure A-9. Calibre Chip Density Results Histogram Bar Highlighted in Layout

Each bar in the histogram results shown in *Figure A-8* on page 106 can be subdivided into another histogram by right-clicking over the bar and selecting **Histogram this bar** or **Histogram this bar by > DV**. Highlighting each new histogram bar in the layout might be sufficient to localize the error.

Local density results, which use a smaller window and step size, can also help localize a chip density error. In *Figure A-10*, a histogram was produced from the diffusion local density RDB file, and the left-most bar was highlighted in the layout to show a more accurate picture of the violation.

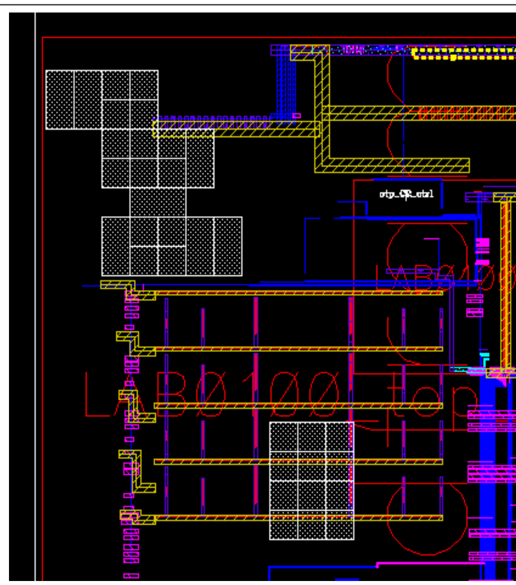


Figure A-10. Calibre Local Density Results Histogram Bar Highlighted in Layout

The layout shown in *Figure A-11* highlights areas of densely packed poly resistors that prevent diffusion predictive fill generation. The poly resistors must be spaced farther apart to permit additional diffusion fill.

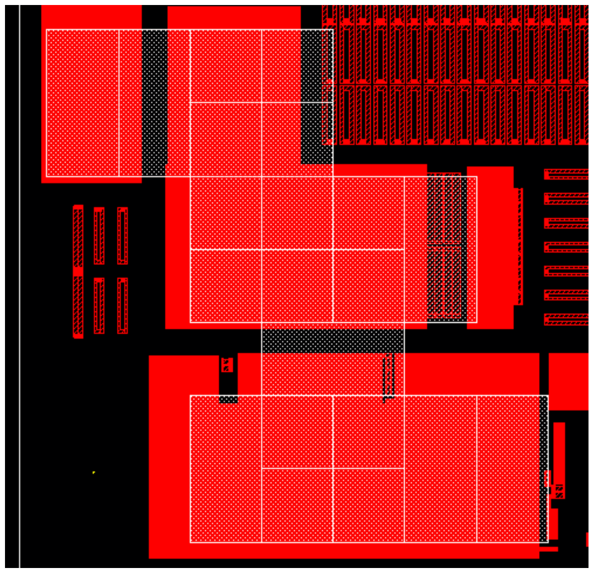


Figure A-11. Calibre Densely Packed Poly Resistors Causing Diffusion Chip Density Violations

A.2 PVS DRC Density Checking

A.2.1 Density Checks Configuration Menu

To run the density checking rules, open the **Configurator** tab in the PVS DRC Run Submission Form. Turn off **SKIP DENSITY CHECKS?** and turn on both **DENSITY – GENERATE LOCAL FILL RESULTS FOR DENSITY CHECKS?** and **DENSITY – GENERATE PREDICTIVE FILL BEFORE DENSITY CHECKS?**, as shown in *Figure A-12* on page 109.

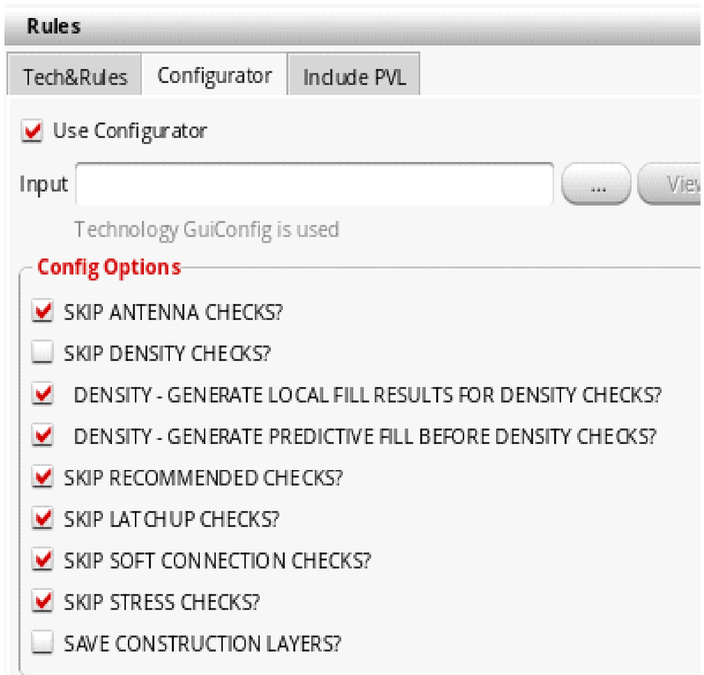


Figure A-12. PVS DRC Density Checks Configuration Options

A.2.2 Density DRC Setup

Before starting the DRC run, set the **Report Limit** to “ALL” in the **Output** panel of the DRC Run Submission Form, as shown in *Figure A-13*. Click **Submit** to start the run.

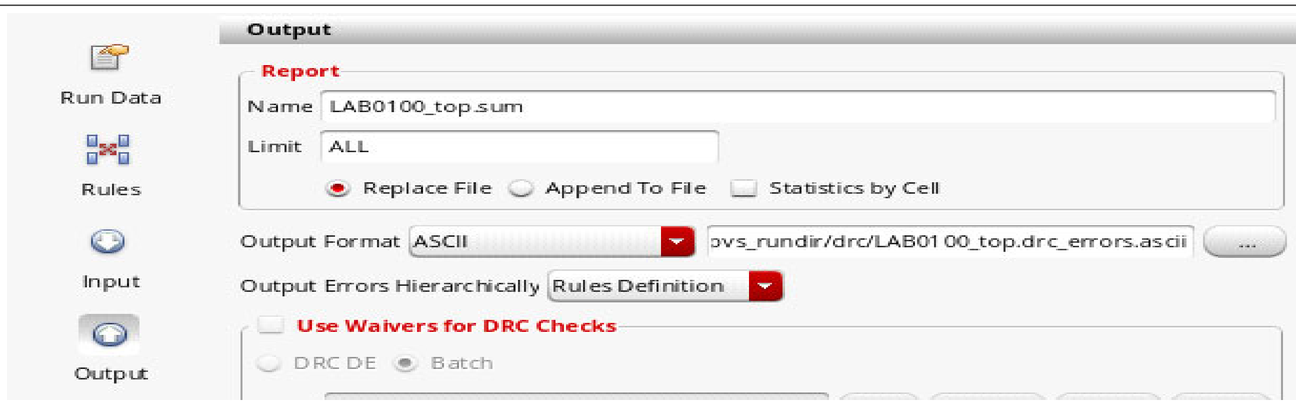


Figure A-13. PVS DRC Run Submission Form

The PVS density rule file outputs a large number of results as it displays predictive fill rectangles. The number of results *always* exceeds both the “total number of errors” and the “total number of errors to display per rule” limits set by Cadence. After the run completes, users must override the default values and reload the errors before proceeding. In the Results Viewer, select **Options > Preferences** to open the Preferences form shown in *Figure A-14*.

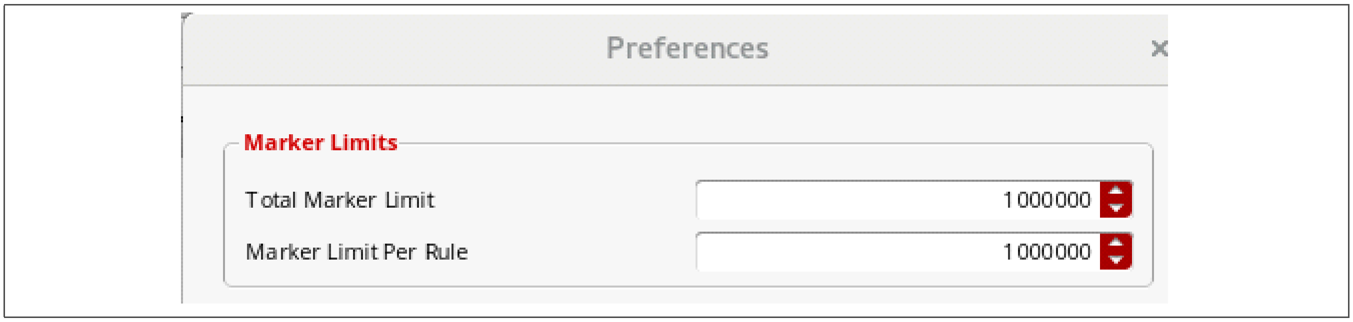


Figure A-14. PVS Results Viewer Preferences Form

Set the **Total Marker Limit** and **Marker Limit Per Rule** values to much larger numbers, like 100,000,000 each. Then reload the results by executing the **File > Reload DRC Results** command from the Results Viewer menu.

A.2.3 Density Check Results

There are two types of PVS DRC density checks:

- Chip-level density checks run in a 700 µm × 700 µm window stepped at 70 µm. They are the foundry-required checks.
- Local density checks run in a 100 µm × 100 µm window stepped at 50 µm. They can be used to localize areas that cause chip-level density violations; otherwise, they can be ignored.

Density check results enable users to view predictive fill for each layer. A large number of results, like those shown in the example in *Figure A-15*, can take a long time to view. In this example, only local density violations are listed, and they can be ignored because there are no corresponding chip-level violations.

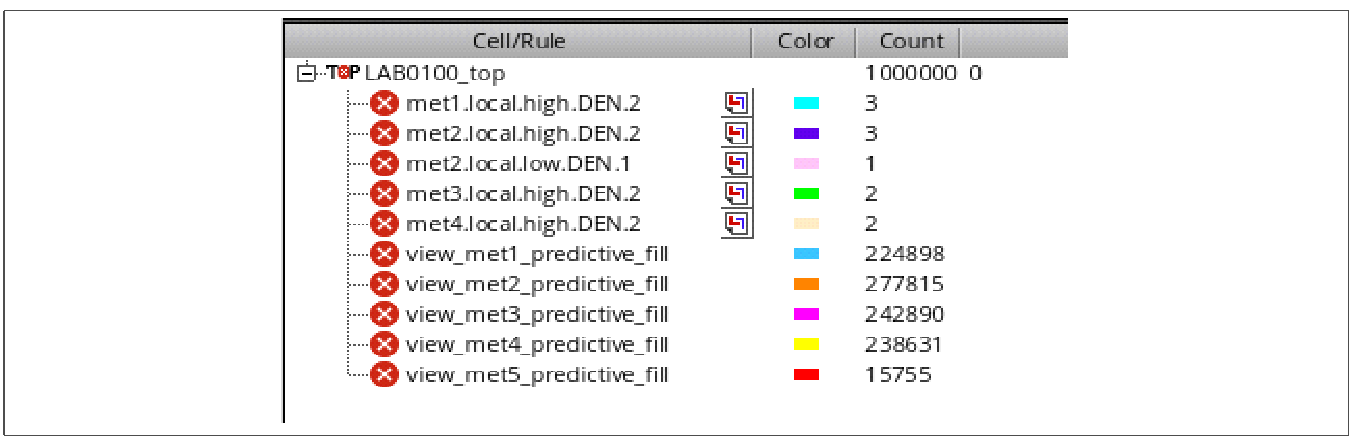


Figure A-15. PVS Met1–Met4 Density Check Results Example

PVS refers to the ASCII results database as an RDB file. The density results can be viewed by executing the **File > Open ASCII DRC Results…** command in the Results Viewer window. For a low diffusion local density violation, select the `diff_local_low_density.rdb` file from the list of available files. *Figure A-16* shows a new tab for this RDB file in the DRC Results Viewer form.

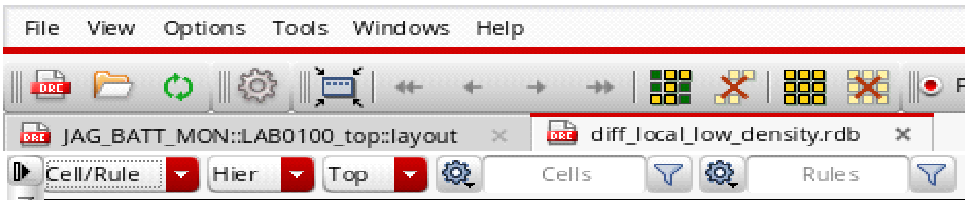


Figure A-16. PVS DRC Results Viewer RDB File Tab

Click each listed density violation to open a pane with more information. For example, *Figure A-17* shows details about the diff.local.low.DEN.1 local density rule violation.

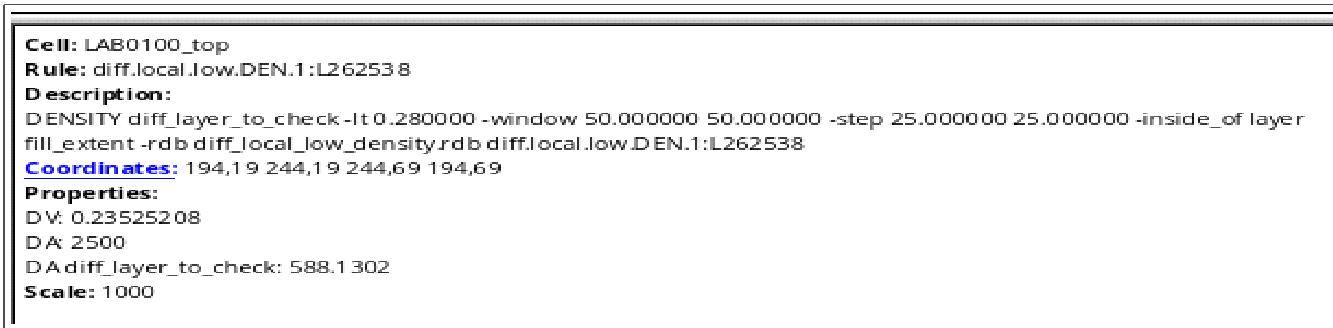


Figure A-17. PVS Diffusion Local Density Violation Details

In this example, the DV value (times 100) indicates the violation is a local density value of 23.52508%. The *S130 Design Rule Manual* lists a minimum density of 28% for the diff.local.low.DEN.1 design rule. Note that the level of accuracy implied by five decimal places is misleading because the check used predictive fill, which is estimated.

View the details for each violation. Any result within 5% of the target can likely be ignored.

A.2.4 Density Check Results for Poly and Diffusion

Simulated chip density fill favors diffusion density over poly density (the two cannot overlap). For S130, the diffusion first-pass fill size of 4.08 μm^2 tends to inhibit the production of simulated poly fill, which has a first-pass fill size of only 0.72 μm^2. Consequently, predictive fill for poly is less reliable than predictive fill for other layers. The poly density target is also a narrow range, from 30–40%.

A.2.5 Density Check Results—Histogram Examples

A histogram can be produced for any density result by right-clicking over the violation in the Results Viewer window and selecting **Histogram**. *Figure A-18* on page 112 is an example of a histogram for a low diffusion local density result. This histogram shows a diffusion density range of 1.75% to 28%.

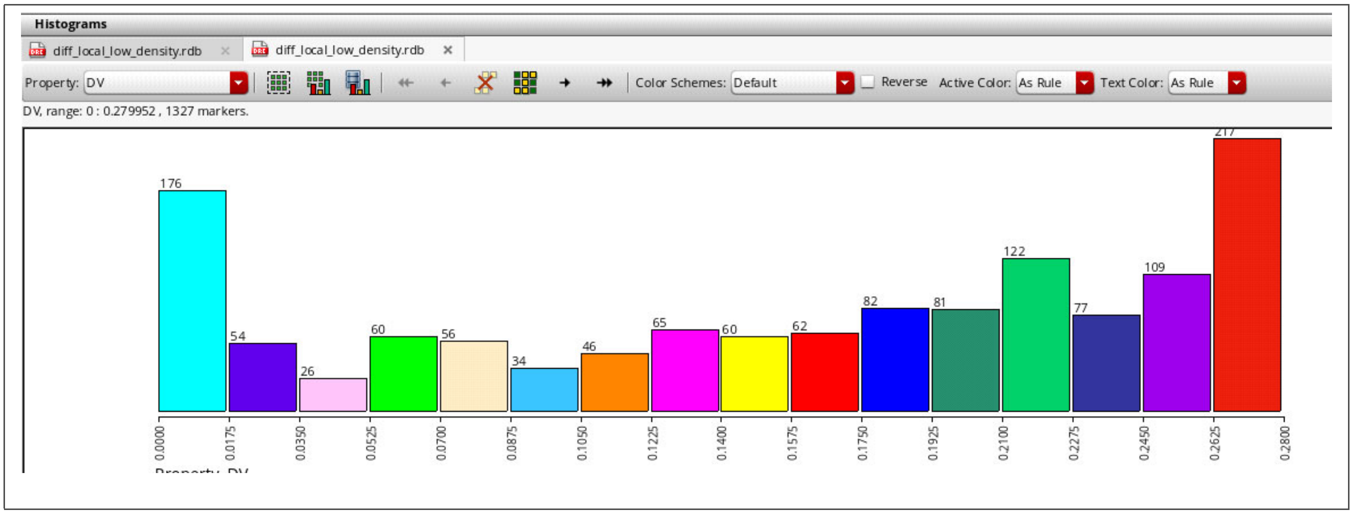


Figure A-18. PVS Low Diffusion Local Density Histogram Example 1

Each bar in the histogram results shown in *Figure A-18* can be subdivided into another histogram by clicking the **Build histogram from selection** toolbar button above the histogram. Select a bar by left-clicking on the bar until it has a dashed outline, as shown in *Figure A-19*.

Figure A-19. PVS Build Histogram from Selection Toolbar Button

In addition, each bar of the histogram results can be highlighted in the layout by selecting the bar and clicking the toolbar button shown in *Figure A-20* on page 113.

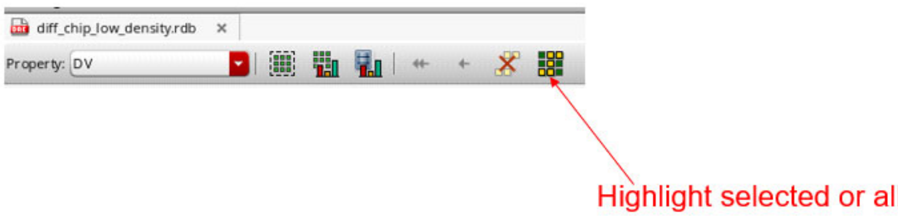


Figure A-20. PVS Histogram Results Highlight in Layout Toolbar Button

Highlighting the first histogram bar in the layout, as shown in *Figure A-21*, does little to localize the density error.

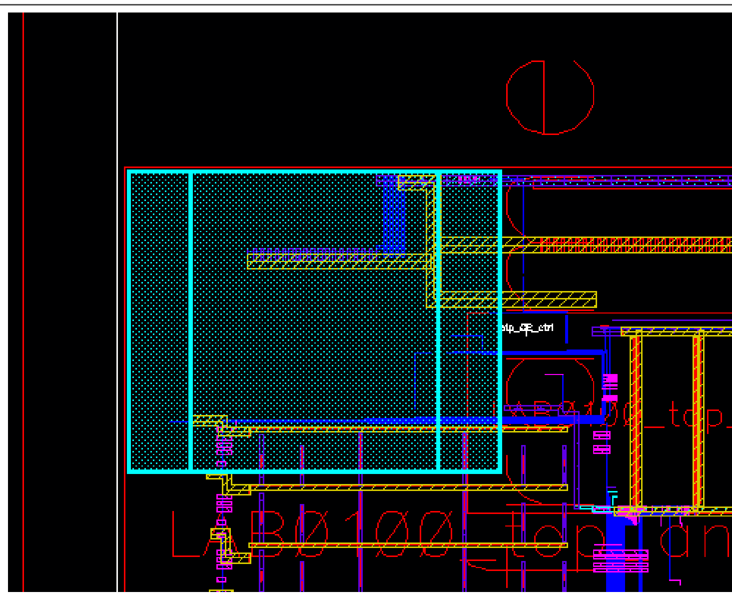


Figure A-21. PVS Density Results Histogram Bar Highlighted in Layout

Local density results, which use a smaller window and step size, can help localize a chip density error. *Figure A-22* on page 114 is a histogram generated from the diffusion local density RDB file.

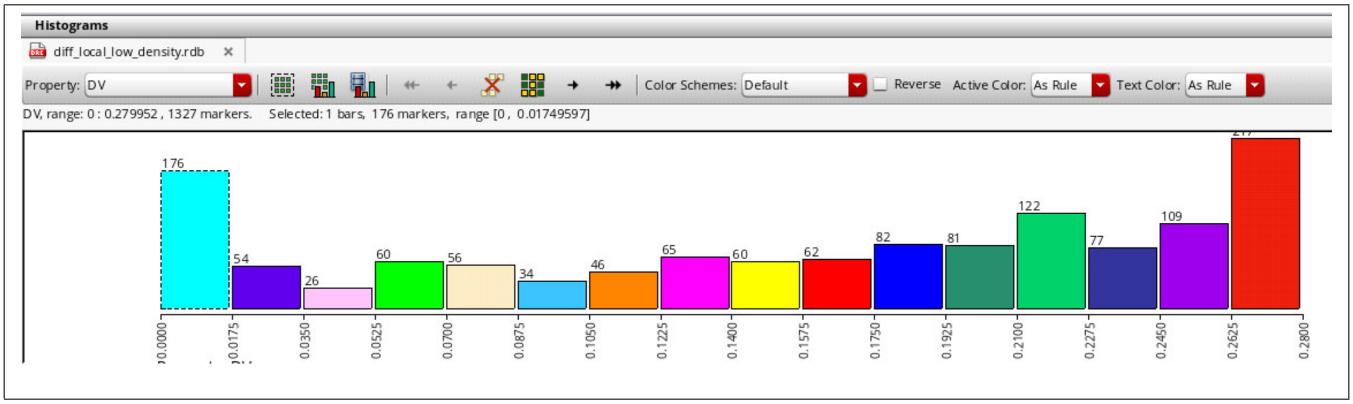


Figure A-22. PVS Low Diffusion Local Density Histogram Example 2

In *Figure A-23*, the first bar shown in *Figure A-22* is highlighted in the layout to help determine the cause of a chip-level density violation.

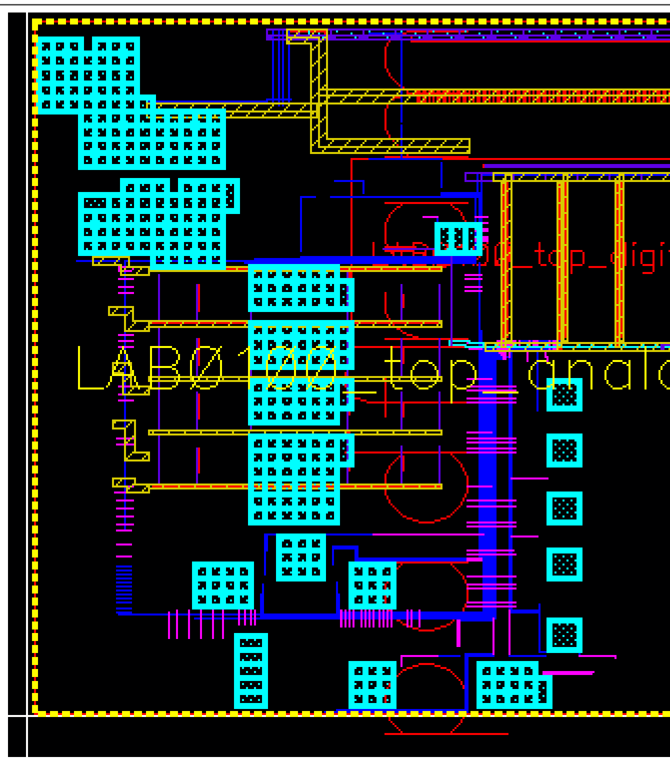


Figure A-23. PVS Local Density Results Histogram Bar Highlighted in Layout